

Estimating Causal Effects of Discrete and Continuous Treatments with Binary Instruments*

Victor Chernozhukov Iván Fernández-Val

Sukjin Han Kaspar Wüthrich

September 7, 2026

Abstract

We propose an instrumental variable framework for identifying and estimating causal effects of discrete and continuous treatments with binary instruments. The basis of our approach is a local copula representation of the joint distribution of the potential outcomes and unobservables determining treatment assignment. This representation allows us to introduce an identifying assumption, so-called *constant local dependence*, that restricts the local dependence of the copula with respect to the treatment propensity. We show that constant local dependence identifies treatment effects for the entire population and other subpopulations such as the treated. The identification results are constructive and lead to practical estimation and inference procedures based on distribution regression. An application to estimating the effect of sleep on well-being uncovers interesting patterns of heterogeneity.

JEL Numbers: C14, C21, C31.

Keywords: Quantile treatment effects, endogeneity, binary instruments, copula.

*The authors respectively represent MIT, BU, U of Bristol, and U of Michigan. Wüthrich is also affiliated with CESifo. For helpful comments, we are grateful to the editor Keisuke Hirano, three anonymous referees, Yingying Dong, Dalia Ghanem, Jinyong Hahn, Stella Hong, Desire Kedagni, Michael Knaus, Eric Mbakop, Ulrich Mueller, Whitney Newey, Jack Porter, Andres Santos, Ed Vytlacil, Martin Weidner, Daniel Wilhelm, and participants in seminars at WashU, Indiana, MSU, Yale, Zhejiang, HKUST, UCSD, Northwestern, Stanford, University of Washington, University of Wisconsin-Madison, LMU, Mannheim, Bonn, Erasmus, Amsterdam, UC3M, Aarhus, Cantabria, and participants at the Asian Meeting of the Econometric Society, the Bristol Econometric Study Group, Munich Econometrics Workshop, KU Leuven Summer Event, California Conference, UCLA Econometrics Mini Conference and IAAE 2026. We also appreciate Matt Hong, Meng Hsuan Hsieh, ChatGPT, Claude, and Refine for excellent research assistance. This research was supported in part through computational resources and services provided by Advanced Research Computing at the University of Michigan, Ann Arbor. Wüthrich gratefully acknowledges funding from a University of Michigan start-up package from the William Haber Economics Faculty Fund. The usual disclaimer applies.

1 Introduction

Endogeneity and heterogeneity are key challenges in causal inference. Endogeneity arises because most treatments and policies of interest are the result of decisions made by economic agents. Unobserved heterogeneity also arises naturally as many of the agents' characteristics are unobserved to the researcher. Accounting for endogeneity and heterogeneity in treatment effects estimation is crucial to answer policy questions, such as how to allocate social resources and combat inequalities. This paper proposes a flexible instrumental variable (IV) modeling framework for identifying heterogeneous treatment effects under endogeneity, which yields practical estimation and inference procedures.

Without additional assumptions, IV strategies cannot point-identify meaningful treatment effects in the presence of heterogeneity. The literature has proposed different solutions to deal with this challenge that exhibit trade-offs between adding structure to the treatment assignment mechanism and potential outcomes. One line of research has restricted the structure and heterogeneity of the potential outcomes while allowing for flexible treatment assignment mechanisms (e.g., [Chernozhukov and Hansen \(2005\)](#) with binary treatments and [Newey and Powell \(2003\)](#) with continuous treatments). Another line of research has shown the usefulness of restricting the treatment assignment while being flexible regarding how the potential outcomes are formed (e.g., [Imbens and Angrist \(1994\)](#) with binary treatments and [Imbens and Newey \(2009\)](#) with continuous treatments).

We explore an intermediate route that imposes structure on the relationship between the treatment assignment and the potential outcomes to achieve point identification of treatment effects. The basis of this approach is a local Gaussian representation of the copula capturing the dependence between the potential outcomes and the unobservable determinants of treatment assignment. This representation is fully *nonparametric*, that is, it does not require that potential outcomes and treatment unobservables are jointly or marginally Gaussian. Indeed, the bivariate Gaussian structure always holds locally by treating the correlation parameter as an implicit function that equates the bivariate Gaussian copula with the copula of the potential outcome and selection unobservables. We use this representation to introduce an assumption that has not been previously considered for identification of treatment effects. This assumption, so-called *constant local dependence* (CLD), restricts the local dependence of the copula with respect to treatment propensity, and thus it restricts the form of endogeneity. We provide a range of statistical and economic interpretations of CLD, which clarify the empirical content of the assumption.

We show that, even with a binary IV, CLD identifies quantile and average treatment effects (QTE and ATE) of binary and ordered discrete treatments and quantile and average

structural functions (QSF and ASF) of continuous treatments for the entire population. The results for ordered discrete and continuous treatments have particular empirical relevance, as previous identification results for global treatment parameters are scarce and typically rely on rich instrument variation; see the literature review below.¹ As a byproduct, we also identify the dependence function, which captures the direction and magnitude of endogenous selection and may be of interest in applications. When covariates are available, we impose CLD conditional on them, allowing for an additional source of heterogeneity in our model.

The CLD-based identification strategy is constructive and leads to practical semiparametric estimation procedures based on distribution regression for both discrete and continuous treatments.² We establish the asymptotic Gaussianity of the estimators of the potential outcome distributions, dependence function, and functionals such as the unconditional QSF and QTE. We show that the bootstrap is valid for estimating the limiting laws and provide an explicit algorithm for constructing uniform confidence bands.

The proposed method adds to the IV toolkit by expanding the directions of modeling trade-offs in identifying treatment effects. We allow for richer patterns of effect heterogeneity, compared to [Chernozhukov and Hansen \(2005\)](#) and [Newey and Powell \(2003\)](#), and more heterogeneity in the treatment assignment, compared to [Imbens and Angrist \(1994\)](#) and [Imbens and Newey \(2009\)](#), while imposing more restrictions on the dependence structure (i.e., the form of endogeneity), as we detail below.

We apply our method to estimate the distributional effects of sleep on well-being. In this case, sleep time is treated as a continuous treatment. We use data from the experimental analysis by [Bessone et al. \(2021b\)](#), who studied the effects of randomized interventions to increase the sleep time of low-income adults in India. A simple two-stage least squares analysis suggests that sleep has moderate average effects on well-being. Using our method, we document interesting patterns of heterogeneity across the distributions of sleep time and well-being that are missed by average effects. For example, the quantile treatment effects of increasing sleep on well-being at the lower tail, which are particularly policy-relevant, are larger than the corresponding average effect. The estimates are pointwise significant in the tails, but the uniform confidence band includes zero at all quantiles, which is unsurprising given the small sample size. Our estimates also suggest greater heterogeneity

¹There is a wide range of empirical studies estimating the effects of continuous and ordered endogenous variables using IVs with limited variation. Prominent examples include work on intergenerational mobility (e.g., [Black and Devereux, 2011](#)), studies of the returns to schooling (e.g., [Angrist and Imbens, 1995](#)), analyses of the impact of air pollution (e.g., [Chay and Greenstone, 2005](#)), demand analyses with discrete IVs (e.g., [Angrist et al., 2000](#)), and studies where discrete experimental treatments are used as IVs for ordered discrete and continuous treatments (e.g., [Bessone et al., 2021b](#)).

²We refer to these estimators as “semiparametric” because distribution regression models involve function-valued parameters.

than estimates obtained under conditional exogeneity or based on the IV quantile regression model (Chernozhukov and Hansen, 2005, 2006).

Related Literature. The literature on identification of heterogeneous treatment effects with endogeneity is vast. We focus the review on approaches that do not impose parametric distributional assumptions to achieve point identification.

A first strand of the literature imposed assumptions on the generation of the potential outcomes, such as rank invariance (RI) and rank similarity (RS). RI imposes that the outcome equation is strictly monotonic in a scalar unobservable that is the same for all potential outcomes. This convenient assumption allows for identification of QTE and ATE with discrete and continuous treatments (Chernozhukov and Hansen, 2005). Additive error structures which imply RI can also be used to identify the ASF with continuous treatments (Newey and Powell, 2003; Blundell et al., 2007).³ Chernozhukov and Hansen (2005)’s RS is slightly weaker as it does not necessarily restrict the unobservable to be the same across potential outcomes, although both produce the same testable restriction (Chernozhukov and Hansen, 2013). Vuong and Xu (2017) further showed that RI and strict monotonicity identify individual treatment effects under suitable regularity conditions. This literature remains flexible about the treatment selection process.

The proposed CLD assumption and RS or RI are non-nested: the former concerns the dependence between potential outcomes and selection unobservables across instrument levels, whereas the latter restricts this dependence across *treatment levels*. Indeed, we show that RS can be viewed as another form of constant local dependence. However, our CLD allows for more general patterns of treatment effect heterogeneity. Also, approaches for point identification based on RI and RS rely on strict monotonicity of the outcome equation, which can only hold for continuous outcomes, and on full rank and completeness conditions that rule out, for example, ordered discrete and continuous treatments with a binary instrument. When treatments are continuous, they typically lead to ill-posedness. By contrast, the proposed method does not suffer from ill-posedness and accommodates discrete and mixed discrete-continuous outcomes and ordered and continuous treatments with a binary instrument.

A second strand focuses on assumptions imposed on the treatment assignment. Imbens and Angrist (1994) and Heckman and Vytlacil (2005) combined a scalar unobservable in the treatment assignment with monotonicity of potential treatments in a binary instrument to identify local average and marginal effects of binary treatments. Abadie et al. (2002) and Carneiro and Lee (2009) extended this approach to local quantile effects, and Vytlacil (2006)

³Ai and Chen (2003) and Chen and Pouzo (2009, 2012, 2015) considered a general framework that nests these models.

to ordered discrete treatments. [Newey et al. \(1999\)](#) and [Imbens and Newey \(2009\)](#) imposed strict monotonicity of the treatment selection equation in a scalar unobservable and large support of the instrument (ruling out discrete instruments) to identify global effects using a control variable approach, which only applies to continuous treatments. [Newey and Stouli \(2021\)](#) avoided the large support requirement of [Imbens and Newey \(2009\)](#) by assuming a parametric conditional expectation of the treatment that enables extrapolation outside the instrument support. Compared to this strand, CLD restricts the relationship between potential outcomes and treatment assignment, but identifies global effects of discrete and continuous treatments with discrete instruments, without relying on a scalar unobservable in the assignment mechanism.

There are also approaches that combine or modify the assumptions of the previous strands. [Chesher \(2003\)](#) showed identification of the quantile effect of continuous treatments on continuous outcomes with continuous instruments assuming strict monotonicity of the outcome and treatment selection equations in scalar unobservables. Under similar assumptions, [D’Haultfoeulle and Février \(2015\)](#) and [Torgovitsky \(2015\)](#) found that quantile effects can be identified with discrete instruments, and [Torgovitsky \(2017\)](#) proposed a two-step minimum distance estimator for the finite-dimensional parameter of the outcome function. Again, these restrictions are not nested with CLD and, for continuous treatments, our identification strategy is constructive, yielding straightforward plug-in estimators. There are also partial identification solutions imposing less structure on the treatment assignment and potential outcomes (e.g., [Manski \(1990\)](#) and [Balke and Pearl \(1997\)](#) for earlier references, and [Chesher and Rosen \(2020\)](#) for a more recent survey).

The copula has been previously employed in econometrics for identification and estimation. For example, [Chen et al. \(2006\)](#) used a parametric copula to achieve efficient estimation in a class of multivariate distributions.⁴ In a semiparametric triangular model with binary dependent variables, [Han and Vytlacil \(2017\)](#) introduced a class of single-parameter copulas for the dependence between the unobservables and established a condition—satisfied by many well-known copulas, including the Gaussian—under which the parameters are identified. With a binary treatment and binary outcome, the current paper’s framework is relevant to [Han and Vytlacil \(2017\)](#). However, while they assumed a parametric copula, we assume CLD, of which their Gaussian copula is a special case corresponding to a local dependence function that is constant everywhere. [Han and Lee \(2019\)](#) developed sieve estimation and inference methods based on [Han and Vytlacil \(2017\)](#), and [Han et al. \(2026\)](#) extended them to semiparametric models for dynamic treatment effects. [Mourifié and Wan \(2025\)](#) used

⁴In time series, [Chen and Fan \(2006b,a\)](#), [Beare \(2010\)](#), [Chen et al. \(2021, 2022\)](#), and [Fan et al. \(2023\)](#), among others, used copulas to model temporal dependence.

copulas as a channel to impose assumptions in characterizing identified sets for marginal treatment effects.

[Arellano and Bonhomme \(2017\)](#) and [Chernozhukov et al. \(2025a\)](#) studied IV identification of selection models using assumptions on the copula between the latent outcome and selection unobservable: the former assumed a real analytic copula and required continuous instruments, while the latter used the local Gaussian representation and copula exclusion, a special case of CLD, with a binary instrument. Therefore, our framework with a binary treatment is related to their setup but differs in several dimensions. First, our setting requires two-way sample selection due to the switching of treatment status. Second, we introduce a general threshold-crossing selection model with a multivariate unobservable, allowing for rich selection patterns. Third, these features make the local representations and constant local dependence assumptions specific to treatment status and the value of the IV. More importantly, the use of local Gaussian representation and CLD for ordered and continuous treatments is completely new to this paper, with the two cases requiring distinct identification strategies and proof techniques.

Finally, in the difference-in-differences setup, [Athey and Imbens \(2006\)](#) showed that average and quantile treatment effects on the treated (and the untreated) can be identified when the unobservable determinant of the untreated potential outcome is independent of time within groups. [Ghanem et al. \(2023\)](#) provided general identification results under a time invariance assumption on the copula between the potential outcomes and the group indicator, which they show is equivalent to the assumptions in [Athey and Imbens \(2006\)](#) with continuous outcomes. When interpreted through the lens of a local Gaussian representation, their approach restricts the time dependence of the local dependence parameter.

Organization of the Paper. Section [2.1](#) introduces the key variables and parameters of interest, Section [2.2](#) states the local Gaussian representation, and Section [2.3](#) posits the main identifying assumptions that will be used throughout the analyses. We devote Sections [3.1–3.3](#) to the identification analyses with binary, ordered discrete, and continuous treatments, respectively. Section [4](#) discusses CLD in further detail, and Section [5](#) deals with estimation and inference. Section [6](#) provides an empirical illustration. Section [7](#) presents concluding remarks. The Supplemental Appendix (SA) contains the proofs and additional results ([Chernozhukov et al., 2026](#)).

Notation. For scalar random variables (r.v.s) X and Y and a possibly multivariate r.v. Z , $F_{X,Y|Z}$ denotes the joint distribution of (X, Y) conditional on Z , $F_{X|Z}$ denotes the (marginal) distribution of X conditional on Z , and F_Z denotes the marginal (joint) distribution of Z . We use calligraphic letters for support sets of r.v.s (e.g., $\mathcal{Z}$ denotes the support of Z). The

symbol $\perp$ denotes (stochastic) independence. The interior of the set $\mathcal{D}$ is denoted by $\mathring{\mathcal{D}}$.

2 Setup and Assumptions

2.1 Preliminaries

Let $Y \in \mathcal{Y} \subseteq \mathbb{R}$ denote the scalar outcome and $D \in \mathcal{D} \subseteq \mathbb{R}$ denote the scalar treatment. We consider binary, discrete ordered, and continuous treatments with $\mathcal{D} = \{0, 1\}$, $\mathcal{D} = \{0, 1, \dots, K\}$, and $\mathcal{D}$ equal to an uncountable set, respectively. The outcome is not restricted; it can be continuous, discrete or mixed continuous-discrete. Let $Z \in \{0, 1\}$ be the binary IV. We focus on a binary Z as the most challenging case; the analysis readily extends to multi-valued or multivariate Z . Let Y_d denote the potential outcome given $d \in \mathcal{D}$ and D_z the potential treatment given $z \in \{0, 1\}$. They are related to the observed outcome and treatment through $Y = Y_D$ and $D = D_Z$.⁵ Let $X \in \mathcal{X} \subseteq \mathbb{R}^{d_x}$ be a vector of covariates. All the assumptions and analyses are conditional on X , which we suppress to lighten the notation. We make it explicit when we discuss estimation and inference in Section 5.

We consider a general treatment assignment equation:

$$D_z = h(z, V_z), \quad z \in \{0, 1\}, \quad (2.1)$$

where $v \mapsto h(z, v)$ is weakly increasing, and we normalize $V_z \sim U(0, 1)$. We provide examples of the function h for each type of treatment below. The variable V_z can be interpreted as an unobserved treatment propensity ranking, in the sense that high values of V_z are associated with high treatment intensity. By allowing for a different unobservable V_z at each value of z , we essentially permit different propensity rankings for each value of z and therefore D to be a function of the *vector* of unobservables (V_0, V_1) .

We are interested in identifying the distribution of Y_d , F_{Y_d} , for $d \in \mathcal{D}$, and functionals of F_{Y_d} , such as QSFs and ASFs. Thus, by using appropriate operators:

$$QSF_\tau(d) := Q_{Y_d}(\tau) = \mathcal{Q}_\tau(F_{Y_d}), \quad ASF(d) := E[Y_d] = \mathcal{E}(F_{Y_d}),$$

where $\mathcal{Q}_\tau(F) := \inf\{y \in \mathcal{Y} : F(y) \geq \tau\}$ and $\mathcal{E}(F) := \int y dF(y)$. QTE and ATE can be expressed as $\{QSF_\tau(d) - QSF_\tau(d')\}/(d - d')$ and $\{ASF(d) - ASF(d')\}/(d - d')$, $d, d' \in \mathcal{D}$ with $d' \neq d$, and for continuous treatments we can also consider $\partial QSF_\tau(d)/\partial d$ and $\partial ASF(d)/\partial d$, $d \in \mathcal{D}$. When the treatment is binary, we may also be interested in the distribution of Y_d in subpopulations such as the treated, $F_{Y_d|D}(\cdot | 1)$, and untreated, $F_{Y_d|D}(\cdot | 0)$, for $d \in \{0, 1\}$,

⁵When D is continuous, we require that Y_d is suitably measurable (Hirano and Imbens, 2004).

and functionals of these distributions. More generally, we may be interested in these objects for subpopulations defined by values of the covariates in X .

2.2 Local Gaussian Representation

Treatment endogeneity can be captured by the joint distribution of the potential outcome and unobservable treatment propensity ranking in equation (2.1). We use the local Gaussian representation (LGR) to represent this joint distribution. This representation is the basis of our identification and estimation strategies. Throughout the paper, let $C(u_1, u_2; \rho)$ denote the Gaussian copula with correlation coefficient ρ , that is, $C(u_1, u_2; \rho) := \Phi_2(\Phi^{-1}(u_1), \Phi^{-1}(u_2); \rho)$ where $\Phi_2(\cdot, \cdot; \rho)$ is the standard bivariate Gaussian distribution with parameter ρ and Φ is the standard univariate Gaussian distribution.

The following lemma shows that the distribution of any bivariate r.v. has a local Gaussian representation (dos Anjos and Kolev, 2005; Kolev et al., 2006; Chernozhukov et al., 2025a).

Lemma 2.1 (LGR). *For any r.v.s Y and V , the joint distribution of Y and V admits the following representation: for (y, v) such that $0 < F_Y(y), F_V(v) < 1$,*

$$F_{Y,V}(y, v) \equiv C(F_Y(y), F_V(v); \rho_{Y,V}(y, v)),$$

where $\rho_{Y,V}(y, v)$ is the unique solution in ρ to $F_{Y,V}(y, v) = C(F_Y(y), F_V(v); \rho)$.⁶

In the lemma, the LGR is fully nonparametric and is not imposing Gaussianity on the joint or marginal distributions. The distribution is equated to the Gaussian copula (with the nonparametric marginal distributions) by adjusting the value of the correlation coefficient ρ for each evaluation point (y, v) . Note that the solution $\rho_{Y,V}(y, v)$ depends on both the dependence structure and marginals. Lemma A.1 in Chernozhukov et al. (2025b) shows that $\rho_{Y,V}(y, v)$ has the same sign as $\text{Cov}(1\{Y \leq y\}, 1\{V \leq v\})$.

Gaussianity in the local representation in Lemma 2.1 is convenient for introducing identifying assumptions, interpreting these assumptions using joint normality as a benchmark, and developing estimators. However, Gaussianity is not essential for the local representation, and other comprehensive single-parameter copulas could be used; see SA B.4.

2.3 Assumptions

We maintain the following assumptions.⁷

⁶For completeness, we set $\rho_{Y,V}(y, v) := 0$ when one of the marginals is equal to 0 or 1.

⁷We impose the assumptions for all values of $d \in \mathcal{D}$. If the assumptions only hold for a subset of values $\mathcal{D}_A \subset \mathcal{D}$, then our analysis applied to $d \in \mathcal{D}_A$ yields identification of F_{Y_d} , $d \in \mathcal{D}_A$, and of its functionals.

Assumption EX (Independence). For $d \in \mathcal{D}$ and $z \in \{0, 1\}$, $Z \perp (Y_d, V_z)$.

Assumption REL (Relevance). (i) $Z \in \{0, 1\}$; (ii) $0 < \Pr[Z = 1] < 1$; and (iii) if $\mathcal{D} = \{0, 1\}$, $\Pr[D = 1 \mid Z = 1] \neq \Pr[D = 1 \mid Z = 0]$ and $0 < \Pr[D = 1 \mid Z = z] < 1$, $z \in \{0, 1\}$; if $\mathcal{D} = \{0, 1, \dots, K\}$, $F_{D|Z}(d \mid 0) > F_{D|Z}(d \mid 1)$, $d \in \mathcal{D} \setminus \{K\}$,⁸ and $\Pr[D = d \mid Z = z] > 0$, $(z, d) \in \{0, 1\} \times \mathcal{D}$; and if $\mathcal{D}$ is uncountable, $F_{D|Z}(d \mid 1) \neq F_{D|Z}(d \mid 0)$ and $0 < F_{D|Z}(d \mid z) < 1$, $(z, d) \in \{0, 1\} \times \mathring{\mathcal{D}}$.

EX is a standard exogeneity condition in IV strategies that imposes that the distribution of Y_d and V_z conditional on Z is the same as the (unconditional) distribution of Y_d and V_z . It is weaker than $Z \perp (\{Y_d\}_{d \in \mathcal{D}}, V_0, V_1)$ because it does allow the relationship between Y_d and $Y_{d'}$, $d, d' \in \mathcal{D}$, and between V_0 and V_1 to depend on Z . Also, in **EX**, a standard exclusion restriction is implicit in the notation: $Y_d = Y_{d,z}$ almost surely (a.s.), where $Y_{d,z}$ is the potential outcome given (d, z) . **REL**(ii)–(iii) are the usual IV relevance and non-degeneracy conditions. **REL**(iii) for $\mathcal{D} = \{0, 1\}$ can be formulated as a special case of **REL**(iii) for $\mathcal{D} = \{0, 1, \dots, K\}$ with $K = 1$, but we state it separately for clarity.

Let $\pi_d(z) := F_{D|Z}(d \mid z)$. We refer to $\pi_d(z)$ as the d -treatment threshold induced by $Z = z$ as it gives the marginal value of the treatment propensity ranking V_z for which the treatment changes from $D = d$ to the next level when $Z = z$. For example, when $\mathcal{D} = \{0, 1\}$, the treatment changes from 0 to 1 when V_z passes the threshold $\pi_0(z) = F_{D|Z}(0 \mid z) = \Pr[D = 0 \mid Z = z]$. We make the following assumption about the local dependence parameter of the LGR of (Y_d, V_z) :

Assumption CLD (Constant Local Dependence). Consider the following condition:

$$\rho_{Y_d, V_0}(y, \pi_d(0)) = \rho_{Y_d, V_1}(y, \pi_d(1)) =: \rho_{Y_d}(y) \in (-1, 1). \quad (2.2)$$

For $y \in \mathcal{Y}$, (i) if $\mathcal{D} = \{0, 1\}$, (2.2) holds for $d = 0$ and

$$\rho_{Y_1, V_0}(y, \pi_0(0)) = \rho_{Y_1, V_1}(y, \pi_0(1)) =: \rho_{Y_1}(y) \in (-1, 1), \quad (2.3)$$

(ii) if $\mathcal{D} = \{0, \dots, K\}$ with $K \geq 2$, (2.2) holds for all $d \in \{0, \dots, K - 1\}$ and

$$\rho_{Y_K, V_0}(y, \pi_{K-1}(0)) = \rho_{Y_K, V_1}(y, \pi_{K-1}(1)) =: \rho_{Y_K}(y) \in (-1, 1), \quad (2.4)$$

$$\rho_{Y_d, V_z}(y, \pi_d(z)) = \rho_{Y_d, V_z}(y, \pi_{d-1}(z)), \quad d \in \{1, \dots, K - 1\}, \quad z \in \{0, 1\}, \quad (2.5)$$

⁸The direction of dominance is up to relabeling the levels of Z ; see related discussions in SA A.1.

(iii) if $\mathcal{D}$ is uncountable, (2.2) holds, $v \mapsto F_{Y_d, V_z}(y, v)$ is differentiable,⁹ and

$$\left. \frac{\partial \rho_{Y_d, V_z}(y, v)}{\partial v} \right|_{v=\pi_d(z)} = 0, \quad d \in \mathring{\mathcal{D}}. \quad (2.6)$$

CLD is a high-level condition that imposes a shape restriction on the dependence between Y_d and V_z . In particular, it imposes that the local dependence of Y_d and V_z is constant at the d -treatment thresholds induced by the two levels of the instrument when D is binary and also around these thresholds when D is discrete ordered or continuous. A sufficient condition for **CLD** is that $\rho_{Y_d, V_0}(y, v) = \rho_{Y_d, V_1}(y, v) = \rho_{Y_d}(y)$ for all $v \in \mathcal{V}$, where $\mathcal{V} \subseteq (0, 1)$ is an open interval containing the set of possible values for $\pi_d(z)$, $z \in \{0, 1\}$ (with $\mathcal{V} = (0, 1)$ this condition is equivalent to a Gaussian copula, see SA C.3). Under selection rank invariance (SRI) (i.e., $V_0 = V_1 = V$ a.s.), **CLD** holds trivially when the copula of (Y_d, V) is Gaussian because $\rho_{Y_d, V}(y, v)$ is constant everywhere. Moreover, we show in SA A.2 that **CLD** does not require SRI and in SA A.3 that it can be compatible with dependence between Y_d and V_z (i.e., endogeneity) even with parametric data generating processes (DGPs).

Section 4 provides further discussions of the statistical and economic content of **CLD** and a comparison to identification under RS. We note here that restrictions on the shape of the copula parameter, such as **CLD**, are inherently specific to the copula chosen for the local representation. Moreover, for the LGR to represent a valid distribution function, the local dependence parameter must satisfy certain restrictions (see, e.g., Nelsen, 2006). These restrictions could, in principle, be exploited to develop a specification test for **CLD**.¹⁰

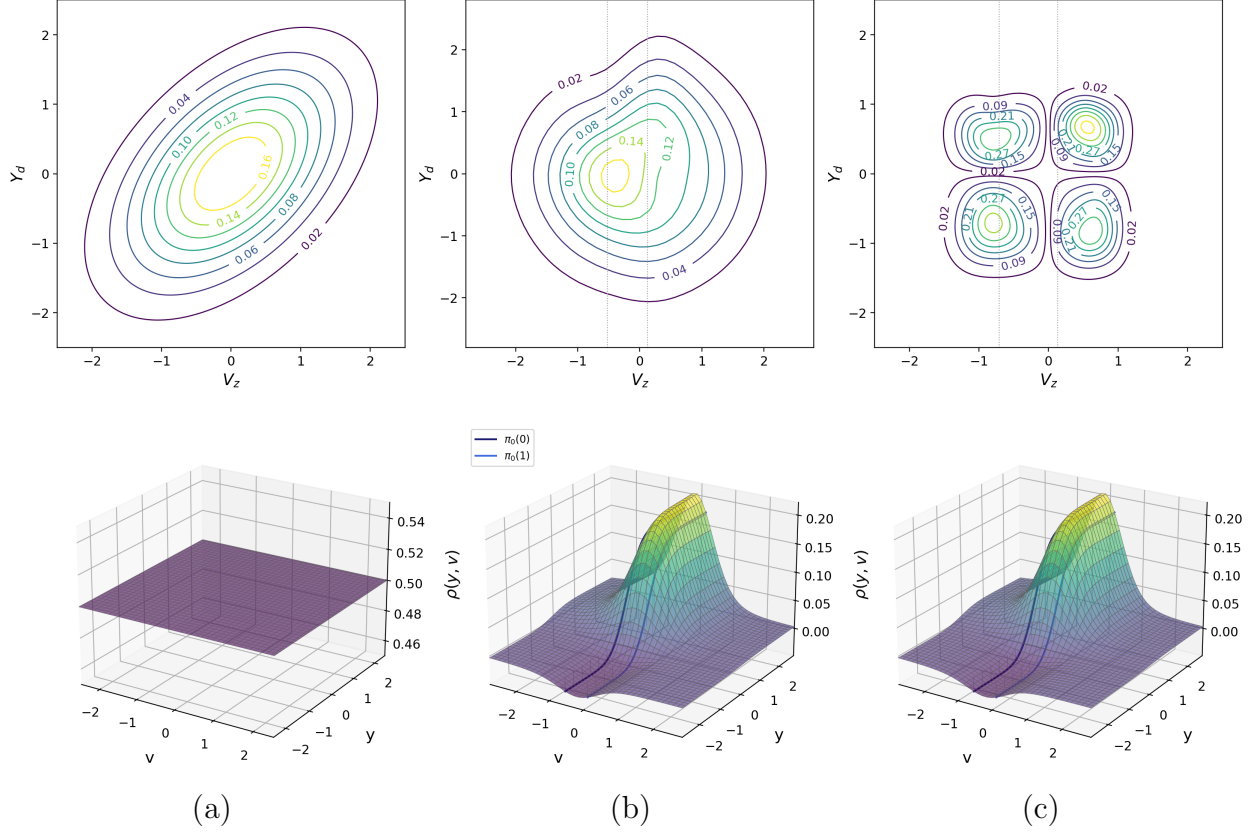
Figure 1 shows examples of joint distributions of (Y_d, V_z) that satisfy **CLD**. Panel (a) corresponds to a bivariate Gaussian distribution (i.e., a Gaussian copula and univariate Gaussian marginals); Panel (b) corresponds to a non-Gaussian copula and standard univariate Gaussian marginals that we introduce later in Section 4.3; and Panel (c) corresponds to the copula in (b) coupled with non-Gaussian marginals. The upper figures display contour plots of the joint distribution, and the lower figures display the corresponding local dependence functions. These examples showcase that **CLD** is compatible with very diverse shapes for the joint distribution, including multimodality and asymmetry.¹¹ Moreover, even though the Gaussian copula does not have tail dependence, **CLD** allows for it because the local dependence parameter can change with the level of Y_d .

⁹ Differentiability of $v \mapsto \rho_{Y_d, V_z}(y, v)$ follows by the differentiability of $v \mapsto F_{Y_d, V_z}(y, v)$ and $v \mapsto C(\cdot, v; \cdot)$ and the implicit function theorem.

¹⁰For example, in the binary treatment case with $d = 0$, these restrictions are that $y \mapsto C(F_{Y_0}(y), \pi_0(z); \rho_{Y_0}(y))$ and $y \mapsto F_{Y_0}(y) - C(F_{Y_0}(y), \pi_0(z); \rho_{Y_0}(y))$ be nondecreasing on $\mathcal{Y}$ for $z \in \{0, 1\}$.

¹¹We refer to Liu et al. (2009) and Chernozhukov et al. (2025a) for more examples.

Figure 1: Examples of Distributions that Satisfy CLD



Notes: Panel (a): Bivariate Gaussian distribution (i.e., Gaussian copula and univariate Gaussian marginals) with correlation 0.5; Panel (b): a two-channel-deviation (non-Gaussian) copula with $\rho_{Y_d, V_0}(y, \pi_0(0)) = \rho_{Y_d, V_1}(y, \pi_0(1)) = \rho_{Y_d}(y)$ for all y , with $\rho_{Y_d, V_z}(y, v)$ non-constant for all other (y, v) , and univariate Gaussian marginals (see Section 4.3); Panel (c): the copula in (b) and non-Gaussian marginals $\Phi(a(\text{sgn}(u) + 1(u > 0)/2)u^2 + b)$ for both Y_d and V_z , where a and b are calibrated such that the distribution has zero mean and unit variance. The upper figures display contour plots of the joint distribution and the lower figures display the corresponding local dependence functions. All the figures are drawn in terms of Φ^{-1} -transformed V_z for readability.

3 Identification Analysis

3.1 Binary Treatment

We start by considering the identification of the causal effects of a binary treatment $D \in \mathcal{D} = \{0, 1\}$. To reflect this, we consider a treatment selection equation

$$D_z = h(z, V_z) = 1\{V_z > \pi_0(z)\}, \quad z \in \{0, 1\}, \quad (3.1)$$

with $\Pr[D = 0 \mid Z = z] = \Pr[D_z = 0 \mid Z = z] = \Pr[V_z \leq \pi_0(z)] = \pi_0(z)$ by EX and the normalization $V_z \sim U(0, 1)$. Note that V_1 and V_0 are two potentially distinct unobservables

that do not restrict the relationship between D_0 and D_1 .¹²

For the identification analysis, consider

$$\begin{aligned}\Pr[Y \leq y, D = 0 \mid Z = z] &= \Pr[Y_0 \leq y, D_z = 0 \mid Z = z] = \Pr[Y_0 \leq y, V_z \leq \pi_0(z) \mid Z = z] \\ &= C(F_{Y_0}(y), \pi_0(z); \rho_{Y_0, V_z}(y, \pi_0(z))), \quad (y, z) \in \mathcal{Y} \times \{0, 1\},\end{aligned}\quad (3.2)$$

where the second equality uses equation (3.1), and the last equality follows from EX and Lemma 2.1. For each $y \in \mathcal{Y}$, this is a system of two equations in three unknowns, $F_{Y_0}(y)$, $\rho_{Y_0, V_0}(y, \pi_0(0))$ and $\rho_{Y_0, V_1}(y, \pi_0(1))$. CLD(i) ((2.2) in particular) reduces the number of unknowns to two: $(F_{Y_0}(y), \rho_{Y_0}(y))$. The following theorem shows that the nonlinear system of equations (3.2) has a unique solution under CLD(i). This result follows from a global univalence theorem of Gale and Nikaido (1965), because the Jacobian of the system of equations is a P-matrix under REL(iii), the map defined by the system is differentiable (as the copula is differentiable) and the parameter space $(0, 1) \times (-1, 1)$ for $(F_{Y_0}(y), \rho_{Y_0}(y))$ is open and rectangular.¹³ A similar argument shows that the distribution of Y_1 and the local dependence parameter of the LGR of Y_1 and V_z are identified under (2.3) in CLD(i).¹⁴ Let $\bar{\mathcal{Y}}_d := \{y \in \mathbb{R} : 0 < F_{Y_d}(y) < 1\}$.

Theorem 3.1 (Identification for Binary Treatment). *Suppose $D_z \in \{0, 1\}$ satisfies (3.1) for $z \in \{0, 1\}$. Under EX, REL, and CLD(i), the functions $y \mapsto F_{Y_d}(y)$ and $y \mapsto \rho_{Y_d}(y)$ are identified on $y \in \bar{\mathcal{Y}}_d$, for $d \in \{0, 1\}$.¹⁵*

The proof of Theorem 3.1 is contained in SA B.1. It is interesting to compare Theorem 3.1 to Chernozhukov and Hansen (2005), who also establish identification of F_{Y_d} when D and Z are binary. Compared to them, we do not impose RS and thus, as we show in SA C.2, allow for richer effect heterogeneity. As a trade-off, we restrict the form of endogeneity by imposing CLD. For a more detailed comparison, see Section 4.4.

Remark 3.1 (Identification of $F_{Y_1|D}$ and $F_{Y_0|D}$). *We focus on identification for the untreated, $D = 0$; identification for the treated, $D = 1$, follows by a similar argument. The distribution of Y_0 is trivially identified from $F_{Y_0|D}(y \mid 0) = F_{Y|D}(y \mid 0)$. Identification of the distribution of Y_1 follows from $F_{Y_1|D}(y \mid 0) = (F_{Y_1}(y) - \Pr[D = 1]F_{Y|D}(y \mid 1))/\Pr[D = 0]$, where $F_{Y_1}(y)$ is identified by Theorem 3.1.* $\square$

¹²This is in contrast to the LATE monotonicity assumption (Imbens and Angrist, 1994), which imposes either $D_1 \geq D_0$ or $D_0 \geq D_1$, a.s., which corresponds to $V_1 = V_0$, a.s. (Vytlačil, 2002). See SA A.2 for further discussions.

¹³We discuss identification using other local representations in SA B.4.

¹⁴The system (3.2) becomes $\Pr[Y \leq y, D = 1 \mid Z = z] = C(F_{Y_1}(y), 1 - \pi_0(z); -\rho_{Y_1, V_z}(y, \pi_0(z)))$, $(y, z) \in \mathcal{Y} \times \{0, 1\}$.

¹⁵The set $\bar{\mathcal{Y}}_d$ is identified because $F_{Y_d}(y) = 1$ if and only if $\Pr(Y \leq y \mid D = d, Z = z) = 1$, $z \in \{0, 1\}$, and $F_{Y_d}(y) = 0$ if and only if $\Pr(Y \leq y \mid D = d, Z = z) = 0$, $z \in \{0, 1\}$.

Remark 3.2 (One-sided non-compliance). *Under one-sided non-compliance, where $D = 1$ whenever $Z = 1$ (i.e., $\Pr[D = 1 \mid Z = 1] = 1$), [REL](#)(iii) is violated. In this case $F_{Y_0}(y)$ is no longer identified because one of the equations in (3.2) becomes uninformative as $\pi_0(1) = 0$. We can still identify $F_{Y_1}(y) = F_{Y|Z}(y \mid 1)$. Therefore, we can also identify $F_{Y_1|D}(y \mid d)$, $d \in \{0, 1\}$, as in [Remark 3.1](#); and $F_{Y_0|D}(y \mid 0) = F_{Y|D}(y \mid 0)$. $\square$*

Remark 3.3 (Overidentification with non-binary instruments). *It is clear from the identification analysis that, when the IV takes more than two values (or equivalently, when there are multiple binary IVs), the resulting system of equations produces overidentifying restrictions. These restrictions can be used to test the model specification, and [CLD](#) in particular. For example, if we have multiple binary IVs, we can combine the first-order conditions of the estimators for each IV in [Section 5](#) through a GMM approach and use the standard J -statistic to test for overidentification. Multi-valued IVs can also be used to make the model more flexible by allowing the local dependence parameter to partially vary with the value of V_z as [CLD](#) only needs to be satisfied for two values. Although we shall not repeat it, this remark also applies to the subsequent cases of discrete ordered and continuous D . $\square$*

Remark 3.4 (Relaxing [CLD](#)). *Here, we outline how one can relax [CLD](#) and analyze the impact of failures of [CLD](#) on the identification of F_{Y_0} and F_{Y_1} . We focus on F_{Y_0} ; the analysis for F_{Y_1} is symmetric. Consider the following relaxation of [CLD](#)(i)*

$$|\rho_{Y_0, V_0}(y, \pi_0(0)) - \rho_{Y_0, V_1}(y, \pi_0(1))| \leq r_0 \quad (3.3)$$

for some $0 < r_0 \leq 2$. Under this relaxation, we can build on [Chernozhukov et al. \(2025b, Appendix A.4\)](#) to characterize the identified set for $F_{Y_0}(y)$ as a function of r_0 . To describe the resulting identified set, consider the following system of equations:

$$\Pr[Y \leq y, D = 0 \mid Z = z] = C(F_0, \pi_0(z); \rho_{0,z}), \quad z \in \{0, 1\}. \quad (3.4)$$

For $z \in \{0, 1\}$, denote by $S_{0,z}$ the set of all solutions to (3.4). Then, [Chernozhukov et al. \(2025b, Theorem A.1\)](#) implies that the identification region for $F_{Y_0}(y)$ is $\text{Proj}_1 R_0(r_0)$ where Proj_1 is the projection operator with respect to the first coordinate and

$$R_0(r_0) := \{(F_0, \rho_{0,0}, \rho_{0,1}) : (F_0, \rho_{0,z}) \in S_{0,z}, z \in \{0, 1\}\} \cap \{(F_0, \rho_{0,0}, \rho_{0,1}) : |\rho_{0,0} - \rho_{0,1}| \leq r_0\}.$$

In practice, $\text{Proj}_1 R_0(r_0)$ needs to be computed numerically. By varying r_0 , one can perform sensitivity and breakdown analyses. $\square$

3.2 Ordered Discrete Treatment

We consider identification of the causal effects of a multi-valued ordered treatment $D \in \mathcal{D} = \{0, \dots, K\}$ using a binary instrument $Z \in \{0, 1\}$. We assume a threshold-crossing model for the treatment selection equation, which can be viewed as a natural extension of model (3.1) from two to multiple treatment levels,

$$D_z = h(z, V_z) = \sum_{d \in \mathcal{D}} d \mathbf{1}\{\pi_{d-1}(z) < V_z \leq \pi_d(z)\} = \begin{cases} 0, & \pi_{-1}(z) < V_z \leq \pi_0(z) \\ 1, & \pi_0(z) < V_z \leq \pi_1(z) \\ \vdots & \vdots \\ K, & \pi_{K-1}(z) < V_z \leq \pi_K(z) \end{cases}, \quad (3.5)$$

where $\pi_{-1}(z) = 0$ and $\pi_K(z) = 1$. Equation (3.5) generalizes the model in Section 7.2 of Heckman and Vytlacil (2007) by allowing for two unobservables (V_0, V_1) and a different impact of the instrument on the different cutoffs; see SA B.3. It is not fully general, however, as it imposes that the unobservables are the same for all the treatment levels.¹⁶

Under the normalization $V_z \sim U(0, 1)$ and EX, the threshold functions $\pi_d(z)$ are identified by the distribution of the observed treatment conditional on the instrument, $\pi_d(z) = F_{D|Z}(d | z)$ for $d \in \mathcal{D}$. For the identification analysis, consider

$$\begin{aligned} \Pr[Y \leq y, D = d | Z = z] &= \Pr[Y_d \leq y, \pi_{d-1}(z) < V_z \leq \pi_d(z) | Z = z] \\ &= C(F_{Y_d}(y), \pi_d(z); \rho_{Y_d, V_z}(y, \pi_d(z))) \\ &\quad - C(F_{Y_d}(y), \pi_{d-1}(z); \rho_{Y_d, V_z}(y, \pi_{d-1}(z))), \quad (y, d, z) \in \mathcal{Y} \times \mathcal{D} \times \{0, 1\}, \end{aligned} \quad (3.6)$$

where the first equality follows from (3.5) and the second equality from EX and Lemma 2.1. For each $d \in \mathcal{D}$ and $y \in \mathcal{Y}$, (3.6) is a system of two equations in five unknowns: $F_{Y_d}(y)$, $\rho_{Y_d, V_0}(y, \pi_{d-1}(0))$, $\rho_{Y_d, V_0}(y, \pi_d(0))$, $\rho_{Y_d, V_1}(y, \pi_{d-1}(1))$, and $\rho_{Y_d, V_1}(y, \pi_d(1))$. REL(iii) guarantees that the two equations of the system are not redundant.¹⁷

For $d \in \{0, K\}$, one of the terms on the right hand side drops out because either $\pi_{d-1}(z) = 0$ or $\pi_d(z) = 1$, yielding a system of two equations in three unknowns. Consequently, the distribution of the potential outcome and local dependence parameter can be identified by combining REL(iii) and (2.2) or (2.4). For $d \in \{1, \dots, K-1\}$, (2.2) reduces the number of unknowns only to four. We impose additionally (2.5) to further reduce the number of

¹⁶Cunha et al. (2007) showed that ordered choice models with multivariate unobservables are point identified under strong assumptions.

¹⁷A weaker version of REL(iii) (i.e., $F_{D|Z}(d | 1) \neq F_{D|Z}(d | 0)$) is sufficient for nonredundancy, but REL(iii) is used for establishing uniqueness of the solution below.

unknowns to two: $F_{Y_d}(y)$ and $\rho_{Y_d}(y)$. The Jacobian of the resulting system of equations, however, does not satisfy the conditions to apply the global univalence results of [Gale and Nikaido \(1965\)](#) even under [REL](#)(iii). We show uniqueness of the solution using an alternative proof strategy described below.

We summarize the identification result in the following theorem:

Theorem 3.2 (Identification for Ordered Treatment). *Suppose D_z , $z \in \{0, 1\}$, satisfies (3.5). Under [EX](#), [REL](#), and [CLD](#)(ii), the functions $y \mapsto F_{Y_d}(y)$ and $y \mapsto \rho_{Y_d}(y)$ are identified on $y \in \bar{\mathcal{Y}}_d$, for $d \in \mathcal{D}$.*

In the proof of Theorem 3.2, contained in SA [B.2](#), we proceed as follows to show that (3.6) has a unique solution. First, we use the monotonicity in the first coordinate $F_{Y_d}(y)$ to profile out this coordinate using the first equation of the system. Second, we show that the profiled second equation is strictly monotonic in the remaining coordinate $\rho_{Y_d}(y)$. Here, we use the fact that the Jacobian of the system has full rank everywhere in the parameter space (by [REL](#)).¹⁸ Third, by combining the monotonicity in the first and second steps, we conclude that the solution of the system is unique everywhere in the parameter space.

3.3 Continuous Treatment

Suppose $D \in \mathcal{D} \subseteq \mathbb{R}$, where $\mathcal{D}$ is uncountable and $v \mapsto F_{D|Z}^{-1}(v | z)$ is strictly increasing on $(0, 1)$ for $z \in \{0, 1\}$. Then, by [EX](#) and $V_z \sim U(0, 1)$, the treatment selection equation is

$$D_z = h(z, V_z) = F_{D|Z}^{-1}(V_z | z), \quad z \in \{0, 1\}. \quad (3.7)$$

For the identification analysis, consider

$$F_{Y|D,Z}(y | d, z) = F_{Y_d|D_z,Z}(y | d, z) = F_{Y_d|V_z,Z}(y | \pi_d(z), z), \quad (3.8)$$

where the second equality follows from equation (3.7) and a change of variables. By the properties of the conditional distribution, Lemma 2.1, and [EX](#),

$$\begin{aligned} F_{Y_d|V_z,Z}(y | v, z) &= \frac{(\partial/\partial v)F_{Y_d,V_z|Z}(y, v | z)}{(\partial/\partial v)F_{V_z|Z}(v | z)} = C_2(F_{Y_d}(y), v; \rho_{Y_d,V_z}(y, v)) \\ &\quad + C_\rho(F_{Y_d}(y), v; \rho_{Y_d,V_z}(y, v))(\partial/\partial v)\rho_{Y_d,V_z}(y, v), \end{aligned} \quad (3.9)$$

¹⁸We rely on the stochastic dominance between $\pi_d(0) = F_{D|Z}(d | 0)$ and $\pi_d(1) = F_{D|Z}(d | 1)$ in [REL](#)(iii) to establish full rank of the Jacobian. In contrast, a weaker relevance condition $\pi_d(0) \neq \pi_d(1)$ is not sufficient in general.

where C_2 and C_ρ are the derivatives of $C(\cdot, v; \rho)$ with respect to v and ρ , respectively, and the differentiability follows from [CLD\(iii\)](#). Then, by the properties of Gaussian copula and [CLD\(iii\)](#), combining [\(3.8\)](#) and [\(3.9\)](#) yields

$$\Phi^{-1}(F_{Y|D,Z}(y | d, z)) = a_{d,y} + b_{d,y}\Phi^{-1}(\pi_d(z)), \quad z \in \{0, 1\}, \quad (3.10)$$

where $a_{d,y} := \Phi^{-1}(F_{Y_d}(y))/\sqrt{1 - \rho_{Y_d}(y)^2}$ and $b_{d,y} := -\rho_{Y_d}(y)/\sqrt{1 - \rho_{Y_d}(y)^2}$. Equation [\(3.10\)](#) yields a linear system of two equations in two unknowns, $a_{d,y}$ and $b_{d,y}$, which has solution

$$\begin{aligned} a_{d,y} &= \frac{\Phi^{-1}(F_{Y|D,Z}(y | d, 0))\Phi^{-1}(\pi_d(1)) - \Phi^{-1}(F_{Y|D,Z}(y | d, 1))\Phi^{-1}(\pi_d(0))}{\Phi^{-1}(\pi_d(1)) - \Phi^{-1}(\pi_d(0))}, \\ b_{d,y} &= \frac{\Phi^{-1}(F_{Y|D,Z}(y | d, 1)) - \Phi^{-1}(F_{Y|D,Z}(y | d, 0))}{\Phi^{-1}(\pi_d(1)) - \Phi^{-1}(\pi_d(0))}, \end{aligned} \quad (3.11)$$

under [REL](#).¹⁹

This discussion implies the following identification result.

Theorem 3.3 (Identification for Continuous Treatment). *Suppose D_z , $z \in \{0, 1\}$, satisfies [\(3.7\)](#). Under [EX](#), [REL](#), and [CLD\(iii\)](#), the functions $y \mapsto F_{Y_d}(y)$ and $y \mapsto \rho_{Y_d}(y)$ are identified on $y \in \bar{\mathcal{Y}}_d$, for $d \in \bar{\mathcal{D}}$, by*

$$F_{Y_d}(y) = \Phi\left(\frac{a_{d,y}}{\sqrt{1 + b_{d,y}^2}}\right), \quad \rho_{Y_d}(y) = \frac{-b_{d,y}}{\sqrt{1 + b_{d,y}^2}},$$

where $a_{d,y}$ and $b_{d,y}$ are defined in [\(3.11\)](#).

It is interesting to compare this identification result to [Imbens and Newey \(2009\)](#) and [Torgovitsky \(2010, 2015\)](#), who also provide identification results for nonseparable models with continuous D . Unlike [Imbens and Newey \(2009\)](#), our approach does not require an instrument with large support or SRI, but instead imposes [CLD](#). Unlike [Torgovitsky \(2010\)](#), our approach does not require RI nor SRI, but again imposes [CLD](#).

4 Discussions on Constant Local Dependence

To further understand [CLD](#), we provide an equivalent condition and interpret its implications for treatment selection (Sections [4.1](#) and [4.2](#)), showcase its flexibility in the context of Roy

¹⁹In [SA B.4](#), we provide another representation of the joint distribution of (Y_d, V_z) under which identification is achieved with the same set of assumptions.

models (Section 4.3), compare it with RS (Section 4.4), and explain how it extrapolates to achieve identification for the overall population (Section 4.5).

4.1 CLD is Equivalent to a Local Single Index Condition

We start with a representation of conditional probabilities that follows from the LGR in Lemma 2.1. We shall use this representation to provide a local single index condition equivalent to CLD and to interpret CLD in terms of treatment selection. Although the representation holds generally, we specialize it to our case with $V \sim U(0, 1)$ for concreteness.

Lemma 4.1 (LGR for Conditional Probabilities). *If $V \sim U(0, 1)$, $\rho_{Y,V}(y, v)$ is the local dependence parameter of the LGR $F_{Y,V}(y, v) \equiv C(F_Y(y), v; \rho_{Y,V}(y, v))$ if and only if*

$$\Pr[Y \leq y \mid V \leq v] = \frac{1}{v} \int_0^v \Phi(a_{Y,V}(y, v) + b_{Y,V}(y, v)\Phi^{-1}(\tilde{v})) d\tilde{v},$$

where

$$a_{Y,V}(y, v) := \frac{\Phi^{-1}(F_Y(y))}{\sqrt{1 - \rho_{Y,V}(y, v)^2}}, \quad b_{Y,V}(y, v) := -\frac{\rho_{Y,V}(y, v)}{\sqrt{1 - \rho_{Y,V}(y, v)^2}}.$$

Lemma 4.1 shows that any conditional probability has an aggregated local single index representation with coefficients that depend on the values of y and v . The following assumption on the coefficients of this representation for Y_d and V_z is equivalent to CLD, as shown in Theorem 4.1 below.

Assumption SI (Local Single Index). *Let $a_{Y_d, V_0}(y, \pi_d(0)) = a_{Y_d, V_1}(y, \pi_d(1)) =: a_{d,y}$ and $b_{Y_d, V_0}(y, \pi_d(0)) = b_{Y_d, V_1}(y, \pi_d(1)) =: b_{d,y}$, so that*

$$\Pr[Y_d \leq y \mid V_z \leq \pi_d(z)] = \frac{1}{\pi_d(z)} \int_0^{\pi_d(z)} \Phi(a_{d,y} + b_{d,y}\Phi^{-1}(v)) dv, \quad z \in \{0, 1\}. \quad (4.1)$$

For $y \in \mathcal{Y}$, (i) if $\mathcal{D} = \{0, 1\}$, (4.1) holds for $d = 0$, $a_{Y_1, V_0}(y, \pi_0(0)) = a_{Y_1, V_1}(y, \pi_0(1))$ and $b_{Y_1, V_0}(y, \pi_0(0)) = b_{Y_1, V_1}(y, \pi_0(1))$; (ii) if $\mathcal{D} = \{0, \dots, K\}$, $K \geq 2$, (4.1) holds for all $d \in \{0, \dots, K-1\}$, and $a_{Y_K, V_0}(y, \pi_{K-1}(0)) = a_{Y_K, V_1}(y, \pi_{K-1}(1))$, $b_{Y_K, V_0}(y, \pi_{K-1}(0)) = b_{Y_K, V_1}(y, \pi_{K-1}(1))$, $a_{Y_d, V_z}(y, \pi_{d-1}(z)) = a_{d,y}$ and $b_{Y_d, V_z}(y, \pi_{d-1}(z)) = b_{d,y}$ for $z \in \{0, 1\}$ and $d \in \{1, \dots, K-1\}$, so that, for $z \in \{0, 1\}$,

$$\Pr[Y_d \leq y \mid \pi_{d-1}(z) < V_z \leq \pi_d(z)] = \frac{1}{\pi_d(z) - \pi_{d-1}(z)} \int_{\pi_{d-1}(z)}^{\pi_d(z)} \Phi(a_{d,y} + b_{d,y}\Phi^{-1}(v)) dv;$$

or (iii) if $\mathcal{D}$ is uncountable, (4.1) holds, $v \mapsto a_{Y_d, V_z}(y, v)$ and $v \mapsto b_{Y_d, V_z}(y, v)$ are differentiable and $(\partial/\partial v)a_{Y_d, V_z}(y, v) = (\partial/\partial v)b_{Y_d, V_z}(y, v) = 0$ at $v = \pi_d(z)$ for all $z \in \{0, 1\}$ and $d \in \mathring{\mathcal{D}}$, so

that

$$F_{Y_d|V_z}(y \mid \pi_d(z)) = \Phi(a_{d,y} + b_{d,y}\Phi^{-1}(\pi_d(z))), \quad z \in \{0, 1\}, \quad d \in \mathring{\mathcal{D}}.$$

The first condition of [SI](#) in (4.1) imposes that the index coefficients of the representation of the conditional probabilities are the same at the two d -thresholds induced by the instrument. [SI\(ii\)](#) additionally imposes that the conditional probability in between thresholds of consecutive treatment levels has an aggregated single index form with the same coefficients across instrument levels, whereas [SI\(iii\)](#) additionally imposes that the distribution of Y_d conditional on V_z has a single index form locally at the d -thresholds.

Theorem 4.1 ([CLD](#) $\iff$ [SI](#)). *[SI](#) is a necessary and sufficient condition for [CLD](#).*

We focus on [SI\(iii\)](#) as a way to understand [CLD\(iii\)](#). [CLD\(i\)](#) is discussed in the subsequent sections and [CLD\(ii\)](#) can be viewed as an intermediate case. We show that [SI\(iii\)](#) is weaker than Gaussianity of the copula of Y_d and V_z . We make the comparison under SRI, $V_1 = V_0 = V$ a.s., because this assumption is usually maintained together with Gaussianity. Under this assumption, [SI\(iii\)](#) becomes

$$F_{Y_d|V}(y \mid \pi_d(z)) = \Phi(a_{d,y} + b_{d,y}\Phi^{-1}(\pi_d(z))), \quad z \in \{0, 1\}. \quad (4.2)$$

Gaussianity imposes that

$$F_{Y_d|V}(y \mid v) = \Phi(a_{d,y} + b_d\Phi^{-1}(v)), \quad v \in (0, 1). \quad (4.3)$$

Compared to (4.2), (4.3) requires that the single index restriction be *global* in $v \in (0, 1)$ with the same index coefficient b_d for every value of y ; we show in Section 4.2 that this restriction implies monotone treatment selection. Based on this comparison and Lemma 4.1, we can interpret [SI\(iii\)](#) as imposing Gaussian behavior only locally at the d -thresholds induced by the instrument, with the added flexibility that this local behavior can exhibit heterogeneous treatment selection patterns at different levels of y . Moreover, [SI\(iii\)](#) does not require SRI, which we have imposed here only for simplicity. Finally, we note that given equation (3.10), $b_{d,y}$ in [SI\(iii\)](#) can be viewed as a control function. While we have $b_{d,y} = 0$ without endogeneity (i.e., $\rho_{Y_d}(y) = 0$), $b_{d,y}$ corrects for endogeneity as $\rho_{Y_d}(y)$ departs from zero. We provide additional examples and counterexamples of [CLD/SI](#) in SA C.1.

4.2 Treatment Selection Patterns Under [CLD](#)

In this section, we discuss the implications of [CLD](#) for treatment selection. Assume SRI for simplicity and $\pi_d(0) > \pi_d(1)$ without loss of generality. We can assess the sign of the

treatment selection by comparing the distribution of Y_d conditional on $V = v$ at different values of v . In particular, the function $v \mapsto \Pr[Y_d \leq y \mid V = v]$ measures the proportion of individuals with $Y_d \leq y$ at each treatment propensity v . It is constant under random treatment assignment, increasing under negative treatment selection, that is, when Y_d and V are negatively related, and decreasing under positive treatment selection. Moreover, the sign of the derivative of $v \mapsto \Pr[Y_d \leq y \mid V = v]$ determines minus the sign of the selection locally at each value of v , provided that the function is differentiable.

When the copula of Y_d and V is Gaussian with correlation ρ , $v \mapsto \Pr[Y_d \leq y \mid V = v] = \Phi(a_{d,y} + b_d \Phi^{-1}(v))$ is globally constant, increasing or decreasing for all the values of v and y depending on the sign of b_d (i.e., $-\rho$). Gaussianity therefore imposes monotone treatment selection, that is, a first-order stochastic dominance relationship in the distribution of $Y_d \mid V = v$ at different values of v . [CLD](#)(iii) imposes restrictions on treatment selection *locally* at the two d -treatment thresholds induced by Z ; see (4.2). Thus, the sign of $\partial \Pr[Y_d \leq y \mid V = v] / \partial v$ is the same at $\pi_d(z)$, $z \in \{0, 1\}$ for each y . However, this sign does not need to be the same for all the values of y because the parameter $\rho_{Y_d}(y)$ can change with y , allowing for non-monotone treatment selection. For example, if $\rho_{Y_d}(y) > 0$, there is positive selection at y , but there might be adverse selection at y' if $\rho_{Y_d}(y') < 0$. In other words, while switching Z from 0 to 1 changes the treatment propensity from $\pi_d(0)$ to $\pi_d(1)$, it does not change the sorting mechanism locally, but the mechanism and extent of the sorting can vary at different parts of the distribution of Y_d . Hence, [CLD](#) does not impose any stochastic order relationship between the distribution of Y_d conditional on $V = \pi_d(0)$ and on $V = \pi_d(1)$.²⁰

4.3 Roy Models and Control Functions Under [CLD](#)

Figure 1 in Section 2.3 demonstrates the flexibility of [CLD](#) in modeling the joint distribution of (Y_d, V_z) . Our analyses show how point identification can be achieved under substantially weaker modeling restrictions than Gaussianity, which can be appealing to empirical researchers. To illustrate this further, consider a generalized Roy model for two sectors indexed by binary indicator D . Let the potential log wage in sector d be $Y_d = E[Y_d] + \epsilon_d$ for $d \in \{0, 1\}$, and suppose that individuals choose sector $D = 1$ if the benefit $Y_1 - Y_0$ exceeds the cost C_z where $Z = z$ is a cost shifter. This implies that $D_z = 1\{V_z > \pi_0(z)\}$, where $V_z := F_{Y_1 - Y_0 - C_z}(Y_1 - Y_0 - C_z)$ and $\pi_0(z) := F_{Y_1 - Y_0 - C_z}(0)$.²¹ Suppose $Z \perp (Y_0, Y_1, C_z)$. Then, $E[Y \mid D = 0, Z = z] = E[Y_0] + E[\epsilon_0 \mid V_z \leq \pi_0(z)]$, where $E[\epsilon_0 \mid V_z \leq \pi_0(z)]$ is the *control*

²⁰Note that the sign of $\Pr[Y_d \leq y \mid V = \pi_d(0)] - \Pr[Y_d \leq y \mid V = \pi_d(1)]$ is determined by the sign of $b_{d,y}$ (i.e., $-\rho_{Y_d}(y)$).

²¹This model is more flexible than the standard generalized Roy model as the cost C_z can include a z -specific unobservable, which our framework accommodates.

function that captures endogenous selection into sector $d = 0$; we can obtain a similar expression for $d = 1$. For ease of illustration, focus on sector 0, and let $E[Y_0] = 0$, and impose SRI, so that the control function simply becomes $E[\epsilon_0 \mid V_z \leq \pi_0(z)] = E[Y_0 \mid V \leq \pi_0(z)]$.

When (Y_0, V) follows a Gaussian copula with Gaussian marginal for Y_0 , $E[Y_0 \mid V \leq \pi_0(z)]$ is proportional to the inverse Mills ratio (Heckman, 1979). This function has a specific shape: it is strictly increasing (decreasing) with respect to the threshold $\pi_0(z)$ when the correlation coefficient is positive (negative), as depicted in Figure 2(b) below.²² By contrast, CLD allows for general non-monotone selection patterns in the control function, analogous to the non-monotonicity of the conditional distribution in Section 4.2.

To illustrate, we introduce a particular class of DGPs that is consistent with CLD(i) and still yields an economically interpretable closed-form expression for the control function with potentially opposite selection patterns at the two threshold values. Recall that $\pi_0(z)$ ($z \in \{0, 1\}$) are the two threshold values for which CLD(i) holds with $d = 0$. Let $U_0 := F_{Y_0}(Y_0)$ denote the sector-0 wage rank assuming a continuous Y_0 . We consider a model in which latent traits shift the distribution of wage rank U_0 . For $z \in \{0, 1\}$, let $C(u, \pi_0(z); r(u))$ be the LGR of (U_0, V) under CLD(i). Define its deviation from independence at $\pi_0(z)$ as $\Delta_z(u) := C(u, \pi_0(z); r(u)) - u \pi_0(z) \in (-1/4, 1/4)$.

Assumption LT (Model of Latent Traits). *Let $\Theta = (\Theta_0, \Theta_1)$ be latent traits with*

$$E[\Theta_z] = 0, \quad E[\Theta_z^2] = 1, \quad E[\Theta_0 \Theta_1] = 0, \quad \text{and } U_0 \perp\!\!\!\perp V \mid \Theta. \quad (4.4)$$

Suppose the traits tilt the conditional rank and propensity distributions according to

$$F_{U_0|\Theta}(u \mid \theta) = u + \theta_0 \delta_0(u) + \theta_1 \delta_1(u), \quad F_{V|\Theta}(v \mid \theta) = v + \theta_0 \ell_0(v) + \theta_1 \ell_1(v), \quad (4.5)$$

where $\ell_z(v) := \kappa_z L_z(v)$ and $\delta_z(u) := \Delta_z(u)/\kappa_z$ with nonzero constants (κ_0, κ_1) , and $L_0, L_1 : [0, 1] \rightarrow \mathbb{R}$ are differentiable functions with $L_z(0) = L_z(1) = 0$,²³

$$L_0(\pi_0(0)) = 1, \quad L_0(\pi_0(1)) = 0, \quad L_1(\pi_0(0)) = 0, \quad L_1(\pi_0(1)) = 1. \quad (4.6)$$

In this assumption, (4.4) is motivated by the matching on unobservable factors in Abbring and Heckman (2007, p. 5173); see also Heckman and Vytlacil (2007, pp. 5039–5040). Given $U_0 \perp\!\!\!\perp V \mid \Theta$, the latent traits Θ are the source of endogenous selection under LT.

²²The control function $E[Y_0 \mid V \leq \pi_0(z)]$ remains strictly monotone even when (Y_0, V) follows a more general log-concave distribution (Heckman and Honoré, 1990).

²³ For the expressions in (4.5) to be valid (nondecreasing) CDFs for every realization of Θ , the support of Θ should be bounded, with $1 + \theta_0 \delta'_0(u) + \theta_1 \delta'_1(u) \geq 0$ and $1 + \theta_0 \ell'_0(v) + \theta_1 \ell'_1(v) \geq 0$ over that support. Subject to this restriction, $\Delta_z(u)$ and $L_z(v)$ can take both positive and negative values.

LT implies a particular distribution for (U_0, V) :

$$F_{U_0, V}(u, v) = E[F_{U_0|\Theta}(u | \Theta)F_{V|\Theta}(v | \Theta)] = uv + L_0(v)\Delta_0(u) + L_1(v)\Delta_1(u), \quad (4.7)$$

where the second equality is by (4.4). Note that (4.7) is a valid copula capturing an endogenous deviation from an independence benchmark uv , where the deviation is decomposed into two selection channels.²⁴ Each term $L_z(v)\Delta_z(u)$ is the selection channel generated by latent trait Θ_z , which affects both the sector wage rank, through Δ_z , and the sector-choice propensity, through L_z . Given (4.6), the value of Z determines which trait is responsible for selection at each of the thresholds. We can show that (4.7) and thus LT implies (2.2) in CLD:

Theorem 4.2 (LT $\implies$ CLD (2.2)). *Under LT, the LGR of (Y_0, V) satisfies*

$$\rho_{Y_0, V}(y, \pi_0(0)) = \rho_{Y_0, V}(y, \pi_0(1)) = r(F_{Y_0}(y)) \quad \text{for all } y \in \mathcal{Y}.$$

An assumption analogous to LT but with $d = 1$ implies (2.3) in CLD.

To interpret LT, let sector 0 be the technology sector, sector 1 be the non-technology (e.g., retail) sector, Θ_0 be technical productivity (e.g., coding skills), Θ_1 be organizational capital (e.g., communication skills, match quality), and Z be an indicator for low tech-sector entry costs so that $\pi_0(0) < \pi_0(1)$. The anchoring conditions (4.6) then imply that coding skills are the sole source of selection into technology when entry is restricted ($\pi_0(0)$) and communication skills when entry is easier ($\pi_0(1)$). In model (4.5), if coding skills shift mass toward the upper tail of the tech-sector wage distribution, then $\delta_0(u) < 0$ in the upper tail, so a worker with $\theta_0 > 0$ is shifted upward there. Similarly, if communication skills are valuable for team-lead or client-facing tech jobs, then $\delta_1(u) < 0$ over the relevant upper-middle ranks, and workers with high θ_1 are shifted upward in that region. In terms of sector choice, the tech-sector workforce is positively selected on coding skills where $\ell_0(v) > 0$, and it is negatively selected on communication skills where $\ell_1(v) < 0$.²⁵

In addition, under the latent traits model LT, the control function admits a closed form with a transparent economic structure. Since $Y_0 = Q_{Y_0}(U_0)$ and $U_0 \perp\!\!\!\perp V | \Theta$, we have $Y_0 \perp\!\!\!\perp V | \Theta$, and (4.4) and $F_{V|\Theta}$ in (4.5) yield

$$\lambda_0(v) := E[Y_0 | V \leq v] = \text{Cov}(Y_0, \Theta_0)E[\Theta_0 | V \leq v] + \text{Cov}(Y_0, \Theta_1)E[\Theta_1 | V \leq v]. \quad (4.8)$$

²⁴By LT, the conditions in footnote 23 imply the 2-increasingness $1 + L'_0(v)\Delta'_0(u) + L'_1(v)\Delta'_1(u) \geq 0$ for the copula. This copula can be viewed as a Sarmanov (1966) family, which is a generalization of the Farlie-Gumbel-Morgenstern family.

²⁵When viewed in terms of quantiles, the model (4.5) relates to Chesher (2003) and Matzkin (2003), except that it involves latent explanatory variables.

The derivations of (4.8) and (4.9) below are contained in SA C.4. Each term in (4.8) is the product of two interpretable factors. The selection factor $E[\Theta_z \mid V \leq v] = \ell_z(v)/v$ is the average prevalence of trait z among tech-sector participants with $V \leq v$, i.e., trait-specific selection. The payoff factor $\text{Cov}(Y_0, \Theta_z) = E[Y_0 \Theta_z]$ is the covariance between the tech-sector wage and trait z , interpreted as the mean wage “payoff” to the trait. Hence (4.8) describes the contributions of coding skills and communication skills to the mean tech-sector wage among selected workers, each contribution being the product of what the trait pays and how prevalent it is among participants.

As shown below, the two-channel sorting in (4.8) generates a more general shape of selection patterns than a single sorting under Gaussianity. In particular, the sign of $\lambda'_0(v)$ can flip between $\pi_0(0)$ and $\pi_0(1)$. Based on (4.8), we show in SA C.4 that

$$\begin{aligned} \pi_0(z) \lambda'_0(\pi_0(z)) &= \underbrace{\text{Cov}(Y_0, \Theta_z) (E[\Theta_z \mid V = \pi_0(z)] - E[\Theta_z \mid V \leq \pi_0(z)])}_{=: c_{z,\text{own}}} \\ &\quad + \underbrace{\text{Cov}(Y_0, \Theta_{1-z}) E[\Theta_{1-z} \mid V = \pi_0(z)]}_{=: c_{z,\text{cross}}}. \end{aligned} \quad (4.9)$$

The own-channel term $c_{z,\text{own}}$ is the payoff-weighted change in the prevalence of trait z between marginal entrants and the incumbent workers. The cross-channel term $c_{z,\text{cross}}$ is the payoff-weighted prevalence of the other trait among marginal entrants; it has no incumbent counterpart because the anchoring (4.6) implies $E[\Theta_{1-z} \mid V \leq \pi_0(z)] = 0$. The anchoring thus ties each threshold to a different selection channel in the example: at $\pi_0(0)$, incumbents are selected on coding skills alone, so the slope combines the difference in coding-skill prevalence between marginal entrants and incumbents ($c_{0,\text{own}}$) with the communication-skill prevalence of entrants ($c_{0,\text{cross}}$); at $\pi_0(1)$, the roles of the two traits are reversed. Because the two thresholds are governed by different own/cross configurations, the slope of λ_0 can take opposite signs:

Lemma 4.2 (Non-monotonicity of the control function). *Under LT, suppose*

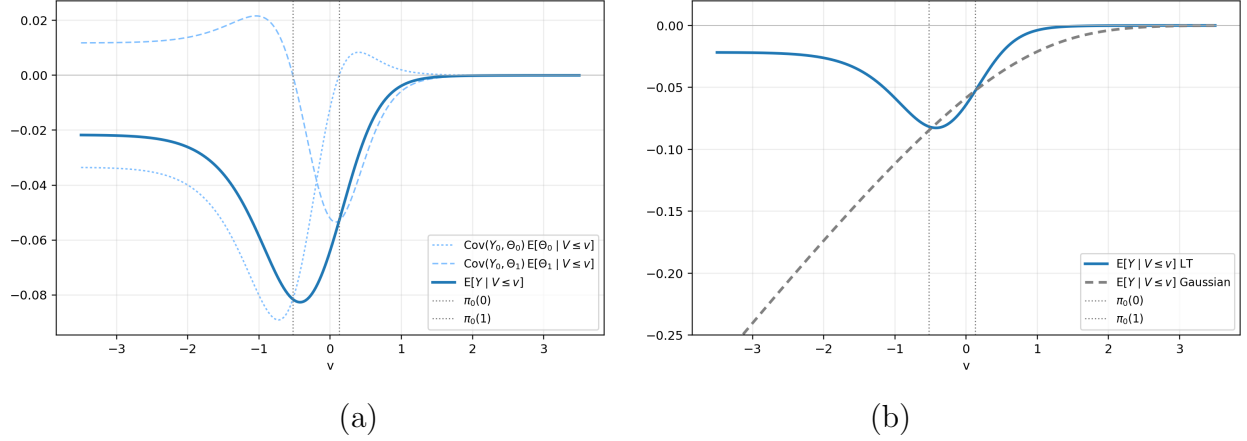
$$c_{0,\text{own}} + c_{0,\text{cross}} < 0, \quad c_{1,\text{own}} + c_{1,\text{cross}} > 0. \quad (4.10)$$

*Then $\lambda'_0(\pi_0(0)) < 0$ and $\lambda'_0(\pi_0(1)) > 0$. In particular, $v \mapsto E[Y_0 \mid V \leq v]$ is non-monotone.*²⁶

Figure 2 is generated from a simulated DGP that satisfies LT and (4.10); the same DGP is used for Figure 1(b) in Section 2.3. The control function in panel (a) is non-monotone, with

²⁶Let $m_0(v) := E[Y_0 \mid V = v]$. (4.10) implies that $v \mapsto m_0(v)$ is also non-monotone: since $\lambda_0(v)$ is the average of m_0 over $[0, v]$, a monotone m_0 would give λ'_0 a constant sign.

Figure 2: Examples of Control Functions That Satisfy CLD



Notes: Figure 1(b) in Section 2.3 depicts the isoquants of the joint density of (Y_0, V) that satisfies LT and (4.10) with $\pi_0(0)$ and $\pi_0(1)$ marked by vertical dotted lines (top) and the correlation function $\rho_{Y_0, V}(y, v)$ with the highlighted slices at $\pi_0(0)$ and $\pi_0(1)$ (bottom). The current figure is based on the same DGP. Panel (a): the control function $E[Y_0 | V \leq v]$ and its decomposition (4.8). Panel (b): the control function under Gaussianity (i.e., inverse-Mills ratio) versus LT (i.e., CLD). All the figures are drawn in terms of Φ^{-1} -transformed V for readability.

$\lambda'_0(\pi_0(0)) < 0$ and $\lambda'_0(\pi_0(1)) > 0$ (Lemma 4.2). Panel (a) also plots its decomposition (4.8) into the payoff-weighted channel contributions $\text{Cov}(Y_0, \Theta_z)E[\Theta_z | V \leq v]$: the coding-skills contribution vanishes and switches sign at $\pi_0(1)$, and the communication-skills contribution vanishes and switches sign at $\pi_0(0)$. As entry costs fall from the high-cost margin $\pi_0(0)$ to the low-cost margin $\pi_0(1)$, the trait mixture of the tech-sector workforce shifts from the coding channel toward the communication channel, and the payoff-weighted composition of marginal entrants moves from less favorable than the incumbents' (the first condition in (4.10)) to more favorable (the second condition), producing the slope reversal.

CLD only constrains $\rho_{Y_0, V}(y, v)$ at the two propensity-score thresholds, leaving various selection channels flexible to redistribute mass across the propensity-score support. The shape of the control function is then driven by the varying composition of different selection-channel forces. In contrast, joint Gaussianity for (Y_0, V) (panel (b)) allows for only a single channel of selection, resulting in a monotone shape of the control function. Note that, compared to Section 4.2, the flexibility exhibited here survives aggregation over y : although $\rho_{Y_0}(y)$ is integrated out in computing the conditional mean, substantial flexibility remains in λ_0 . This flexibility is what allows CLD to rationalize a Roy model and sector choices in which adverse selection at one margin and positive selection at another margin can coexist. The CLD Roy economy does not necessarily imply the conclusions on dispersion and skewness of sector wages in Heckman and Honoré (1990, Theorems 2 and 3) either. In this sense,

economic implications under [CLD](#) are more nuanced than under Gaussianity. Moreover, the analysis in [Section 3.1](#) implies that the parameters in the sector wage equations (as well as $\rho_{Y_0}(y)$) can be identified even with a binary cost shifter.²⁷

Remark 4.1 (Counterexample: Gaussian Mixture). *The primitive model [LT](#) consists of two latent components, which leads to a control function generated by a flexible mixture of latent skills in our running example. Although reminiscent of this mixture DGP, a DGP described by a bivariate Gaussian mixture of subpopulations with different skill sets (i.e., $\omega C(u, v; \rho_1) + (1 - \omega)C(u, v; \rho_2)$ with $0 < \omega < 1$) does not satisfy [CLD](#). The formal statement and its proof are contained in [SA C.5](#). $\square$*

4.4 Comparison to IVQR and RS

When the outcome is continuous, the IV quantile regression (IVQR) model ([Chernozhukov and Hansen, 2005](#)) provides an alternative set of conditions under which QSF_τ is point identified. Here, we discuss how IVQR is related to our approach when D is binary.

The IVQR model is based on the Skorohod representation, $Y_d = Q_{Y_d}(U_d)$, $U_d \sim U(0, 1)$, $d \in \{0, 1\}$. [Chernozhukov and Hansen \(2005\)](#) consider a general selection mechanism, $D = \delta(Z, V)$, where V can be vector-valued and Z satisfies $U_d \perp\!\!\!\perp Z$ (which is implied by [EX](#)). The key assumption of the IVQR model is RS, $U_1 \stackrel{d}{=} U_0 \mid Z, V$. RS weakens the classical RI assumption, which requires $U_1 = U_0$ a.s.

The IVQR model yields the following conditional moment restriction ([Chernozhukov and Hansen, 2005](#), Theorem 1),

$$\tau = \Pr[Y_1 \leq Q_{Y_1}(\tau), D_z = 1 \mid Z = z] + \Pr[Y_0 \leq Q_{Y_0}(\tau), D_z = 0 \mid Z = z], \quad z \in \{0, 1\}. \quad (4.11)$$

Under the selection model [\(3.1\)](#) and [EX](#), [\(4.11\)](#) can be rewritten as

$$\Pr[Y_1 \leq Q_{Y_1}(\tau), V_z \leq \pi_0(z)] = \Pr[Y_0 \leq Q_{Y_0}(\tau), V_z \leq \pi_0(z)], \quad z \in \{0, 1\}.$$

Using [Lemma 2.1](#), we can further rewrite it as

$$C(\tau, \pi_0(z); \rho_{Y_1, V_z}(Q_{Y_1}(\tau), \pi_0(z))) = C(\tau, \pi_0(z); \rho_{Y_0, V_z}(Q_{Y_0}(\tau), \pi_0(z))), \quad z \in \{0, 1\}.$$

²⁷Without any distributional assumptions or assumptions on the dependence structure, one would require IVs with large support to identify the wage parameters ([Eisenhauer et al., 2015](#)).

This shows that, interestingly, IVQR also relies on a version of constant local dependence:

$$\rho_{Y_1, V_z}(Q_{Y_1}(\tau), \pi_0(z)) = \rho_{Y_0, V_z}(Q_{Y_0}(\tau), \pi_0(z)), \quad z \in \{0, 1\}. \quad (4.12)$$

The IVQR model imposes restrictions across potential outcomes for each instrument level, whereas CLD(i) imposes restrictions across instrument levels for each potential outcome.

This discussion shows that the IVQR model and our approach rely on non-nested and complementary constant local dependence assumptions to identify causal effects for the overall population using instruments. Relative to the IVQR model, the proposed approach has three main advantages. First, it does not rely on continuity of the outcome to achieve point identification, and it naturally accommodates discrete and mixed discrete-continuous outcomes. Second, it imposes fewer restrictions on the relationship across potential outcomes and thus effect heterogeneity and selection on gains, compared to RS (see SA C.2). Finally, it does not require additional full rank and completeness conditions for identification.²⁸

4.5 Gaussian Relaxation and Extrapolation

Here, we provide an interpretation of CLD as an approach to extrapolation. We discuss two related aspects. First, we show how our approach can be viewed as a relaxation of Gaussianity by comparing CLD to assuming a Gaussian copula. Second, we shed light on how CLD extrapolates from the compliers to the overall population under the LATE assumptions using a first-order Taylor approximation of the LGR.

Gaussian Relaxation. To understand how our approach extrapolates from the observable potential outcome distributions $F_{Y_d|D_z}(y \mid d) = F_{Y|D,Z}(y \mid d, z)$ (under EX) to $F_{Y_d}(y)$, we begin by illustrating how extrapolation works with a Gaussian copula and then show how CLD relaxes Gaussianity while preserving point identification and extrapolation. We focus on the binary D case and CLD(i) for the purpose of illustration, and on the distribution of Y_0 ; the analysis for the distribution of Y_1 is symmetric.

If (Y_d, V_z) have a Gaussian copula with dependence parameter ρ_{Y_d, V_z} , which can be dif-

²⁸The moment restriction (4.11) is not sufficient for point identification. For point identification with a binary D , Chernozhukov and Hansen (2005) require the Jacobian of (4.11) to be of full rank and continuous. Vuong and Xu (2017) provide weaker conditions, requiring piecewise strict monotonicity of $y \mapsto (-1)^d(\Pr[Y \leq y, D = d \mid Z = 0] - \Pr[Y \leq y, D = d \mid Z = 1])$ for some d (in addition to REL). Both conditions restrict how the distribution of $(Y, D) \mid Z = z$ changes with z . By contrast, under CLD, we only require REL.

ferent for different values of (d, z) , then²⁹

$$\Pr[Y \leq y \mid D = 0, Z = z] = \frac{1}{\pi_0(z)} \int_0^{\pi_0(z)} \Phi \left(\frac{\Phi^{-1}(F_{Y_0}(y)) - \rho_{Y_0, V_z} \Phi^{-1}(\tilde{v})}{\sqrt{1 - \rho_{Y_0, V_z}^2}} \right) d\tilde{v}, \quad z \in \{0, 1\}.$$

If Y_0 takes on $L \geq 2$ values and $\rho_{Y_0, V_z} = \rho_{Y_0}$, then this yields a system of $2 \times (L - 1)$ equations with L unknowns ($L - 1$ values of $F_{Y_0}(y)$ and ρ_{Y_0}). The system is exactly determined if Y_0 is binary ($L = 2$) and overdetermined if $L > 2$.

By contrast, under **CLD**(i), we use a local version of the previous system,

$$\Pr[Y \leq y \mid D = 0, Z = z] = \frac{1}{\pi_0(z)} \int_0^{\pi_0(z)} \Phi \left(\frac{\Phi^{-1}(F_{Y_0}(y)) - \rho_{Y_0}(y) \Phi^{-1}(\tilde{v})}{\sqrt{1 - \rho_{Y_0}(y)^2}} \right) d\tilde{v}, \quad z \in \{0, 1\},$$

which also has $2 \times (L - 1)$ equations, but has $2 \times (L - 1)$ unknowns and therefore it is always exactly determined.³⁰ **CLD** is local in that the Gaussian copula does not need to be the same at every value of y , and the system only needs to hold at the two treatment propensities induced by Z . In this sense, **CLD** can be seen as a relaxation of Gaussianity. It shows how far we can deviate from Gaussianity while preserving point identification.

Extrapolation from Compliers to Overall Population. **CLD** identifies treatment effects for the overall population with a binary IV. This raises the question of how it extrapolates from the compliers to the overall population in the LATE framework of [Imbens and Angrist \(1994\)](#) under SRI ([Vytlacil, 2002](#)). There are several different approaches to extrapolation in the LATE/MTE framework. For example, [Brinch et al. \(2017\)](#) and [Kowalski \(2023\)](#) impose a linear model for $E[Y_d \mid V = v]$ to extrapolate from the compliers to the overall population.³¹ By contrast, as shown in Section 4.1, **CLD** imposes a flexible model for the entire conditional distribution, $F_{Y_d|V}(y \mid v)$, locally at $v \in \{\pi_d(0), \pi_d(1)\}$.

To better understand extrapolation under **CLD**, we relate F_{Y_0} to the potential outcome distribution for the compliers, $F_{Y_0|D_1 > D_0}$, using a first-order Taylor approximation of the Gaussian copula around $\rho = 0$ (i.e., around exogeneity).³² The relationship between F_{Y_1} and

²⁹The notion of Gaussian copula here is weaker than in the treatment effects literature because we do not impose SRI, which restricts selection heterogeneity, and work with separate copula representations for (Y_0, V_1) and (Y_0, V_0) .

³⁰Note that, without **CLD**(i), there are $3 \times (L - 1)$ unknowns ($L - 1$ values of each $F_{Y_0}(y)$, $\rho_{Y_0, V_0}(y, \pi_0(0))$ and $\rho_{Y_0, V_1}(y, \pi_0(1))$) by Lemma 4.1, and therefore the system is underdetermined.

³¹Other approaches are based on structural models (e.g., [Heckman et al., 2003](#)), covariates (e.g., [Angrist and Fernández-Val, 2013](#)), RS (e.g., [Wüthrich, 2020](#)), and the smoothness of the marginal treatment effect function (e.g., [Mogstad et al., 2018](#); [Han and Yang, 2024](#)).

³²This analysis is inspired by [Chiburis et al. \(2011, 2012\)](#), who use a Taylor expansion to relate the LATE to the ATE obtained from a bivariate Probit model.

$F_{Y_1|D_1>D_0}$ can be characterized similarly.³³ Under CLD(i) and using Lemma 2.1, $F_{Y_0|D_1>D_0}(y)$ can be written as

$$F_{Y_0|D_1>D_0}(y) = \frac{C(F_{Y_0}(y), \pi_0(0); \rho_{Y_0}(y)) - C(F_{Y_0}(y), \pi_0(1); \rho_{Y_0}(y))}{\pi_0(0) - \pi_0(1)}.$$

Then, using a first-order Taylor approximation of the Gaussian copula around $\rho = 0$, $C(u, v; \rho) \approx uv + \rho\phi(\Phi^{-1}(u))\phi(\Phi^{-1}(v))$, and rearranging, we get

$$\underbrace{F_{Y_0}(y)}_{\text{population CDF}} \approx \underbrace{F_{Y_0|D_1>D_0}(y)}_{\text{complier CDF}} + \underbrace{\rho_{Y_0}(y)\phi(\Phi^{-1}(F_{Y_0}(y))) \frac{\phi(\Phi^{-1}(\pi_0(1))) - \phi(\Phi^{-1}(\pi_0(0)))}{\pi_0(0) - \pi_0(1)}}_{\text{correction term}}. \quad (4.13)$$

Equation (4.13) sheds light on how our approach extrapolates from the compliers to the overall population. Specifically, it starts with the complier distribution and adds a correction term, which has three components. The first component, $\rho_{Y_0}(y)$, captures the degree of endogeneity, which under CLD is specific to the point y . If $\rho_{Y_0}(y) = 0$, then the correction term is zero, and the complier CDF coincides with the population CDF at y . The second component, $\phi(\Phi^{-1}(F_{Y_0}(y)))$, captures the impact of the point y at which we are extrapolating. It is large around the median where $F_{Y_0}(y) = 0.5$ and smaller in the tails. This reflects the fact that there is more room for variation in the CDF around the median than in the tails given that it is bounded between zero and one. The third component is equal to $E[\Phi^{-1}(V) | D_1 > D_0] - E[\Phi^{-1}(V)]$ and thus captures the average position of the compliers in the distribution of V relative to the overall population. It is equal to zero under perfect compliance ($\pi_0(0) = 1$ and $\pi_0(1) = 0$), consistent with the fact that F_{Y_0} is point identified via $F_{Y_0|D_1>D_0}$ in this case. The third component is also equal to zero if the first stage is symmetric, $\pi_0(0) = 1 - \pi_0(1)$. This provides an interesting connection to Angrist (2004) who shows that the LATE equals the ATE under symmetry of the first stage (combined with symmetry of the latent error distribution); see also Heckman and Vytlacil (2000, Section 7).

5 Estimation and Inference

The identification results in Section 3 are constructive and lead to tractable estimators. Here, we propose semiparametric estimators of F_{Y_d} for the three types of treatments based on distribution regression (DR) and show how to construct estimators of treatment effect parameters using the plug-in rule. We focus on target parameters that yield $\sqrt{n}$ -consistent

³³A similar argument to below can be used to characterize the relationship between $F_{Y_0}(y)$ and $F_{Y_0|V_z}(y | \pi_0(z))$ under CLD(iii).

and asymptotically normal estimators. We do not consider parameters like $\partial QSF_\tau(d)/\partial d$, because estimating such parameters would involve nonparametric methods, which require choosing tuning parameters and exhibit slower convergence rates.

In this section, we make the role of the covariates X explicit. We assume we have access to a random sample of size n from (Y, D, Z, X) , $\{(Y_i, D_i, Z_i, X_i)\}_{i=1}^n$. Let $B(X_i)$, $B(Z_i, X_i)$, and $B(D_i, Z_i, X_i)$ denote vectors of transformations of X_i , (Z_i, X_i) , and (D_i, Z_i, X_i) , respectively. Define the indicators $I_i(y) := 1\{Y_i \leq y\}$ and $J_i(d) := 1\{D_i \leq d\}$. Let $\mathcal{D}_h$ and $\mathcal{Y}_h$ be two finite grids covering $\mathcal{D}$ and $\mathcal{Y}$ with spacing h .³⁴

5.1 Binary Treatments

We consider separate DR models for the conditional potential outcome distributions,

$$F_{Y_d|X}(y|x) = \Phi(B(x)'\beta_d(y)), \quad d \in \{0, 1\}, \quad (5.1)$$

where $y \mapsto \beta_d(y)$ is a function-valued unknown parameter, and a Probit model for the propensity score,

$$1 - \pi_0(z, x) := \Pr[D = 1 \mid Z = z, X = x] = \Phi(B(z, x)'\pi). \quad (5.2)$$

We model the local dependence parameter at the d -treatment thresholds induced by Z as

$$\rho_{Y_d;X}(y; x) = \rho(B(x)'\gamma_d(y)), \quad d \in \{0, 1\}, \quad (5.3)$$

where $\rho(u) = \tanh(u) \in (-1, 1)$, the inverse Fisher transformation, and $y \mapsto \gamma_d(y)$ is a function-valued unknown parameter.³⁵ Together, (5.1), (5.2), and (5.3) imply the bivariate DR model

$$\begin{aligned} \Pr[Y \leq y, D = 1 \mid Z = z, X = x] &= \Phi_2(B(x)'\beta_1(y), B(z, x)'\pi; -\rho(B(x)'\gamma_1(y))), \\ \Pr[Y \leq y, D = 0 \mid Z = z, X = x] &= \Phi_2(B(x)'\beta_0(y), -B(z, x)'\pi; \rho(B(x)'\gamma_0(y))), \end{aligned}$$

where we have used the symmetry properties of the bivariate Gaussian distribution to simplify the previous expressions.

We propose a computationally tractable two-step maximum likelihood estimator, building on Chernozhukov et al. (2025a).

³⁴We can set $\mathcal{Y}_h = \mathcal{Y}$ when $\mathcal{Y}$ is finite and $\mathcal{D}_h = \mathcal{D}$ for binary and ordered D .

³⁵To simplify the notation, we use the same vector of transformations in models (5.1) and (5.3). This is not essential, and one can use different specifications in both models.

Algorithm 5.1 (Estimation with Binary D). *We compute the estimator in two stages:*

1. *Treatment equation: estimate π using a Probit regression*

$$\hat{\pi} \in \arg \max_c \sum_{i=1}^n [D_i \log \Phi(B(Z_i, X_i)'c) + (1 - D_i) \log(1 - \Phi(B(Z_i, X_i)'c))].$$

2. *Outcome equation: for $y \in \mathcal{Y}_h$ and $d \in \{0, 1\}$, $\hat{F}_{Y_d|X}(y|x) = \Phi(B(x)'\hat{\beta}_d(y))$ and $\hat{\rho}_{Y_d;X}(y; x) = \rho(B(x)'\hat{\gamma}_d(y))$, where*

$$\begin{aligned} (\hat{\beta}_1(y), \hat{\gamma}_1(y)) &\in \arg \max_{b,g} \sum_{i=1}^n D_i [I_i(y) \log \Phi_2(B(X_i)'b, B(Z_i, X_i)'\hat{\pi}; -\rho(B(X_i)'g)) \\ &\quad + (1 - I_i(y)) \log \Phi_2(-B(X_i)'b, B(Z_i, X_i)'\hat{\pi}; \rho(B(X_i)'g))], \\ (\hat{\beta}_0(y), \hat{\gamma}_0(y)) &\in \arg \max_{b,g} \sum_{i=1}^n (1 - D_i) [I_i(y) \log \Phi_2(B(X_i)'b, -B(Z_i, X_i)'\hat{\pi}; \rho(B(X_i)'g)) \\ &\quad + (1 - I_i(y)) \log \Phi_2(-B(X_i)'b, -B(Z_i, X_i)'\hat{\pi}; -\rho(B(X_i)'g))]. \end{aligned}$$

Rearrange the estimates $y \mapsto \hat{F}_{Y_d|X}(y | x)$ on $\mathcal{Y}_h$ if needed.

Remark 5.1 (Computation). *The first stage of Algorithm 5.1 is a conventional Probit regression, and estimation can proceed using existing software. The second stage is computationally more expensive since it involves a nonlinear smooth optimization problem. This optimization problem can be solved using standard algorithms such as Newton-Raphson. To improve efficiency, a one-step correction can be added to Algorithm 5.1 (see, e.g., Section 3.4 in Newey and McFadden, 1994).* $\square$

5.2 Ordered Discrete Treatments

As for binary treatments, we model all the components using flexible generalized linear and DR models: $F_{Y_d|X}(y | x) = \Phi(B(x)'\beta_d(y))$, $\pi_d(z, x) = F_{D|Z,X}(d | z, x) = \Phi(B(z, x)'\pi(d))$, and $\rho_{Y_d;X}(y; x) = \rho(B(x)'\gamma_d(y))$ with $\rho(u) = \tanh(u)$.

Algorithm 5.2 (Estimation with Ordered D). *We compute the estimator in two stages:*

1. *Treatment equation: set $\hat{\pi}_{-1}(z, x) = 0$ and $\hat{\pi}_K(z, x) = 1$ for all (z, x) . For $d \in \{0, \dots, K - 1\}$, $\hat{\pi}_d(z, x) = \Phi(B(z, x)'\hat{\pi}(d))$, where*

$$\hat{\pi}(d) \in \arg \max_p \sum_{i=1}^n [J_i(d) \log \Phi(B(Z_i, X_i)'p) + (1 - J_i(d)) \log \Phi(-B(Z_i, X_i)'p)].$$

Rearrange the estimates $d \mapsto \hat{\pi}_d(z, x)$ on $\mathcal{D}$ and trim and renormalize if needed to ensure $\hat{\pi}_d(z, x) > \hat{\pi}_{d-1}(z, x)$. This rearrangement is important to avoid having logarithms of nonpositive numbers in the second stage.³⁶

2. Outcome equation: for $y \in \mathcal{Y}_h$ and $d \in \mathcal{D}$, let $\hat{F}_{Y_d|X}(y | x) = \Phi(B(x)' \hat{\beta}_d(y))$ and $\hat{\rho}_{Y_d|X}(y; x) = \rho(B(x)' \hat{\gamma}_d(y))$, where

$$(\hat{\beta}_d(y), \hat{\gamma}_d(y)) \in \arg \max_{b, g} \sum_{i=1}^n 1\{D_i = d\} [I_i(y) \log g_{d,i}(b, g) + (1 - I_i(y)) \log \bar{g}_{d,i}(b, g)],$$

with $g_{0,i}(b, g) := \Phi_2(B(X_i)'b, \Phi^{-1}(\hat{\pi}_0(Z_i, X_i)); \rho(B(X_i)'g))$, for $d \in \mathcal{D} \setminus \{0, K\}$, $g_{d,i}(b, g) := \Phi_2(B(X_i)'b, \Phi^{-1}(\hat{\pi}_d(Z_i, X_i)); \rho(B(X_i)'g)) - \Phi_2(B(X_i)'b, \Phi^{-1}(\hat{\pi}_{d-1}(Z_i, X_i)); \rho(B(X_i)'g))$, $g_{K,i}(b, g) := \Phi(B(X_i)'b) - \Phi_2(B(X_i)'b, \Phi^{-1}(\hat{\pi}_{K-1}(Z_i, X_i)); \rho(B(X_i)'g))$ and $\bar{g}_{d,i}(b, g) := \hat{\pi}_d(Z_i, X_i) - \hat{\pi}_{d-1}(Z_i, X_i) - g_{d,i}(b, g)$. Rearrange the estimates $y \mapsto \hat{F}_{Y_d|X}(y | x)$ on $\mathcal{Y}_h$ if needed.

Remark 5.2 (Computation). The first stage is a sequence of Probit regressions that can be solved using standard software. The second stage is a nonlinear smooth optimization problem that can be solved using standard algorithms such as Newton-Raphson. As in Algorithm 5.1, a one-step correction can be added to Algorithm 5.2 to improve efficiency. $\square$

5.3 Continuous Treatments

We construct plug-in estimators based on the closed-form solutions in Section 3.3. We consider DR models for $F_{Y|D,Z,X}$ and $F_{D|Z,X}$, $F_{Y|D,Z,X}(y | d, z, x) = \Phi(B(d, z, x)' \beta(y))$ and $F_{D|Z,X}(d | z, x) = \Phi(B(z, x)' \pi(d))$, respectively, where $y \mapsto \beta(y)$ and $d \mapsto \pi(d)$ are unknown function-valued parameters.

Algorithm 5.3 (Estimation with Continuous D). We compute the estimator in two stages:

1. Observable conditional distributions: for $y \in \mathcal{Y}_h$ and $d \in \mathcal{D}_h$, $\hat{F}_{Y|D,Z,X}(y|d, z, x) = \Phi(B(d, z, x)' \hat{\beta}(y))$ and $\hat{F}_{D|Z,X}(d|z, x) = \Phi(B(z, x)' \hat{\pi}(d))$, where

$$\begin{aligned} \hat{\beta}(y) &\in \arg \max_b \sum_{i=1}^n [I_i(y) \log \Phi(B(D_i, Z_i, X_i)'b) + (1 - I_i(y)) \log(1 - \Phi(B(D_i, Z_i, X_i)'b))], \\ \hat{\pi}(d) &\in \arg \max_p \sum_{i=1}^n [J_i(d) \log \Phi(B(Z_i, X_i)'p) + (1 - J_i(d)) \log(1 - \Phi(B(Z_i, X_i)'p))]. \end{aligned}$$

³⁶In the second stage, $g_{d,i}(b, g) > 0$ and $\bar{g}_{d,i}(b, g) > 0$ a.s. if $\hat{\pi}_d(Z_i, X_i) > \hat{\pi}_{d-1}(Z_i, X_i)$ a.s.

2. *Potential outcome distributions: for $y \in \mathcal{Y}_h$ and $d \in \mathcal{D}_h$, $\widehat{F}_{Y_d|X}(y|x) = \Phi(\widehat{a}_{d,y;x}/\sqrt{1 + \widehat{b}_{d,y;x}^2})$ and $\widehat{\rho}_{Y_d|X}(y; x) = -\widehat{b}_{d,y;x}/\sqrt{1 + \widehat{b}_{d,y;x}^2}$, where*

$$\begin{aligned}\widehat{a}_{d,y;x} &= \frac{(B(d, 0, x)' \widehat{\beta}(y))(B(1, x)' \widehat{\pi}(d)) - (B(d, 1, x)' \widehat{\beta}(y))(B(0, x)' \widehat{\pi}(d))}{B(1, x)' \widehat{\pi}(d) - B(0, x)' \widehat{\pi}(d)}, \\ \widehat{b}_{d,y;x} &= \frac{B(d, 1, x)' \widehat{\beta}(y) - B(d, 0, x)' \widehat{\beta}(y)}{B(1, x)' \widehat{\pi}(d) - B(0, x)' \widehat{\pi}(d)}.\end{aligned}$$

Rearrange the estimates $y \mapsto \widehat{F}_{Y_d|X}(y | x)$ on $\mathcal{Y}_h$ if needed.

5.4 Marginal Distributions and Functionals

With covariates, the marginal distributions of the potential outcomes are identified as

$$F_{Y_d}(y) = \int F_{Y_d|X}(y | x) dF_X(x), \quad d \in \mathcal{D},$$

where F_X is the distribution of X . We can use this expression to construct estimators of F_{Y_d} and functionals of interest, such as the QSF and QTE, by plugging in the estimators obtained above. To provide a unified approach to all the treatment cases, we set $\mathcal{D}_h = \mathcal{D}$ when D is binary or discrete ordered.

Algorithm 5.4 (Estimation of F_{Y_d} , QSF and QTE). *Estimation proceeds in two steps.*

1. *Unconditional distribution: for $y \in \mathcal{Y}_h$ and $d \in \mathcal{D}_h$, $\widehat{F}_{Y_d}(y) = \frac{1}{n} \sum_{i=1}^n \widehat{F}_{Y_d|X}(y | X_i)$. Values y between grid points can be obtained by step or linear interpolation, with extension outside the grid satisfying the CDF boundary constraints.*
2. *Unconditional QSF and QTE: $\widehat{QSF}_\tau(d) = \mathcal{Q}_\tau(\widehat{F}_{Y_d})$ and $\widehat{QTE}_\tau(d, d') = \{\widehat{QSF}_\tau(d) - \widehat{QSF}_\tau(d')\}/(d - d')$.*

Remark 5.3 (Estimating ASF and ATE). *Based on Step 1 of Algorithm 5.4, the ASF and ATE can be estimated using the following plug-in estimators, $\widehat{ASF}(d) = \int y d\widehat{F}_{Y_d}(y)$, and $\widehat{ATE}(d, d') = \{\widehat{ASF}(d) - \widehat{ASF}(d')\}/(d - d')$. Implementing these estimators is straightforward when the outcomes and treatments are discrete. With continuous outcomes or treatments, additional computational challenges arise because we approximate $\mathcal{Y}$ and $\mathcal{D}$ using grids $\mathcal{Y}_h$ and $\mathcal{D}_h$. The resulting estimators of the ASF and ATE can be sensitive to the choice of grid points in the tails, especially when the outcome and treatment are unbounded and the sample size is small or moderate.³⁷ In such settings, we recommend focusing on esti-*

³⁷One can also construct plug-in estimators based on other representations of the ASF, such as $ASF(d) = \int_0^1 QSF_\tau(d) d\tau$ or $ASF(d) = \int_0^\infty (1 - F_{Y_d}(y)) dy - \int_{-\infty}^0 F_{Y_d}(y) dy$.

mands that admit more robust estimators, such as the QSF and QTE for $\tau \in \mathcal{T} \subset (c, 1 - c)$, for $c > 0$, as illustrated in our empirical application. $\square$

5.5 Asymptotic Theory and Inference

The target parameters in Section 5.4 are function-valued. Inference on these parameters can be performed using resampling methods. To provide a unified framework, we denote the functional parameters by $u \mapsto \delta_u, u \in \mathcal{U}$, where $\mathcal{U} \subset \tilde{\mathcal{Y}} \times \tilde{\mathcal{D}} \times \mathcal{T}$, with $\mathcal{T} \subset (c, 1 - c)$ for $c > 0$, $\tilde{\mathcal{Y}}$ is a compact subset of $\mathcal{Y}$ when $\mathcal{Y}$ is uncountable and equal to $\mathcal{Y}$ otherwise, and $\tilde{\mathcal{D}}$ is defined analogously. For example, if we are interested in $\tau \mapsto QSF_\tau(d)$ on $[.05, .95]$, then $u = \tau$, $\delta_u = QSF_u(d)$ and $\mathcal{U} = [.05, .95]$ for fixed d . In practice, we approximate $\mathcal{U}$ using a fine grid $\mathcal{U}_h$. We denote the estimator of δ_u obtained from Algorithms 5.1–5.4 as $\hat{\delta}_u$.

We show in SA D that $\sqrt{n}(\hat{\delta}_u - \delta_u)$ converges in distribution to a mean-zero Gaussian process Z_δ and the bootstrap is valid for a wide range of parameters of interest, including F_{Y_d} and its (Hadamard differentiable) functionals. These results imply that the bootstrap algorithm below is theoretically valid. When the outcomes are discrete or mixed discrete-continuous, the estimators of the QSF or QTE are not asymptotically Gaussian in general. In this case, one can use the inference methods proposed by Chernozhukov et al. (2020).

We focus on constructing pointwise confidence intervals (CIs) and uniform confidence bands (CBs), $CI_{(1-\alpha)}(\delta_u)$ and $CB_{(1-\alpha)}(\delta_u)$ respectively, satisfying $\lim_{n \rightarrow \infty} \Pr[\delta_u \in CI_{(1-\alpha)}(\delta_u)] = 1 - \alpha$ and $\lim_{n \rightarrow \infty} \Pr[\delta_u \in CB_{(1-\alpha)}(\delta_u) \text{ for all } u \in \mathcal{U}] = 1 - \alpha$. Uniform CBs can be used to test a variety of hypotheses of interest, such as the hypotheses of no effect or constant effects when applied to QTE, or stochastic dominance.

The following algorithm provides a generic bootstrap construction of $CB_{(1-\alpha)}(\delta_u)$. A similar algorithm can be used for $CI_{(1-\alpha)}(\delta_u)$, which we omit.

Algorithm 5.5 (Uniform Confidence Bands for Functional Parameters³⁸).

1. For $u \in \mathcal{U}_h$, obtain B bootstrap draws of the estimator $\hat{\delta}_u$, $\{\hat{\delta}_u^{(b)} : 1 \leq b \leq B\}$.
2. For $u \in \mathcal{U}_h$, compute the robust standard error, $SE(\hat{\delta}_u) = (\hat{Q}_{0.75}(\hat{\delta}_u) - \hat{Q}_{0.25}(\hat{\delta}_u)) / (\Phi^{-1}(0.75) - \Phi^{-1}(0.25))$, where $\hat{Q}_\tau(\hat{\delta}_u)$ is the τ -quantile of $\{\hat{\delta}_u^{(b)} : 1 \leq b \leq B\}$.
3. Compute $cv(1-\alpha)$ as the $(1-\alpha)$ -quantile of $\{\max_{u \in \mathcal{U}_h} SE(\hat{\delta}_u)^{-1} |\hat{\delta}_u^{(b)} - \hat{\delta}_u| : 1 \leq b \leq B\}$.
4. Compute the $(1 - \alpha)$ uniform CB as $CB_{(1-\alpha)}(\delta_u) = [\hat{\delta}_u \pm cv(1 - \alpha)SE(\hat{\delta}_u)]$, $u \in \mathcal{U}_h$.

The algorithms for binary and ordered treatments (Algorithms 5.1 and 5.2) involve non-linear optimization problems in the second step. Therefore, we recommend using the multiplier bootstrap in Step 1 of Algorithm 5.5, which avoids re-estimating parameters in each

³⁸See, for example, Chernozhukov et al. (2013, 2020, 2025a) for similar algorithms.

of the B bootstrap iterations. The algorithm for continuous treatments (Algorithm 5.3) does not involve a nonlinear optimization problem in the second step and is computationally less expensive than Algorithms 5.1 and 5.2. Therefore, the standard empirical bootstrap is a natural alternative to the multiplier bootstrap, provided that the sample size is not too large, as for example in Section 6.

Remark 5.4 (Additional theoretical complications when estimating the ASF and ATE). *Additional theoretical complications arise when estimating the ASF and ATE with unbounded $\mathcal{Y}$ and $\mathcal{D}$. Standard functional central limit theorems for DR estimators (e.g., Chernozhukov et al., 2013) guarantee convergence only over a compact subset of the support. One approach to establish functional central limit theorems over the entire support is to impose restrictions on the tails of the distribution (e.g., Chernozhukov et al., 2025c).* $\square$

Remark 5.5 (Combining different identification approaches for efficiency). *The proposed method can be combined with alternative identification approaches to obtain more efficient estimators. If the alternative approaches identify the same estimands and yield nonredundant moment conditions, then such moment conditions can be combined with the first-order conditions from our estimators in a GMM framework. The choice of whether to combine different approaches is subject to a trade-off between robustness (i.e., relying on fewer assumptions) and obtaining more efficient estimators. We recommend using overidentification tests whenever multiple approaches are combined.* $\square$

6 Empirical Illustration

We illustrate our methods by estimating the distributional effects of sleep on well-being.³⁹ We reanalyze the data from Bessone et al. (2021b), who studied the effects of randomized interventions to increase sleep time of low-income adults in India.⁴⁰ According to an expert survey by Bessone et al. (2021b), increasing sleep could have large benefits in this empirical context, which is characterized by low baseline levels of sleep per night and sleep efficiency.

Bessone et al. (2021b) considered two main treatments (see their Section III for details): (i) *devices + encouragement* (information, encouragement and sleep trackers, various devices for improving sleep environment) and (ii) *devices + incentives* (same as (i) and payments for each minute of sleep increase). In addition, they cross-randomized a *nap treatment* (the opportunity to nap at work).

The outcome of interest (Y) is an overall index of individual well-being. The treatment (D) is sleep time per night (in hours). We use an indicator for whether an individual received

³⁹The empirical results were obtained using **Stata** (StataCorp, 2025) and **R** (R Core Team, 2026).

⁴⁰We downloaded the data from the Harvard Dataverse replication package (Bessone et al., 2021a).

either of the main treatments as an instrument (Z) for sleep. The vector of covariates (X) includes controls for gender, three age indicators, and the baseline well-being index, as in Bessone et al. (2021b, Table A.XVII). Following Dong and Lee (2023), we restrict the sample to individuals who did not receive the nap treatment, so that the total sample size is $n = 226$.

The treatment D takes on many values in this application and is treated as continuous. We therefore use the estimators for continuous treatments described in Algorithm 5.3. We choose fully saturated (in Z_i) specifications for $B(d, z, x)$ and $B(z, x)$. The resulting DR models are $F_{Y|D,Z,X}(y \mid d, z, x) = \Phi((d, x')\beta_z(y))$ and $F_{D|Z,X}(d \mid z, x) = \Phi(x'\pi_z(d))$ for $z \in \{0, 1\}$. Our flexible estimators allow us to analyze the full distributional impact of sleep on well-being. Our results thus complement the analyses of the average effect of sleep on well-being in Bessone et al. (2021b) and Dong and Lee (2023).

We begin by analyzing the (distributional) first-stage relationship between D and Z to shed light on the plausibility of REL. Figure 3(a) plots $\hat{F}_{D|Z}(\cdot \mid 1) = n^{-1} \sum_{i=1}^n \hat{F}_{D|Z,X}(\cdot \mid 1, X_i)$ and $\hat{F}_{D|Z}(\cdot \mid 0) = n^{-1} \sum_{i=1}^n \hat{F}_{D|Z,X}(\cdot \mid 0, X_i)$. It shows that the instrument shifts the distribution of D : sleep under the experimental treatments first-order stochastically dominates sleep without treatments.

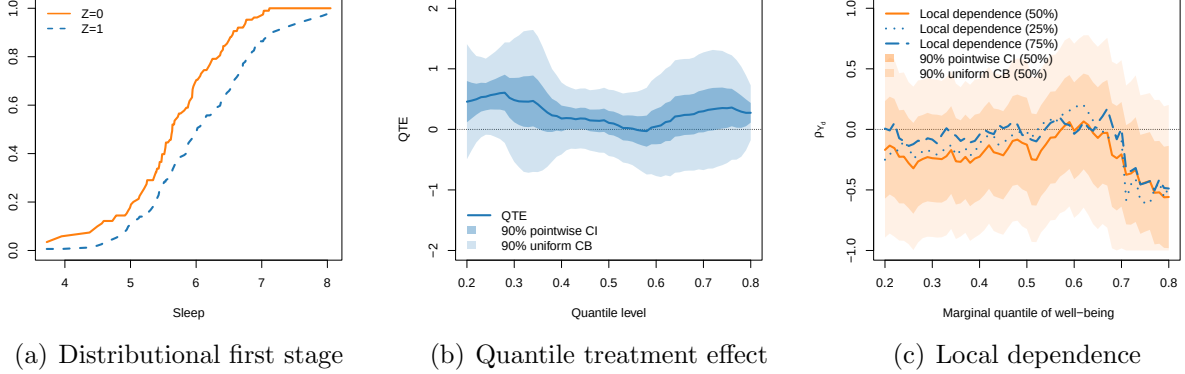
In addition to REL, our method relies on EX and CLD. The experimental random assignment renders the independence assumptions in EX plausible.⁴¹ CLD allows the local dependence between potential well-being and unobserved sleep propensity (i.e., the selection sorting) to depend on the level of well-being (e.g., underlying health), while restricting it to be constant in sleep propensity (e.g., biological sleep factors) at and locally around the d -treatment thresholds. A sufficient condition for CLD to hold is that selection sorting depends on underlying health but does not change with biological sleep factors. CLD would be violated when selection sorting changes with sleep propensity level in an unrestricted fashion. For example, sleep propensity may be negatively related to underlying health for those with high propensity, while positively related for those with low propensity.

Figure 3(b) plots estimates of the QTE, $\{\widehat{QTE}_\tau(d, d') : \tau \in [.2, .8]\}$, including 90% pointwise CIs and uniform CBs computed using the empirical bootstrap. We set $d = \hat{Q}_D(0.75)$ and $d' = \hat{Q}_D(0.25)$, where $\hat{Q}_D(\tau)$ is the τ -quantile of the empirical distribution of sleep. Figure 3(b) suggests interesting effect heterogeneity across the distribution of well-being. The QTEs are the largest and pointwise significant in the tails and smaller and insignificant around the median. While the uniform CB includes zero at all quantiles, it allows us to rule out large negative effects at the lower tail.

An interesting feature of our method is that it allows estimation of the local dependence

⁴¹Note that random assignment does not automatically imply the (implicit) exclusion restriction, which requires that the instrument has no direct effect on well-being.

Figure 3: Quantile Treatment Effects and Local Dependence



Notes: Pointwise CIs and uniform CB for the QTE and local dependence are computed using the empirical bootstrap with 5,000 repetitions. All specifications control for gender, three age indicators, and the baseline well-being index.

parameter $\rho_{Y_d;X}(y; x)$, which can be interpreted as a local measure of endogeneity or self-selection. Figure 3(c) plots the average local dependence parameter, $\tau \mapsto \frac{1}{n} \sum_{i=1}^n \hat{\rho}_{Y_d;X}(\hat{Q}_Y(\tau); X_i)$, for $(d, \tau) \in \{\hat{Q}_D(0.25), \hat{Q}_D(0.50), \hat{Q}_D(0.75)\} \times [0.2, 0.8]$, where $\hat{Q}_Y(\tau)$ is the τ -quantile of the empirical distribution of well-being. The average local dependence is mostly negative, and there is interesting heterogeneity both across well-being and sleep levels. These predominantly negative point estimates suggest negative local dependence between the unobserved sleep propensity and potential well-being. This pattern is consistent with individuals with relatively poor underlying health conditions sleeping more. Figure 3(c) also shows that, at $d = \hat{Q}_D(0.50)$, average local dependence is pointwise significant at the upper tail, but the uniform CB includes zero at all quantiles.

Negative local dependence suggests that estimators ignoring the endogeneity of sleep underestimate the effect of sleep on well-being. Figure 4(a) illustrates this by comparing our (IV) QTE estimates with those under conditional exogeneity, $Y_d \perp\!\!\!\perp D \mid X$ for $d \in \mathcal{D}$.⁴² The estimates under conditional exogeneity are small and insignificant at most quantiles, missing the positive well-being effects of sleep in the tails of the well-being distribution. This highlights the importance of using IV methods in this empirical context.

Figure 4(b) further showcases the value added that our method can bring to standard empirical analyses focusing on average effects. It compares the QTE estimates obtained from our method to standard 2SLS estimates using the same set of covariates. While the 2SLS

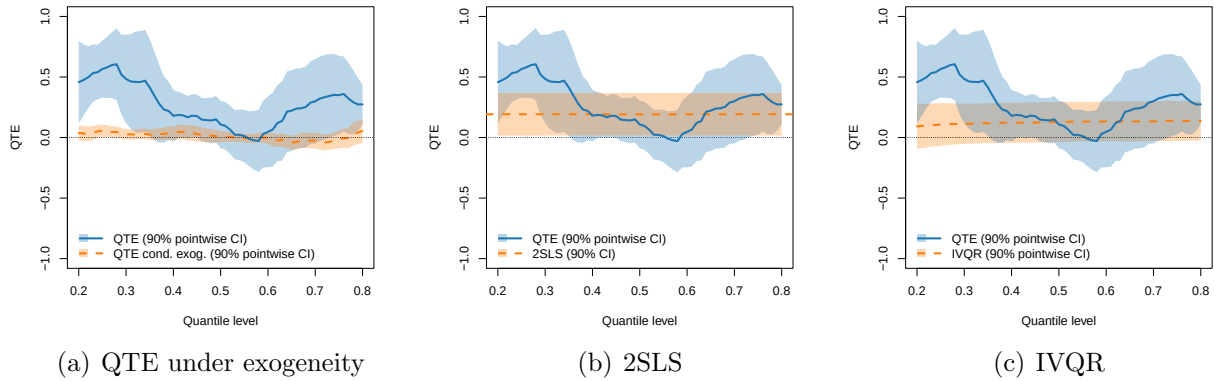
⁴²Under conditional exogeneity, F_{Y_d} is identified as $F_{Y_d}(y) = \int F_{Y|D,X}(y|d, x) dF_X(x)$. For estimation, we consider the DR model $F_{Y|D,X}(y|d, x) = \Phi(d\gamma(y) + x'\beta(y))$. The DR estimator is $\hat{F}_{Y|D,X}(y|d, x) = \Phi(d\hat{\gamma}(y) + x'\hat{\beta}(y))$, where $\hat{\gamma}(y)$ and $\hat{\beta}(y)$ are the coefficients obtained from a Probit regression of $1\{Y_i \leq y\}$ on D_i and X_i . The final estimator of $F_{Y_d}(y)$ under conditional exogeneity is $\hat{F}_{Y_d}(y) = \frac{1}{n} \sum_{i=1}^n \Phi(d\hat{\gamma}(y) + X_i'\hat{\beta}(y))$.

analysis suggests that sleep has moderate average effects on well-being, our method uncovers interesting patterns of heterogeneity in the effect of sleep on well-being.

Finally, we compare our results with estimates from the IVQR model (Chernozhukov and Hansen, 2005). As discussed in Section 4.4, in addition to IV validity, the key identifying assumption underlying IVQR is RS. RS requires the individuals' ranks in the potential well-being distributions to be the same across different levels of sleep, up to “unsystematic deviations,” in the language of Chernozhukov and Hansen (2005), thereby restricting the heterogeneity in the effect of sleep on well-being. For example, RS requires an individual's well-being ranking when sleeping few hours to remain similar when sleeping many hours, up to unsystematic deviations. This discussion shows how CLD and RS are complementary assumptions that impose non-nested restrictions. CLD restricts how the nature of endogeneity or selection can depend on unobservable determinants of sleep, while RS restricts the effect of sleep on well-being.

Figure 4(c) compares the QTE estimates obtained from our method to unconditional QTE estimates based on the IVQR model.⁴³ The IVQR estimates suggest that there is much less heterogeneity across quantiles than our method, and the pointwise CIs include zero at all quantiles.

Figure 4: Comparison to Other Methods



Notes: Pointwise CIs for the QTE are computed using the empirical bootstrap with 5,000 repetitions. All specifications control for gender, three age indicators, and the baseline well-being index.

⁴³To obtain unconditional QTE estimates based on the IVQR model, we proceed as follows. We estimate linear IVQR models using the inverse quantile regression estimator of Chernozhukov and Hansen (2006) and obtain the unconditional QTE estimates as described in Chernozhukov et al. (2017, Section 9.3) using the plug-in approach proposed in Chernozhukov et al. (2013).

7 Concluding Remarks

When identifying the causal effect of endogenous treatments, researchers inevitably face modeling trade-offs. This paper proposes a new direction to deal with these trade-offs by imposing assumptions on the local dependence between potential outcomes and unobservables determining treatment assignment. In doing so, we make minimal assumptions on the equations determining potential outcomes and treatments, thereby allowing for rich heterogeneity in these equations. The proposed framework applies to binary, discrete ordered, and continuous treatments, where point identification is achieved even with instruments that have limited variation (e.g., a binary instrument). In all three cases, the identification analysis leads to practical estimation and inference procedures that might appeal to practitioners.

References

- ABADIE, A., J. ANGRIST, AND G. IMBENS (2002): “Instrumental variables estimates of the effect of subsidized training on the quantiles of trainee earnings,” *Econometrica*, 70, 91–117.
- ABBRING, J. H. AND J. J. HECKMAN (2007): “Chapter 72 Econometric Evaluation of Social Programs, Part III: Distributional Treatment Effects, Dynamic Treatment Effects, Dynamic Discrete Choice, and General Equilibrium Policy Evaluation,” Elsevier, vol. 6 of *Handbook of Econometrics*, 5145–5303.
- AI, C. AND X. CHEN (2003): “Efficient estimation of models with conditional moment restrictions containing unknown functions,” *Econometrica*, 71, 1795–1843.
- ANGRIST, J. D. (2004): “Treatment Effect Heterogeneity in Theory and Practice,” *The Economic Journal*, 114, C52–C83.
- ANGRIST, J. D. AND I. FERNÁNDEZ-VAL (2013): “ExtrapoLATE-ing: External Validity and Overidentification in the LATE Framework,” in *Advances in Economics and Econometrics: Tenth World Congress*, ed. by D. Acemoglu, M. Arellano, and E. Dekel, Cambridge: Cambridge University Press, 401–434.
- ANGRIST, J. D., K. GRADDY, AND G. W. IMBENS (2000): “The Interpretation of Instrumental Variables Estimators in Simultaneous Equations Models with an Application to the Demand for Fish,” *The Review of Economic Studies*, 67, 499–527.
- ANGRIST, J. D. AND G. W. IMBENS (1995): “Two-Stage Least Squares Estimation of Average Causal Effects in Models with Variable Treatment Intensity,” *Journal of the American Statistical Association*, 90, 431–442.

- ARELLANO, M. AND S. BONHOMME (2017): “Quantile selection models with an application to understanding changes in wage inequality,” *Econometrica*, 85, 1–28.
- ATHEY, S. AND G. W. IMBENS (2006): “Identification and Inference in Nonlinear Difference-in-Differences Models,” *Econometrica*, 74, 431–497.
- BALKE, A. AND J. PEARL (1997): “Bounds on treatment effects from studies with imperfect compliance,” *Journal of the American Statistical Association*, 92, 1171–1176.
- BEARE, B. K. (2010): “Copulas and Temporal Dependence,” *Econometrica*, 78, 395–410.
- BESSONE, P., G. RAO, F. SCHILBACH, H. SCHOFIELD, AND M. TOMA (2021a): “Replication Data for: ‘The Economic Consequences of Increasing Sleep among the Urban Poor’,” Harvard Dataverse, <https://doi.org/10.7910/DVN/GJ9QPC>.
- (2021b): “The Economic Consequences of Increasing Sleep Among the Urban Poor,” *The Quarterly Journal of Economics*, 136, 1887–1941.
- BLACK, S. E. AND P. J. DEVEREUX (2011): “Chapter 16 - Recent Developments in Intergenerational Mobility,” Elsevier, vol. 4 of *Handbook of Labor Economics*, 1487–1541.
- BLUNDELL, R., X. CHEN, AND D. KRISTENSEN (2007): “Semi-nonparametric IV estimation of shape-invariant Engel curves,” *Econometrica*, 75, 1613–1669.
- BRINCH, C. N., M. MOGSTAD, AND M. WISWALL (2017): “Beyond LATE with a discrete instrument,” *Journal of Political Economy*, 125, 985–1039.
- CARNEIRO, P. AND S. LEE (2009): “Estimating distributions of potential outcomes using local instrumental variables with an application to changes in college enrollment and wage inequality,” *Journal of Econometrics*, 149, 191–208.
- CHAY, K. AND M. GREENSTONE (2005): “Does Air Quality Matter? Evidence from the Housing Market,” *Journal of Political Economy*, 113, 376–424.
- CHEN, X. AND Y. FAN (2006a): “Estimation and model selection of semiparametric copula-based multivariate dynamic models under copula misspecification,” *Journal of Econometrics*, 135, 125–154.
- (2006b): “Estimation of copula-based semiparametric time series models,” *Journal of Econometrics*, 130, 307–335.
- CHEN, X., Y. FAN, AND V. TSYRENNIKOV (2006): “Efficient estimation of semiparametric multivariate copula models,” *Journal of the American Statistical Association*, 101, 1228–1240.
- CHEN, X., Z. HUANG, AND Y. YI (2021): “Efficient estimation of multivariate semi-nonparametric GARCH filtered copula models,” *Journal of Econometrics*, 222, 484–501.
- CHEN, X. AND D. POUZO (2009): “Efficient estimation of semiparametric conditional moment models with possibly nonsmooth residuals,” *Journal of Econometrics*, 152, 46–60.

- (2012): “Estimation of nonparametric conditional moment models with possibly nonsmooth generalized residuals,” *Econometrica*, 80, 277–321.
- (2015): “Sieve Wald and QLR inferences on semi/nonparametric conditional moment models,” *Econometrica*, 83, 1013–1079.
- CHEN, X., Z. XIAO, AND B. WANG (2022): “Copula-based time series with filtered non-stationarity,” *Journal of Econometrics*, 228, 127–155.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, S. HAN, AND K. WÜTHRICH (2026): “Supplemental Appendix to ‘Estimating Causal Effects of Discrete and Continuous Treatments with Binary Instruments’,” *Working paper*.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, AND S. LUO (2025a): “Distribution Regression with Sample Selection and UK Wage Decomposition,” *Journal of Political Economy*, 133, 3952–3992.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, AND S. LUO (2025b): “Supplementary Material to ‘Distribution Regression with Sample Selection and UK Wage Decomposition’,” *Journal of Political Economy*, 133.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, J. MEIER, A. VAN VUUREN, AND F. VELLA (2025c): “Conditional Rank-Rank Regression,” arXiv:2407.06387.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, AND B. MELLY (2013): “Inference on Counterfactual Distributions,” *Econometrica*, 81, 2205–2268.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, B. MELLY, AND K. WÜTHRICH (2020): “Generic Inference on Quantile and Quantile Effect Functions for Discrete Outcomes,” *Journal of the American Statistical Association*, 115, 123–137.
- CHERNOZHUKOV, V. AND C. HANSEN (2005): “An IV model of quantile treatment effects,” *Econometrica*, 73, 245–261.
- (2006): “Instrumental quantile regression inference for structural and treatment effect models,” *Journal of Econometrics*, 132, 491–525.
- (2013): “Quantile Models with Endogeneity,” *Annual Review of Economics*, 5, 57–81.
- CHERNOZHUKOV, V., C. HANSEN, AND K. WÜTHRICH (2017): “Instrumental Variable Quantile Regression,” in *Handbook of Quantile Regression*, ed. by V. Chernozhukov, X. He, R. Koenker, and L. Peng, CRC Chapman-Hall, 119–143.
- CHESHER, A. (2003): “Identification in nonseparable models,” *Econometrica*, 71, 1405–1441.
- CHESHER, A. AND A. M. ROSEN (2020): “Chapter 1 – Generalized instrumental variable models, methods, and applications,” in *Handbook of Econometrics*, ed. by S. N. Durlauf, L. P. Hansen, J. J. Heckman, and R. L. Matzkin, Elsevier, vol. 7, 1–110.

- CHIBURIS, R. C., J. DAS, AND M. LOKSHIN (2011): “A Practical Comparison of the Bivariate Probit and Linear IV Estimators,” *World Bank Policy Research Working Paper*.
- (2012): “A practical comparison of the bivariate probit and linear IV estimators,” *Economics Letters*, 117, 762–766.
- CUNHA, F., J. J. HECKMAN, AND S. NAVARRO (2007): “The identification and economic content of ordered choice models with stochastic thresholds,” *International Economic Review*, 48, 1273–1309.
- D’HAULTFOEUILLE, X. AND P. FÉVRIER (2015): “Identification of Nonseparable Triangular Models With Discrete Instruments,” *Econometrica*, 83, 1199–1210.
- DONG, Y. AND Y.-Y. LEE (2023): “Nonparametric Doubly Robust Identification of Causal Effects of a Continuous Treatment using Discrete Instruments,” arXiv:2310.18504.
- DOS ANJOS, U. U. AND N. KOLEV (2005): “Representation of bivariate copulas via local measure of dependence,” RT-MAE 2005-3.
- EISENHAUER, P., J. J. HECKMAN, AND E. VYTLACIL (2015): “The generalized Roy model and the cost-benefit analysis of social programs,” *Journal of Political Economy*, 123, 413–443.
- FAN, Y., F. HAN, AND H. PARK (2023): “Estimation and inference in a high-dimensional semiparametric Gaussian copula vector autoregressive model,” *Journal of Econometrics*, 237, 105513.
- GALE, D. AND H. NIKAIDO (1965): “The Jacobian matrix and global univalence of mappings,” *Mathematische Annalen*, 159, 81–93.
- GHANEM, D., D. KÉDAGNI, AND I. MOURIFIÉ (2023): “Evaluating the Impact of Regulatory Policies on Social Welfare in Difference-in-Difference Settings,” arXiv:2306.04494.
- HAN, S. AND S. LEE (2019): “Estimation in a generalization of bivariate probit models with dummy endogenous regressors,” *Journal of Applied Econometrics*, 34, 994–1015.
- HAN, S., S. LEE, AND S. YANG (2026): “Semiparametric Models for Dynamic Treatment Effects and Mediation Analyses with Observational Data,” Working Paper.
- HAN, S. AND E. J. VYTLACIL (2017): “Identification in a generalization of bivariate probit models with dummy endogenous regressors,” *Journal of Econometrics*, 199, 63–73.
- HAN, S. AND S. YANG (2024): “A computational approach to identification of treatment effects for policy evaluation,” *Journal of Econometrics*, 240, 105680.
- HECKMAN, J. J. (1979): “Sample Selection Bias as a Specification Error,” *Econometrica*, 47, 153–161.
- HECKMAN, J. J. AND B. E. HONORÉ (1990): “The Empirical Content of the Roy Model,” *Econometrica*, 58, 1121–1149.

- HECKMAN, J. J., J. L. TOBIAS, AND E. VYTLACIL (2003): “Simple Estimators for Treatment Parameters in a Latent-Variable Framework,” *The Review of Economics and Statistics*, 85, 748–755.
- HECKMAN, J. J. AND E. J. VYTLACIL (2000): “Local Instrumental Variables,” Working Paper t0252, National Bureau of Economic Research.
- (2005): “Structural equations, treatment effects, and econometric policy evaluation,” *Econometrica*, 73, 669–738.
- (2007): “Chapter 71 Econometric Evaluation of Social Programs, Part II: Using the Marginal Treatment Effect to Organize Alternative Econometric Estimators to Evaluate Social Programs, and to Forecast their Effects in New Environments,” in *Handbook of Econometrics*, ed. by J. J. Heckman and E. E. Leamer, Elsevier, vol. 6B, 4875–5143.
- HIRANO, K. AND G. W. IMBENS (2004): “The Propensity Score with Continuous Treatments,” in *Applied Bayesian Modelling and Causal Inference from Incomplete-Data Perspectives*, ed. by A. Gelman and X.-L. Meng, Wiley, 73–84.
- IMBENS, G. W. AND J. D. ANGRIST (1994): “Identification and Estimation of Local Average Treatment Effects,” *Econometrica*, 62, 467–475.
- IMBENS, G. W. AND W. K. NEWEY (2009): “Identification and estimation of triangular simultaneous equations models without additivity,” *Econometrica*, 77, 1481–1512.
- KOLEV, N., U. D. ANJOS, AND B. V. D. M. MENDES (2006): “Copulas: a review and recent developments,” *Stochastic models*, 22, 617–660.
- KOWALSKI, A. E. (2023): “Reconciling seemingly contradictory results from the Oregon health insurance experiment and the Massachusetts health reform,” *Review of Economics and Statistics*, 105, 646–664.
- LIU, H., J. LAFFERTY, AND L. WASSERMAN (2009): “The nonparanormal: Semiparametric estimation of high dimensional undirected graphs,” *Journal of Machine Learning Research*, 10, 2295–2328.
- MANSKI, C. F. (1990): “Nonparametric bounds on treatment effects,” *The American Economic Review*, 80, 319–323.
- MATZKIN, R. L. (2003): “Nonparametric estimation of nonadditive random functions,” *Econometrica*, 71, 1339–1375.
- MOGSTAD, M., A. SANTOS, AND A. TORGOVITSKY (2018): “Using instrumental variables for inference about policy relevant treatment parameters,” *Econometrica*, 86, 1589–1619.
- MOURIFIÉ, I. AND Y. WAN (2025): “Layered policy analysis in program evaluation using the marginal treatment effect,” *Journal of Econometrics*, 251, 106060.
- NELSEN, R. B. (2006): *An introduction to copulas*, Springer.

- NEWKEY, W. K. AND D. MCFADDEN (1994): “Chapter 36 Large sample estimation and hypothesis testing,” in *Handbook of Econometrics*, ed. by R. F. Engle and D. L. McFadden, Elsevier, vol. 4, 2111–2245.
- NEWKEY, W. K. AND J. L. POWELL (2003): “Instrumental variable estimation of nonparametric models,” *Econometrica*, 71, 1565–1578.
- NEWKEY, W. K., J. L. POWELL, AND F. VELLA (1999): “Nonparametric estimation of triangular simultaneous equations models,” *Econometrica*, 67, 565–603.
- NEWKEY, W. K. AND S. STOULI (2021): “Control variables, discrete instruments, and identification of structural functions,” *Journal of Econometrics*, 222, 73–88.
- R CORE TEAM (2026): *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, Austria.
- SARMANOV, O. V. (1966): “Generalized normal correlation and two-dimensional Fréchet classes,” in *Doklady Akademii Nauk*, Russian Academy of Sciences, vol. 168, 32–35.
- STATA CORP (2025): “Stata Statistical Software: Release 19,” College Station, TX; Stata-Corp LLC.
- TORGOVITSKY, A. (2010): “Identification and Estimation of Nonparametric Quantile Regressions with Endogeneity,” Job Market Paper.
- (2015): “Identification of Nonseparable Models Using Instruments With Small Support,” *Econometrica*, 83, 1185–1197.
- (2017): “Minimum distance from independence estimation of nonseparable instrumental variables models,” *Journal of Econometrics*, 199, 35–48.
- VUONG, Q. AND H. XU (2017): “Counterfactual mapping and individual treatment effects in nonseparable models with binary endogeneity,” *Quantitative Economics*, 8, 589–610.
- VYTLACIL, E. J. (2002): “Independence, monotonicity, and latent index models: An equivalence result,” *Econometrica*, 70, 331–341.
- (2006): “Ordered discrete-choice selection models and local average treatment effect assumptions: Equivalence, nonequivalence, and representation results,” *The Review of Economics and Statistics*, 88, 578–581.
- WÜTHRICH, K. (2020): “A Comparison of Two Quantile Models With Endogeneity,” *Journal of Business & Economic Statistics*, 38, 443–456.

Supplemental Appendix to “Estimating Causal Effects of Discrete and Continuous Treatments with Binary Instruments”

Victor Chernozhukov Iván Fernández-Val
Sukjin Han Kaspar Wüthrich

September 7, 2026

Abstract

In this supplemental appendix we provide omitted discussions, results, and proofs by section, in the same order they are referred to in the paper.

A Supplements to Section 2

A.1 Discussion on Stochastic Dominance in REL(iii)

It is interesting to see what type of compliance behavior with respect to D_0 and D_1 is ruled out by the stochastic dominance between $\pi_d(0) = F_{D|Z}(d | 0)$ and $\pi_d(1) = F_{D|Z}(d | 1)$ in REL(iii). To explore this, define the compliers and defiers of order $j \in \mathcal{D} \setminus \{0\}$ as $C_j := \bigcup_{d=0}^{K-j} \{D_0 = d, D_1 = d + j\}$ and $B_j := \bigcup_{d=0}^{K-j} \{D_1 = d, D_0 = d + j\}$.

Lemma A.1 (Compliance Shares). *Suppose V_0 and V_1 are exchangeable, i.e., $F_{V_0, V_1}(v_0, v_1) = F_{V_0, V_1}(v_1, v_0)$ for all $v_0, v_1 \in (0, 1)$. Then, $\pi_d(0) > \pi_d(1)$ (resp. $<$) for all $d \in \mathcal{D} \setminus \{K\}$ implies that the share of all complier groups is larger (resp. smaller) than the share of all defier groups, that is, $\Pr[\bigcup_{j=1}^K C_j] \geq \Pr[\bigcup_{j=1}^K B_j]$ (resp. $\leq$). Moreover, the weak inequality becomes strict either when the joint distribution of (V_0, V_1) has full support on $(0, 1)^2$ or under SRI.*

The exchangeability, which trivially holds if $V_0 = V_1$ a.s., states that the distribution of (V_0, V_1) is symmetric; most known copulas are symmetric. The condition about the share of compliers and defiers is reminiscent of a similar assumption used in De Chaisemartin (2017) in the case of a binary treatment. Another simple interpretation of Lemma A.1 can be made

under the restriction $V_0 = V_1$ a.s.: there are no defier groups (i.e., $\Pr[D_1 < D_0] = 0$) if and only if $F_{D|Z}(d | 0) > F_{D|Z}(d | 1)$ for all $d \in \mathcal{D} \setminus \{K\}$. In general, when V_z is not restricted, the exchangeability of (V_0, V_1) alone does not eliminate compliers or defiers.

A.1.1 Proof of Lemma A.1

Before presenting a formal proof, it is helpful to consider an illustrative example with $K = 2$. In this case we have two complier groups and two defier groups:

$$\begin{aligned} C_1 &:= \{D_0 = 0, D_1 = 1\} \cup \{D_0 = 1, D_1 = 2\}, \\ C_2 &:= \{D_0 = 0, D_1 = 2\}, \\ B_1 &:= \{D_1 = 0, D_0 = 1\} \cup \{D_1 = 1, D_0 = 2\}, \\ B_2 &:= \{D_1 = 0, D_0 = 2\}. \end{aligned}$$

Note that the union of C_1 and C_2 is identical to the multiple-counting union of

$$\begin{aligned} C_1 &:= \{D_0 = 0, D_1 = 1\} \cup \{D_0 = 1, D_1 = 2\}, \\ C_2 &:= \{D_0 = 0, D_1 = 2\}, \\ C_2 &:= \{D_0 = 0, D_1 = 2\}. \end{aligned}$$

Suppose $\pi_d(0) > \pi_d(1)$ for $d = 0, 1$. An individual belongs to a complier group if and only if her treatment crosses at least one of the two treatment thresholds from below. Thus,

$$\begin{aligned} \Pr[C_1 \cup C_2] &= \Pr[\{0 < V_0 \leq \pi_0(0), \pi_0(1) < V_1 \leq 1\} \cup \{0 < V_0 \leq \pi_1(0), \pi_1(1) < V_1 \leq 1\}] \\ &= \Pr[\{0 < V_1 \leq \pi_0(0), \pi_0(1) < V_0 \leq 1\} \cup \{0 < V_1 \leq \pi_1(0), \pi_1(1) < V_0 \leq 1\}] \\ &\geq \Pr[\{0 < V_1 \leq \pi_0(1), \pi_0(0) < V_0 \leq 1\} \cup \{0 < V_1 \leq \pi_1(1), \pi_1(0) < V_0 \leq 1\}] \\ &= \Pr[B_1 \cup B_2], \end{aligned}$$

where the second equality follows from exchangeability and the weak inequality is by $\pi_d(0) > \pi_d(1)$ for all $d = 0, 1$.

The following is the formal proof of the lemma. With $\pi_{-1}(z) = 0$ and $\pi_K(z) = 1$ for all z ,

$$\begin{aligned}
\Pr \left[\bigcup_{j=1}^K C_j \right] &= \Pr \left[\bigcup_{j=1}^K \bigcup_{s=0}^{K-j} \{ \pi_{s-1}(0) < V_0 \leq \pi_s(0), \pi_{s+j-1}(1) < V_1 \leq \pi_{s+j}(1) \} \right] \\
&= \Pr \left[\bigcup_{j=0}^{K-1} \{ 0 < V_0 \leq \pi_j(0), \pi_j(1) < V_1 \leq 1 \} \right] = \Pr \left[\bigcup_{j=0}^{K-1} \{ 0 < V_1 \leq \pi_j(0), \pi_j(1) < V_0 \leq 1 \} \right] \\
&\geq \Pr \left[\bigcup_{j=0}^{K-1} \{ 0 < V_1 \leq \pi_j(1), \pi_j(0) < V_0 \leq 1 \} \right] = \Pr \left[\bigcup_{j=1}^K B_j \right].
\end{aligned}$$

The second equality follows from the same derivation as in the case of $K = 2$, and the third equality is by exchangeability. The weak inequality follows from $\pi_d(0) > \pi_d(1)$ for all $d \in \{0, \dots, K-1\}$. Finally, the weak inequality becomes strict either when the joint distribution of (V_0, V_1) has full support or under SRI (which implies $\Pr[B_j] = 0 \ \forall j$). The proof of the opposite direction of inequality is symmetric.

A.2 CLD and LATE Monotonicity

The LATE framework of [Imbens and Angrist \(1994\)](#) provides conditions for identifying causal effects for the subpopulation of compliers. The compliers are the individuals who react to the instrument so that $D_1 > D_0$. Compared to the assumptions in Section 3.1, the LATE framework restricts the selection model by setting $V_1 = V_0 = V$ a.s. ([Vytlacil, 2002](#)), but does not impose any constant local dependence.

Note that the specification of D_z in (2.1) allows for rich compliance patterns due to the inclusion of two unobservables: V_1 and V_0 . For example with binary D , (3.1) is weaker than LATE monotonicity, and Remark A.1 shows that CLD does not imply $V_0 = V_1$ a.s. Consequently, we can generate any compliance patterns from the joint distribution of (V_1, V_0) :

$$\begin{aligned}
\Pr[D_1 = 1, D_0 = 1] &= \Pr[V_1 > \pi_0(1), V_0 > \pi_0(0)] \\
\Pr[D_1 = 0, D_0 = 1] &= \Pr[V_1 \leq \pi_0(1), V_0 > \pi_0(0)] \\
\Pr[D_1 = 1, D_0 = 0] &= \Pr[V_1 > \pi_0(1), V_0 \leq \pi_0(0)] \\
\Pr[D_1 = 0, D_0 = 0] &= \Pr[V_1 \leq \pi_0(1), V_0 \leq \pi_0(0)]
\end{aligned}$$

In addition to allowing for richer selection patterns, CLD also allows us to identify treatment effects for the entire population rather than the subpopulation of compliers. See Section 4.5 for a discussion of how CLD extrapolates from the compliers to the overall population.

Remark A.1 (CLD and SRI). *One might wonder if CLD implies SRI, $V_0 = V_1$, a.s. The following example shows that this is not the case. Let $\tilde{V}_0 := \Phi^{-1}(V_0)$ and $\tilde{V}_1 := \Phi^{-1}(V_1)$ be the normal scores corresponding to V_0 and V_1 ,¹ and*

$$\begin{pmatrix} Y_d \\ \tilde{V}_0 \\ \tilde{V}_1 \end{pmatrix} \sim \mathcal{N}_3 \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho_{Y_d, V_0} & \rho_{Y_d, V_1} \\ \rho_{Y_d, V_0} & 1 & \rho_{V_0, V_1} \\ \rho_{Y_d, V_1} & \rho_{V_0, V_1} & 1 \end{bmatrix} \right).$$

Under CLD, $\rho_{Y_d, V_0} = \rho_{Y_d, V_1}$, so that $(Y_d, \tilde{V}_0)$ and $(Y_d, \tilde{V}_1)$ have the same distribution. Moreover, the covariance matrix is positive definite if $\rho_{V_0, V_1} > 2\rho_{Y_d, V_0}^2 - 1$. In other words, $\rho_{Y_d, V_0} = \rho_{Y_d, V_1}$ does not imply that $\rho_{V_0, V_1} = 1$ and therefore $V_1 = V_0$, if $\rho_{Y_d, V_0}^2 < 1$. $\square$

A.3 CLD and (In)dependence

Consider applying CLD to the LGR. CLD allows for dependence between Y_d and V_z (i.e., endogeneity) when the DGP is a Gaussian copula and $\rho_{Y_d, V_z}(y, v) = \rho_{Y_d}$. Here, we formally show that CLD is compatible with dependence for many different non-Gaussian DGPs when the outcome is binary. We also provide an example of a restricted class of DGPs under which CLD implies independence.

We consider the case in which D and Y are binary and focus on the dependence between Y_0 and V_z ; the argument for the dependence between Y_1 and V_z is analogous. In this case, the object of interest is $F_{Y_0}(0) \in (0, 1)$, and the relevant restriction in CLD is

$$\rho_{Y_0, V_0}(0, \pi_0(0)) = \rho_{Y_0, V_1}(0, \pi_0(1)). \quad (\text{A.1})$$

If Y_0 and V_z are independent, then $F_{Y_0, V_z}(0, v) = F_{Y_0}(0)v$. Suppose that the dependence between Y_0 and V_z can be modeled using a parametric copula, $\tilde{C}_z(\cdot, \cdot; \theta_z)$, $F_{Y_0, V_z}(0, v; \theta_z) = \tilde{C}_z(F_{Y_0}(0), v; \theta_z)$, and define θ_z^0 to be the parameter at which independence is achieved, i.e., $\tilde{C}_z(u, v; \theta_z^0) = uv$. Define the following map

$$m_z(\theta_z) := \text{unique } \rho \text{ s.t. } \tilde{C}_z(F_{Y_0}(0), \pi_0(z); \theta_z) = C(F_{Y_0}(0), \pi_0(z); \rho).$$

Using this definition, (A.1) requires that $m_0(\theta_0) = m_1(\theta_1)$. A sufficient condition for (A.1) to be compatible with dependence is that there exists an $m^* \neq 0$ in $\text{Range}(m_0(\cdot)) \cap \text{Range}(m_1(\cdot))$. We refer to this condition as the *range-overlap condition*. If such m^* exists, then one can choose θ_1^* and θ_0^* such that $m_0(\theta_0^*) = m_1(\theta_1^*) = m^* \neq 0$ and (A.1) holds. But

¹Note that it is equivalent to work with the normal scores because they are a one-to-one transformation of V_0 and V_1 .

since $m^* \neq 0$ and $C(F_{Y_0}(0), \pi_0(z); m^*) \neq F_{Y_0}(0)\pi_0(z)$, we have that $\tilde{C}_z(F_{Y_0}(0), \pi_0(z); \theta_z^*) = F_{Y_0, V_z}(0, \pi_0(z); \theta_z^*) \neq F_{Y_0}(0)\pi_0(z)$, i.e., Y_0 and V_z are not independent.

The range-overlap condition holds for many parametric copula families. A simple sufficient condition is the existence of an admissible path from independence to positive (or negative) dependence, referred to as *positive (or negative) dependence path*. Here we focus on positive dependence paths; an analogous argument applies to negative dependence paths. Suppose that, for $z \in \{0, 1\}$, there exists a path $\theta_z(t)$, $t \in [0, \epsilon]$, starting at θ_z^0 , such that $\tilde{C}_z(F_{Y_0}(0), \pi_0(z); \theta_z(t)) > F_{Y_0}(0)\pi_0(z)$ for $t > 0$. Because $\rho \mapsto C(F_{Y_0}(0), \pi_0(z); \rho)$ is strictly increasing by the properties of the Gaussian copula, $m_z(\theta_z(t)) > 0$ for $t > 0$. If $t \mapsto m_z(\theta_z(t))$ is continuous, the interval $[0, m_z(\theta_z(\bar{t}_z))]$ is contained in $\text{Range}(m_z(\cdot))$ for some $\bar{t}_z > 0$.² Consequently, the ranges of m_1 and m_0 overlap away from zero, and the range-overlap condition is satisfied.

Positive (and negative) dependence paths exist for many copula families, including FGM and Fréchet-Mardia copulas, as we show in the following examples.

Example 1 (FGM copula: $\tilde{C}_z(u, v; \theta_z) = uv + \theta_z uv(1 - u)(1 - v)$). Take $\theta_z(t) = t$ with $t \in [0, \epsilon]$ and $\epsilon > 0$ sufficiently small. For $t > 0$, we have that

$$\tilde{C}_z(F_{Y_0}(0), \pi_0(z); t) - F_{Y_0}(0)\pi_0(z) = tF_{Y_0}(0)(1 - F_{Y_0}(0))\pi_0(z)(1 - \pi_0(z)) > 0.$$

Moreover, the map $t \mapsto \tilde{C}_z(F_{Y_0}(0), \pi_0(z); t)$ is continuous. Thus, a positive dependence path exists and the range-overlap condition holds.

Example 2 (Fréchet-Mardia copula: $\tilde{C}_z(u, v; a_z, b_z) = a_z M(u, v) + (1 - a_z - b_z)uv + b_z W(u, v)$, where $M(u, v) = \min\{u, v\}$ and $W(u, v) = \max\{u + v - 1, 0\}$). Take $(a_z(t), b_z(t)) = (t, 0)$ with $t \in [0, \epsilon]$ and $\epsilon > 0$ sufficiently small. For $t > 0$, we have that

$$\tilde{C}_z(F_{Y_0}(0), \pi_0(z); t, 0) - F_{Y_0}(0)\pi_0(z) = t(M(F_{Y_0}(0), \pi_0(z)) - F_{Y_0}(0)\pi_0(z)) > 0.$$

Moreover, the map $t \mapsto \tilde{C}_z(F_{Y_0}(0), \pi_0(z); t, 0)$ is continuous. Thus, a positive dependence path exists and the range-overlap condition holds.

The Clayton and Frank copula families also satisfy the range-overlap condition, once they are defined via their continuous extensions at independence.³

²This is implied by continuity of $t \mapsto \tilde{C}_z(F_{Y_0}(0), \pi_0(z); \theta_z(t))$, which holds for many parametric copulas, since $\rho \mapsto C(F_{Y_0}(0), \pi_0(z); \rho)$ is continuous and strictly increasing.

³For a single-parameter copula $\tilde{C}$, the independence copula is the unique continuous extension at $\theta = 0$ if and only if $\lim_{\theta \rightarrow 0} \tilde{C}(u_1, u_2; \theta) = u_1 u_2$ for every $(u_1, u_2) \in [0, 1]^2$. The latter holds for both Clayton and Frank copulas, whose expressions are shown in SA B.4, although they are mentioned in that section as examples for local representation rather than DGPs.

This discussion demonstrates that **CLD** is compatible with dependence and endogeneity for many parametric copula families. At the same time, it is possible to construct examples of classes of DGPs for which **CLD** implies independence. Suppose that $\tilde{C}_0$ is an FGM family that only allows for positive dependence, $\tilde{C}_0(u, v; \theta_0) = uv + \theta_0 uv(1 - u)(1 - v)$, $\theta_0 \in [0, 1]$, and $\tilde{C}_1$ is an FGM family that only allows for negative dependence, $\tilde{C}_1(u, v; \theta_1) = uv + \theta_1 uv(1 - u)(1 - v)$, $\theta_1 \in [-1, 0]$. Note that since $\tilde{C}_0(F_{Y_0}(0), \pi_0(0); \theta_0) - F_{Y_0}(0)\pi_0(0) \geq 0$ and $\tilde{C}_0(F_{Y_0}(0), \pi_0(0); \theta_0) - F_{Y_0}(0)\pi_0(0) = 0$ if and only if $\theta_0 = 0$, it follows that $m_0(\theta_0) \geq 0$ with $m_0(\theta_0) = 0$ if and only if $\theta_0 = 0$ because the Gaussian copula is strictly increasing in ρ . Similarly, we have that $m_1(\theta_1) \leq 0$ with $m_1(\theta_1) = 0$ if and only if $\theta_1 = 0$. Therefore, $m_0(\theta_0) = m_1(\theta_1)$ can only occur when $\theta_1 = \theta_0 = 0$ in which case $Y_0 \perp V_0$ and $Y_0 \perp V_1$. Thus, **CLD** implies independence in this sign-separated FGM class.

The formal argument above is specific to binary outcomes. With multi-valued discrete or continuous outcomes, **CLD** needs to hold at more values of y and the same pair (θ_0, θ_1) must satisfy all of these restrictions, which can be more restrictive. For example, one can show that **CLD** requires independence with FGM copulas as DGPs when the outcome is continuous. Non-Gaussian DGPs satisfying **CLD** with multi-valued discrete or continuous outcomes, such as the latent-traits model of Section 4.3, therefore generally rely on richer dependence structures than those allowed by one-parameter copula families.

B Supplements to Section 3

B.1 Proof of Theorem 3.1

Note that $\pi_0(z)$ is identified as a reduced-form parameter. Let $F_{d,y} := F_{Y_d}(y)$ and $\rho_{d,y} := \rho_{Y_d}(y)$ be the structural parameters of interest. Consider the following mapping between the structural and reduced-form parameters:

$$F_{Y|D,Z}(y | 0, 0)\pi_0(0) = C(F_{0,y}, \pi_0(0); \rho_{0,y}), \quad (\text{B.1})$$

$$F_{Y|D,Z}(y | 0, 1)\pi_0(1) = C(F_{0,y}, \pi_0(1); \rho_{0,y}), \quad (\text{B.2})$$

which we can express as $\pi_y = G(\theta_y)$, where $\theta_y := (F_{0,y}, \rho_{0,y})'$, $\pi_y := (F_{Y|D,Z}(y|0, 0)\pi_0(0), F_{Y|D,Z}(y|0, 1)\pi_0(1))'$, and $G : (0, 1) \times (-1, 1) \rightarrow (0, 1)^2$. Let C_1 , C_2 and C_ρ denote the derivatives of copula $C(u_1, u_2; \rho)$ with respect to u_1 , u_2 and ρ , respectively. Consider the

Jacobian of the system of nonlinear equations (B.1)–(B.2):

$$J = \frac{\partial G}{\partial \theta_y} = \begin{bmatrix} C_1(F_{0,y}, \pi_0(0); \rho_{0,y}) & C_\rho(F_{0,y}, \pi_0(0); \rho_{0,y}) \\ C_1(F_{0,y}, \pi_0(1); \rho_{0,y}) & C_\rho(F_{0,y}, \pi_0(1); \rho_{0,y}) \end{bmatrix}.$$

The matrix has full rank if and only if

$$\frac{C_\rho(F_{0,y}, \pi_0(1); \rho_{0,y})}{C_1(F_{0,y}, \pi_0(1); \rho_{0,y})} \neq \frac{C_\rho(F_{0,y}, \pi_0(0); \rho_{0,y})}{C_1(F_{0,y}, \pi_0(0); \rho_{0,y})}, \quad (\text{B.3})$$

which is true by Assumption REL and Lemma 4.1 in Han and Vytlačil (2017) as the Gaussian copula satisfies the stochastically increasing ordering condition (Assumption 5 in Han and Vytlačil (2017)). Moreover, the Jacobian is a P-matrix for all $(F_{0,y}, \rho_{0,y}) \in (0, 1) \times (-1, 1)$ by the proof of Theorem 2.1 in Chernozhukov et al. (2025). Moreover, the domain of the mapping G (i.e., $(0, 1) \times (-1, 1)$) is open and rectangular and G is differentiable as $C(u_1, u_2; \rho)$ is differentiable with respect to (u_1, ρ) . Therefore, one can apply (Gale and Nikaido, 1965, Theorem 4w)’s global univalence theorem, which identifies θ_y .⁴ Analogously, we have

$$\begin{aligned} F_{Y|D,Z}(y | 1, z)(1 - \pi_0(z)) &= \Pr[Y_1 \leq y | Z = z] - \Pr[Y_1 \leq y, V_z \leq \pi_0(z) | Z = z] \\ &= \Pr[Y_1 \leq y | Z = z] - C(F_{Y_1|Z}(y|z), \pi_0(z); \rho_{Y_1, V_z}(y, \pi_0(z))) \\ &= \Pr[Y_1 \leq y] - C(F_{Y_1}(y), \pi_0(z); \rho_{Y_1}(y)) \end{aligned}$$

and

$$F_{Y|D,Z}(y | 1, 0)(1 - \pi_0(0)) = F_{Y_1}(y) - C(F_{Y_1}(y), \pi_0(0); \rho_{1,y}), \quad (\text{B.4})$$

$$F_{Y|D,Z}(y | 1, 1)(1 - \pi_0(1)) = F_{Y_1}(y) - C(F_{Y_1}(y), \pi_0(1); \rho_{1,y}), \quad (\text{B.5})$$

and the mapping has a unique solution for $\tilde{\theta}_y := (F_{Y_1}(y), \rho_{1,y})'$ by a similar argument as above.

B.2 Proof of Theorem 3.2

Consider $d \in \{1, \dots, K-1\}$. Write $F_{d,y} := F_{Y_d}(y)$. Recall from the text that we additionally impose constant local dependence between a pair of levels:

$$\rho_{Y_d, V_z}(y, \pi_d(z)) = \rho_{Y_d, V_z}(y, \pi_{d-1}(z)) =: \rho_{Y_d}(y) =: \rho_{d,y}.$$

⁴An alternative identification strategy would be to apply a similar argument as in the proof for the case of ordered D (SA B.2). In particular, if the Jacobian has full rank everywhere in the parameter space that is open, Steps 1–3 of that proof can be applied to the system of equations (B.1)–(B.2) here.

For fixed (d, y) and for $z \in \{0, 1\}$, define $G_z(F, \rho) := C(F, \pi_d(z); \rho) - C(F, \pi_{d-1}(z); \rho)$ and $q_z := \Pr[Y \leq y, D = d \mid Z = z]$. Then (3.6) can be written as a system of two equations in $(F, \rho) = (F_{d,y}, \rho_{d,y})$:

$$q_z = G_z(F, \rho), \quad z \in \{0, 1\}. \quad (\text{B.6})$$

Step 1. For every $\rho \in (-1, 1)$, $F \mapsto G_0(F, \rho)$ is strictly increasing because

$$\frac{\partial G_0(F, \rho)}{\partial F} = C_1(F, \pi_d(0); \rho) - C_1(F, \pi_{d-1}(0); \rho) > 0.$$

Moreover, $\lim_{F \downarrow 0} G_0(F, \rho) = 0$ and $\lim_{F \uparrow 1} G_0(F, \rho) = \pi_d(0) - \pi_{d-1}(0)$. Hence, for every $q_0 \in (0, \pi_d(0) - \pi_{d-1}(0))$ and every $\rho \in (-1, 1)$, there exists *globally* a unique $F_{q_0}(\rho) \in (0, 1)$ such that $G_0(F_{q_0}(\rho), \rho) = q_0$, and the global map $\rho \mapsto F_{q_0}(\rho)$ is continuously differentiable on $(-1, 1)$ by the implicit function theorem, with $F'_{q_0}(\rho) = -G_{0,\rho}/G_{0,F}$.

Step 2. It follows that

$$\frac{d}{d\rho} G_1(F_{q_0}(\rho), \rho) = G_{1,\rho} - G_{1,F} \frac{G_{0,\rho}}{G_{0,F}} = \frac{G_{0,F} G_{1,\rho} - G_{0,\rho} G_{1,F}}{G_{0,F}} = \frac{\det J_d}{G_{0,F}}$$

with all derivatives and J_d in the display being evaluated at $(F_{q_0}(\rho), \rho)$ and J_d being the Jacobian for (B.6) displayed below, where we show REL(iii) implies that $\det J_d \neq 0$ throughout $(F, \rho) \in (0, 1) \times (-1, 1)$. Since $\det J_d$ is continuous and its domain is connected, it has the same sign everywhere. Because $G_{0,F} > 0$, the function $\rho \mapsto G_1(F_{q_0}(\rho), \rho)$ is therefore strictly monotone.

Step 3. Suppose (F, ρ) and $(\tilde{F}, \tilde{\rho})$ solve (B.6). The first equation with $z = 0$ implies that $F = F_{q_0}(\rho)$ and $\tilde{F} = F_{q_0}(\tilde{\rho})$ for their common value q_0 . The second equation with $z = 1$ and strict monotonicity of $\rho \mapsto G_1(F_{q_0}(\rho), \rho)$ implies $\rho = \tilde{\rho}$, and therefore $F = \tilde{F}$. A solution exists because, under EX, REL, and CLD(ii), the data are generated by the model at $(F_{Y_d}(y), \rho_{Y_d}(y))$, so that $q_0 \in (0, \pi_d(0) - \pi_{d-1}(0))$ whenever $F_{Y_d}(y) \in (0, 1)$. Therefore $F_{Y_d}(y)$ and $\rho_{Y_d}(y)$ are identified for every y with $F_{Y_d}(y) \in (0, 1)$.

The cases $d = 0$ and $d = K$ reduce to the binary-treatment system and follow from the argument in SA B.1. This completes the proof.

Now, it remains to prove the full rank of the Jacobian of (B.6) over $(F, \rho) \in (0, 1) \times (-1, 1)$:

$$J_d = \begin{bmatrix} C_1(F, \pi_d(0); \rho) - C_1(F, \pi_{d-1}(0); \rho) & C_\rho(F, \pi_d(0); \rho) - C_\rho(F, \pi_{d-1}(0); \rho) \\ C_1(F, \pi_d(1); \rho) - C_1(F, \pi_{d-1}(1); \rho) & C_\rho(F, \pi_d(1); \rho) - C_\rho(F, \pi_{d-1}(1); \rho) \end{bmatrix}.$$

For any $(F, \rho) \in (0, 1) \times (-1, 1)$, this Jacobian has full rank if and only if

$$\frac{C_\rho(F, \pi_d(1); \rho) - C_\rho(F, \pi_{d-1}(1); \rho)}{C_1(F, \pi_d(1); \rho) - C_1(F, \pi_{d-1}(1); \rho)} \neq \frac{C_\rho(F, \pi_d(0); \rho) - C_\rho(F, \pi_{d-1}(0); \rho)}{C_1(F, \pi_d(0); \rho) - C_1(F, \pi_{d-1}(0); \rho)}. \quad (\text{B.7})$$

Showing this is more involved than showing (B.3) with the binary treatment, because the equality can arise due to two points on the indifference curve. Nonetheless, the full-rank condition (B.7) can be expressed as $\lambda(0) \neq \lambda(1)$, where

$$\lambda(z) := \frac{\phi(r_d(z)) - \phi(r_{d-1}(z))}{\Phi(r_d(z)) - \Phi(r_{d-1}(z))}, \quad r_\ell(z) := \frac{\Phi^{-1}(\pi_\ell(z)) - \rho\Phi^{-1}(F)}{\sqrt{1 - \rho^2}}.$$

To interpret this condition, we note that it can be related to the mean of a truncated Gaussian random variable: for $A \sim N(\mu, \sigma^2)$,

$$\mathbb{E}[A \mid l < A < u] = \mu - \left(\frac{\phi\left(\frac{u-\mu}{\sigma}\right) - \phi\left(\frac{l-\mu}{\sigma}\right)}{\Phi\left(\frac{u-\mu}{\sigma}\right) - \Phi\left(\frac{l-\mu}{\sigma}\right)} \right) \sigma.$$

Therefore, the full-rank condition $\lambda(0) \neq \lambda(1)$ can be equivalently expressed as

$$\mathbb{E}[A \mid \pi_{d-1}(0) < \Phi(A) < \pi_d(0)] \neq \mathbb{E}[A \mid \pi_{d-1}(1) < \Phi(A) < \pi_d(1)]$$

with $\mu = \rho\Phi^{-1}(F)$ and $\sigma^2 = 1 - \rho^2$. For example, this holds when threshold functions are such that $\pi_{d-1}(0) > \pi_{d-1}(1)$ and $\pi_d(0) > \pi_d(1)$, which is guaranteed by REL(iii).

B.3 Further Details on Ordered Choice Models

It is interesting to discuss Heckman and Vytlacil (2007)'s model (i.e., $D_z = k$ if $\pi_{k-1} < \mu(z) + V \leq \pi_k$ for $k \in \{0, \dots, K\}$ with $\pi_{-1} = -\infty$, $\pi_K = \infty$ and $V \mid Z \sim N(0, 1)$) in comparison to our ordered choice models (3.5) in Section 3.2 *with the normalization* $V_z \mid Z \sim N(0, 1)$. The cutoffs of the two models are related as $\pi_\ell(z) = \pi_\ell - \mu(z)$. Under this model, we have $r_\ell(z) = \frac{\pi_\ell - \mu(z) - \rho_{d,y}\Phi^{-1}(F_{d,y})}{\sqrt{1 - \rho_{d,y}^2}}$, which is particularly easy to interpret because $r_d(z) - r_{d-1}(z) = \frac{\pi_d - \pi_{d-1}}{\sqrt{1 - \rho_{d,y}^2}}$. On the other hand, in the general model, we have $r_d(z) - r_{d-1}(z) = \frac{\pi_d(z) - \pi_{d-1}(z)}{\sqrt{1 - \rho_{d,y}^2}}$. Note that in the simplified model, the full-rank condition holds by construction. Specifically, since $r_d(z) - r_{d-1}(z)$ does not depend on z , we can write $r_d(z) = r_{d-1}(z) + c$ for $c > 0$. Therefore, we can rewrite $\lambda(z)$ as $\frac{\phi(r_{d-1}(z)+c) - \phi(r_{d-1}(z))}{\Phi(r_{d-1}(z)+c) - \Phi(r_{d-1}(z))}$. This function is monotonically decreasing in $r_{d-1}(z)$. Thus, as long as $\mu(1) \neq \mu(0)$ so that $r_{d-1}(1) \neq r_{d-1}(0)$, the full rank condition holds.

B.4 Identification under Other Local Representations

In the main text, we work with a local Gaussian representation. Here, we ask what other single-parameter copulas can be used for local representation and whether identification can be similarly achieved.

First, consider the case of binary D . Let $C(u_1, u_2; \rho)$ denote the copula which the local representation analogous to Lemma 2.1 is based on. For such a representation to work, the copula needs to be a *comprehensive copula*; see below for examples. We want to show that, for a given copula, the Jacobian of the system (3.2) has full rank. The Jacobian having full rank is a necessary condition for it to be a P-matrix in applying Gale and Nikaido (1965). Moreover, one can show that, if the Jacobian has full rank everywhere in the parameter space that is open, an identification strategy analogous to that in SA B.2 (i.e., Steps 1–3) can be alternatively applied. In showing full rank, the following quantity typically arises once the Jacobian is transformed using elementary operations: fixing $y \in \mathcal{Y}$,

$$\frac{C_\rho(F_{Y_0}(y), \pi_0(0); \rho_{Y_0}(y))}{C_1(F_{Y_0}(y), \pi_0(0); \rho_{Y_0}(y))} - \frac{C_\rho(F_{Y_0}(y), \pi_0(1); \rho_{Y_0}(y))}{C_1(F_{Y_0}(y), \pi_0(1); \rho_{Y_0}(y))},$$

where C_ρ and C_1 are the derivatives with respect to ρ and the first argument, respectively. Therefore, any copula that satisfies

$$\frac{C_\rho(F_{Y_0}(y), \pi_0(0); \rho_{Y_0}(y))}{C_1(F_{Y_0}(y), \pi_0(0); \rho_{Y_0}(y))} \neq \frac{C_\rho(F_{Y_0}(y), \pi_0(1); \rho_{Y_0}(y))}{C_1(F_{Y_0}(y), \pi_0(1); \rho_{Y_0}(y))} \quad (\text{B.8})$$

for $\pi_0(0) \neq \pi_0(1)$ (REL(iii)) will yield the full rank Jacobian. Han and Vytlačil (2017) show that any single-parameter copula that follows the ordering of stochastic increasingness with respect to ρ satisfies (B.8). Let C_L , C_U and C_I denote the lower and upper Fréchet-Hoeffding copula bounds and independence copula. The following Archimedean copulas are comprehensive copulas:

Example 3 (Clayton copula).

$$C(u_1, u_2; \rho) = \max\{u_1^{-\rho} + u_2^{-\rho} - 1, 0\}^{-1/\rho}, \quad \rho \in [-1, \infty) \setminus \{0\}$$

and $C \rightarrow C_L$ when $\rho \rightarrow -1$, $C \rightarrow C_U$ when $\rho \rightarrow \infty$, and $C \rightarrow C_I$ when $\rho \rightarrow 0$.

Example 4 (Frank copula).

$$C(u_1, u_2; \rho) = -\frac{1}{\rho} \ln \left(1 + \frac{(e^{-\rho u_1} - 1)(e^{-\rho u_2} - 1)}{e^{-\rho} - 1} \right), \quad \rho \in (-\infty, \infty) \setminus \{0\}$$

and $C \rightarrow C_L$ when $\rho \rightarrow -\infty$, $C \rightarrow C_U$ when $\rho \rightarrow \infty$, and $C \rightarrow C_I$ when $\rho \rightarrow 0$.

Next, consider the case of continuous D . Following Proposition 1 of [dos Anjos and Kolev \(2005\)](#), consider the following local representation for an arbitrary copula $\tilde{C}(u_1, u_2)$

$$\tilde{C}(u_1, u_2) = u_1 u_2 + \rho(u_1, u_2) \sqrt{u_1 u_2 (1 - u_1)(1 - u_2)}. \quad (\text{B.9})$$

Then, the local dependence function satisfies

$$\rho(u_1, u_2) = \frac{\tilde{C}(u_1, u_2) - u_1 u_2}{\sqrt{u_1 u_2 (1 - u_1)(1 - u_2)}},$$

which captures the normalized deviation from the independent copula. In fact, $\rho(u_1, u_2)$ can be interpreted as a local Spearman correlation coefficient that satisfies many interesting properties ([dos Anjos and Kolev, 2005](#), Sections 3.1 and 3.2). For the identification analysis, by the properties of the conditional distribution, (B.9), EX, and CLD applied to the representation in (B.9), that is $\rho_{Y_d, V_0}(y, \pi_d(0)) = \rho_{Y_d, V_1}(y, \pi_d(1)) = \rho_{Y_d}(y)$ and $\partial \rho_{Y_d, V_z}(y, v) / \partial v|_{v=\pi_d(z)} = 0$, we have for $z \in \{0, 1\}$

$$F_{Y|D,Z}(y | d, z) = F_{Y_d}(y) + \frac{\rho_{Y_d}(y)}{2} w(d, z) \sqrt{F_{Y_d}(y)(1 - F_{Y_d}(y))}, \quad (\text{B.10})$$

where $w(d, z) := (1 - 2\pi_d(z)) / \sqrt{\pi_d(z)(1 - \pi_d(z))}$, which is assumed to be nonzero. Note that

$$\frac{F_{Y|D,Z}(y | d, 1) - F_{Y_d}(y)}{w(d, 1)} = \frac{F_{Y|D,Z}(y | d, 0) - F_{Y_d}(y)}{w(d, 0)},$$

which implies that

$$F_{Y_d}(y) = \frac{F_{Y|D,Z}(y | d, 1)w(d, 0) - F_{Y|D,Z}(y | d, 0)w(d, 1)}{w(d, 0) - w(d, 1)}.$$

Also, combining this result with (B.10) gives

$$\rho_{Y_d}(y) = 2 \frac{F_{Y|D,Z}(y | d, z) - F_{Y_d}(y)}{w(d, z) \sqrt{F_{Y_d}(y)(1 - F_{Y_d}(y))}}.$$

B.5 Alternative Identification Strategies

For the case of binary D , we show there can be other identification strategies using alternative versions of constant local dependence. The analysis can be extended to the ordered treatment case. Here we assume EX and REL and the treatment selection equation $D_z = 1\{V_z > \pi_0(z)\}$ where $V_z \sim U(0, 1)$, and focus on local identification. It might be possible to establish global

identification under additional assumptions using global univalence results.

B.5.1 Restrictions Within Treatment Levels

We focus here on the treatment level $d = 0$. A similar analysis follows for $d = 1$. For $y \in \mathcal{Y}$, consider the LGR of the observed probabilities, that is

$$\Pr[Y_0 \leq y, D = 0 \mid Z = z] = C(F_{Y_0}(y), \pi_0(z); \rho_0(y, \pi_0(z))), \quad z \in \{0, 1\},$$

where $\rho_0(y, \pi_0(z)) := \rho_{Y_0, V_z}(y, \pi_0(z))$. The identification problem is that we have two probabilities to identify three parameters: $F_{Y_0}(y)$, $\rho_0(y, \pi_0(0))$ and $\rho_0(y, \pi_0(1))$. So far, we have reduced the number of parameters by imposing the condition $\rho_0(y, \pi_0(0)) = \rho_0(y, \pi_0(1))$. This restriction was imposed separately for each value of y . However, it is also possible to impose restrictions across values of y . Assume that there exists $y' \in \mathcal{Y}$ such that $F_{Y_0}(y) \neq F_{Y_0}(y')$ and

$$\rho_0(y, \pi_0(z)) = \rho_0(y', \pi_0(z)), \quad z \in \{0, 1\}. \quad (\text{B.11})$$

This condition leads to the following system of four equations with four unknowns:

$$\begin{aligned} \Pr[Y_0 \leq y, D = 0 \mid Z = z] &= C(F_{Y_0}(y), \pi_0(z); \rho_0(y, \pi_0(z))), \quad z \in \{0, 1\}, \\ \Pr[Y_0 \leq y', D = 0 \mid Z = z] &= C(F_{Y_0}(y'), \pi_0(z); \rho_0(y, \pi_0(z))), \quad z \in \{0, 1\}. \end{aligned}$$

Then, it is possible to find conditions under which the solution to this system exists and is locally unique. (B.11) is appealing in that it does not impose restrictions across levels of z .

Let C_1 , C_2 , and C_ρ denote the partial derivatives of the Gaussian copula C with respect to the first and second arguments and ρ . The Jacobian of the system of equations is

$$J(y, y') = \begin{pmatrix} C_1(y, 1) & 0 & C_\rho(y, 1) & 0 \\ C_1(y, 0) & 0 & 0 & C_\rho(y, 0) \\ 0 & C_1(y', 1) & C_\rho(y', 1) & 0 \\ 0 & C_1(y', 0) & 0 & C_\rho(y', 0) \end{pmatrix},$$

where $C_j(k, z) := C_j(F_{Y_0}(k), \pi_0(z); \rho_0(k, \pi_0(z))) > 0$ for $j \in \{1, \rho\}$, $k \in \{y, y'\}$ and $z \in \{0, 1\}$. By the Laplace expansion, the Jacobian determinant is

$$\det(J(y, y')) = C_1(y, 0)C_1(y', 1)C_\rho(y, 1)C_\rho(y', 0) - C_1(y, 1)C_1(y', 0)C_\rho(y', 1)C_\rho(y, 0),$$

which does not vanish if

$$\frac{C_\rho(y, 1) C_\rho(y', 0)}{C_1(y, 1) C_1(y', 0)} \neq \frac{C_\rho(y, 0) C_\rho(y', 1)}{C_1(y, 0) C_1(y', 1)}.$$

Let

$$\lambda(k, z) := \frac{\phi(u(k, z))}{\Phi(u(k, z))}, \quad u(k, z) := \frac{\Phi^{-1}(\pi_0(z)) - \rho_0(k, \pi_0(z))\Phi^{-1}(F_{Y_0}(k))}{\sqrt{1 - \rho_0(k, \pi_0(z))^2}}.$$

Then, using that $C_\rho(k, z)\sqrt{1 - \rho_0(k, \pi_0(z))^2} = \lambda(k, z)\phi(\Phi^{-1}(F_{Y_0}(k)))C_1(k, z)$, the previous condition can be expressed as

$$\frac{\lambda(y, 1)}{\lambda(y, 0)} \neq \frac{\lambda(y', 1)}{\lambda(y', 0)} \quad \text{or} \quad \frac{\lambda(y, 1)}{\lambda(y', 1)} \neq \frac{\lambda(y, 0)}{\lambda(y', 0)}, \quad (\text{B.12})$$

that is, the change in the conditional inverse Mills ratio from $z = 0$ to $z = 1$ is different at y and y' , or the change in the conditional inverse Mills ratio from y to y' is different at $z = 0$ and $z = 1$. For example, if the identification condition holds locally for $y' = y + dy$, then the condition becomes $(\partial/\partial y) \log \lambda(y, 1) \neq (\partial/\partial y) \log \lambda(y, 0)$.

Theorem B.1. *Suppose Assumptions [EX](#) and [REL](#) hold, and $D_z = 1[V_z > \pi_0(z)]$ with $V_z \mid Z = z \sim U[0, 1]$. Given $y \in \mathcal{Y}$, suppose that there exists $y' \in \mathcal{Y}$ such that [\(B.11\)](#), [\(B.12\)](#) and the analogous conditions for $d = 1$ hold. Then, $F_{Y_d}(y)$ and $\rho_{Y_d, V_z}(y, \pi_0(z))$ are locally identified for $d \in \{0, 1\}$ and $z \in \{0, 1\}$.*

The proof of this theorem follows from the arguments appearing before the theorem. In general, let d_y be the number of values of Y that we use to construct the system of equations. Then, we have $2d_y$ equations and $3d_y$ unknowns. Therefore, we need to reduce the number of parameters by d_y using some combination of [\(B.11\)](#) and the alternative assumptions.

B.5.2 Restrictions Between Treatment Levels

As an alternative to the previous subsection, we can impose restrictions involving parameters for different treatment levels. This strategy is based on the system of equations

$$\begin{aligned} \Pr[Y \leq y, D = 0 \mid Z = z] &= C(F_{Y_0}(y), \pi_0(z); \rho_0(y, \pi_0(z))), \quad z \in \{0, 1\}, \\ \Pr[Y \leq y, D = 1 \mid Z = z] &= C(F_{Y_1}(y), 1 - \pi_0(z); -\rho_1(y, \pi_0(z))), \quad z \in \{0, 1\}, \end{aligned}$$

where $\rho_d(y, \pi_0(z)) := \rho_{Y_d, V_z}(y, \pi_0(z))$. Assume that the local dependence function is the same across treatment levels,

$$\rho_0(y, \pi_0(z)) = \rho_1(y, \pi_0(z)), \quad z \in \{0, 1\}. \quad (\text{B.13})$$

This condition is similar to the RS condition in (4.12), but does not require the potential outcomes to be continuous. It leads to the following system of four equations with four unknowns:

$$\begin{aligned}\Pr[Y \leq y, D = 0 \mid Z = z] &= C(F_{Y_0}(y), \pi_0(z); \rho_0(y, \pi_0(z))), \quad z \in \{0, 1\}, \\ \Pr[Y \leq y, D = 1 \mid Z = z] &= C(F_{Y_1}(y), 1 - \pi_0(z); -\rho_0(y, \pi_0(z))), \quad z \in \{0, 1\}.\end{aligned}$$

Then, it is possible to find conditions under which the solution to this system exists and is locally unique. Like (B.11), (B.13) is appealing in that it does not impose restrictions across levels of z . Let C_1 , C_2 , and C_ρ denote the partial derivatives of the Gaussian copula C with respect to the first and second arguments and ρ . Also let $C_j(d, z) := C_j(F_{Y_d}(y), \pi_0(z); \rho_d(y, \pi_0(z))) > 0$ and $\bar{C}_j(d, z) := C_j(F_{Y_d}(y), 1 - \pi_0(z); -\rho_d(y, \pi_0(z))) > 0$ for $j \in \{1, \rho\}$, $d \in \{0, 1\}$ and $z \in \{0, 1\}$. The Jacobian of the system of equations is

$$J = \begin{pmatrix} C_1(0, 1) & 0 & C_\rho(0, 1) & 0 \\ C_1(0, 0) & 0 & 0 & C_\rho(0, 0) \\ 0 & \bar{C}_1(1, 1) & -\bar{C}_\rho(1, 1) & 0 \\ 0 & \bar{C}_1(1, 0) & 0 & -\bar{C}_\rho(1, 0) \end{pmatrix}.$$

By the Laplace expansion, the Jacobian determinant is

$$\det(J) = C_1(0, 1)\bar{C}_1(1, 0)C_\rho(0, 0)\bar{C}_\rho(1, 1) - C_1(0, 0)\bar{C}_1(1, 1)C_\rho(0, 1)\bar{C}_\rho(1, 0),$$

which does not vanish if

$$\frac{C_\rho(0, 1)\bar{C}_\rho(1, 0)}{C_1(0, 1)\bar{C}_1(1, 0)} \neq \frac{C_\rho(0, 0)\bar{C}_\rho(1, 1)}{C_1(0, 0)\bar{C}_1(1, 1)}.$$

Let

$$\lambda(d, z) := \frac{\phi(u(d, z))}{\Phi(u(d, z))} \quad \text{and} \quad \bar{\lambda}(d, z) := \frac{\phi(u(d, z))}{\Phi(-u(d, z))},$$

with $u(d, z) := \{\Phi^{-1}(\pi_0(z)) - \rho_0(y, \pi_0(z))\Phi^{-1}(F_{Y_d}(y))\} / \sqrt{1 - \rho_0(y, \pi_0(z))^2}$. Then, using that $C_\rho(0, z)\sqrt{1 - \rho_0(y, \pi_0(z))^2} = \lambda(0, z)\phi(\Phi^{-1}(F_{Y_0}(y)))C_1(0, z)$ and $\bar{C}_\rho(1, z)\sqrt{1 - \rho_0(y, \pi_0(z))^2} = \bar{\lambda}(1, z)\phi(\Phi^{-1}(F_{Y_1}(y)))\bar{C}_1(1, z)$, the previous condition can be expressed as

$$\frac{\lambda(0, 1)}{\lambda(0, 0)} \neq \frac{\bar{\lambda}(1, 1)}{\bar{\lambda}(1, 0)} \quad \text{or} \quad \frac{\lambda(0, 1)}{\bar{\lambda}(1, 1)} \neq \frac{\lambda(0, 0)}{\bar{\lambda}(1, 0)}, \quad (\text{B.14})$$

that is, the change in the conditional inverse Mills ratio from $z = 0$ to $z = 1$ is different at $d = 1$ and $d = 0$, or the change in the conditional inverse Mills ratio from $d = 1$ to $d = 0$ is

different at $z = 0$ and $z = 1$.

Theorem B.2. *Suppose Assumptions [EX](#) and [REL](#) hold, and $D_z = 1[V_z > \pi_0(z)]$ with $V_z \mid Z = z \sim U[0, 1]$. Assume also that [\(B.13\)](#) and [\(B.14\)](#) hold for all $y \in \mathcal{Y}$. Then, $y \mapsto F_{Y_d}(y)$ and $y \mapsto \rho_{Y_d, V_z}(y, \pi_0(z))$ are locally identified on $\mathcal{Y}$, for $d \in \{0, 1\}$ and $z \in \{0, 1\}$.*

The proof of this theorem follows from the arguments appearing before the theorem.

C Supplements to Section 4

C.1 Additional Examples and Counterexamples of [CLD](#)

Assume first that SRI holds and focus on $d = 0$. Let $\tilde{V} := \Phi^{-1}(V)$, $v_d(z) := \Phi^{-1}(\pi_d(z))$ and $Y_d \mid \tilde{V} = v \sim N(\mu_d + \beta_d \operatorname{sgn}(|v| - c)v, \sigma_d^2 - \beta_d^2)$, where sgn is the sign function and $\beta_d \neq 0$, $\sigma_d > 0$ and $c \geq 0$ are constants such that $|\beta_d| < \sigma_d$. In this case $Y_d \sim N(\mu_d, \sigma_d^2)$ because $v \mapsto \operatorname{sgn}(|v| - c)v$ preserves normality and $F_{Y_d, \tilde{V}}(y, v) = \Phi_2((y - \mu_d)/\sigma_d, v; \beta_d/\sigma_d)$ for $|v| \geq c$, so that $\rho_{Y_d, \tilde{V}}(y, v) = \beta_d/\sigma_d$.⁵ Hence [CLD](#) holds if $|v_d(0)| \geq c$ and $|v_d(1)| \geq c$ (for instance if $c = 0$, i.e. Gaussian copula, or $v_d(0) < -c < c < v_d(1)$).⁶ For $|v| < c$ the local dependence parameter differs from β_d/σ_d at every y , so [CLD](#) fails if exactly one threshold lies in $(-c, c)$.

The conditions for [CLD](#) change without SRI, which we do not assume in the paper. In this case, the analog of the previous example is $Y_d \mid \tilde{V}_z = v \sim N(\mu_d + \beta_d \operatorname{sgn}(|v| - c_z)v, \sigma_d^2 - \beta_d^2)$. This covers the previous example as a special case with $c_0 = c_1 = c$. Then, [CLD](#) holds if $|v_d(0)| \geq c_0$ and $|v_d(1)| \geq c_1$.

C.2 Rank Similarity Restricts Effect Heterogeneity

In Section 4.4, we compared our approach to [Chernozhukov and Hansen \(2005\)](#)'s IVQR approach based on RS. Unlike [CLD](#), IVQR imposes constant local dependence assumptions across potential outcomes. Here we illustrate that these restrictions across potential outcomes restrict the treatment effect heterogeneity in an example.

For ease of notation, we consider a slightly different version of the treatment selection equation [\(3.1\)](#), $D_z = 1\{\tilde{V}_z > q_0(z)\}$, where $\tilde{V}_z \sim \mathcal{N}(0, 1)$ is an alternative normalization such that $q_0(z) := \Phi^{-1}(\pi_0(z))$.⁷ Suppose that the LATE assumptions hold, $\tilde{V}_1 = \tilde{V}_0 = \tilde{V}$ and

⁵Note that $F_{Y_d, \tilde{V}}(y, v) = \int_{-\infty}^v \Phi\left(\frac{y - \mu_d - \beta_d \operatorname{sgn}(|\tau| - c)\tau}{\sqrt{\sigma_d^2 - \beta_d^2}}\right) \phi(\tau) d\tau$. If $v \leq -c$, $\operatorname{sgn}(|\tau| - c) = 1$ for $\tau < v$. If $v \geq c$, $\operatorname{sgn}(|\tau| - c) = -1$ for $-c < \tau < c \leq v$, but the integral over $\tau \in (-c, c)$ does not change.

⁶[CLD\(iii\)](#) does not hold when either $|v_d(0)| = c$ or $|v_d(1)| = c$ because $v \mapsto F_{Y_d, \tilde{V}}(y, v)$ is not differentiable at $v \in \{-c, c\}$, unless $c = 0$.

⁷This is because of the normalization $V_z \mid Z \sim U[0, 1]$ and thus $\tilde{V}_z = \Phi^{-1}(V_z) \mid Z \sim \mathcal{N}(0, 1)$.

$(Y_1, Y_0, \tilde{V}) \perp\!\!\!\perp Z$, and suppose further that $(Y_1, Y_0, \tilde{V})$ are jointly normal with zero means and unit variances, i.e.,

$$\begin{pmatrix} Y_0 \\ Y_1 \\ \tilde{V} \end{pmatrix} \mid Z = z \sim \mathcal{N}_3 \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_{01} & \rho_{0V} \\ \rho_{01} & 1 & \rho_{1V} \\ \rho_{0V} & \rho_{1V} & 1 \end{pmatrix} \right).$$

Here, **CLD** holds by construction, and therefore does not restrict treatment effect heterogeneity since ρ_{01} and hence the relationship between Y_1 and Y_0 is unrestricted. By contrast, RS imposes that $\rho_{0V} = \rho_{1V} = \rho_V$, which restricts treatment effect heterogeneity.

To see the last point, note that the joint distribution of the treatment effect $Y_1 - Y_0$ and treatment propensity $\tilde{V}$ is

$$\begin{pmatrix} Y_1 - Y_0 \\ \tilde{V} \end{pmatrix} \mid Z = z \sim \mathcal{N}_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2(1 - \rho_{01}) & \rho_{1V} - \rho_{0V} \\ \rho_{1V} - \rho_{0V} & 1 \end{pmatrix} \right).$$

Under RS, $\rho_{1V} - \rho_{0V} = 0$ and therefore $Y_1 - Y_0 \perp\!\!\!\perp \tilde{V}$, that is, there is no selection on gains.⁸ Indeed, when selection is based on gains, then RS requires that the potential outcomes are perfectly correlated under joint normality. To see this, suppose that $\tilde{V} \propto Y_1 - Y_0$ and

$$\begin{pmatrix} Y_0 \\ Y_1 \\ Y_1 - Y_0 \end{pmatrix} \sim \mathcal{N}_3 \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_{01} & \rho_{01} - 1 \\ \rho_{01} & 1 & 1 - \rho_{01} \\ \rho_{01} - 1 & 1 - \rho_{01} & 2(1 - \rho_{01}) \end{pmatrix} \right).$$

In this case **CLD** holds trivially, whereas RS requires that $\rho_{01} - 1 = 1 - \rho_{01}$, so that $\rho_{01} = 1$, which corresponds to RI.⁹ Given that the means and variances of Y_1 and Y_0 are the same in this example, $\rho_{01} = 1$ further implies that $Y_1 = Y_0$ a.s.

C.3 **CLD** and Gaussianity

Gaussianity of the copula of (Y_d, V_z) with the same dependence parameter for $z \in \{0, 1\}$ is sufficient for **CLD** but not necessary. Here we show that the sufficient condition for **CLD** mentioned in Section 2.3 is equivalent to Gaussianity when imposed on $\mathcal{V} = (0, 1)$. To see this, suppose that $\rho_{Y_d, V_z}(y, v) = \rho_{Y_d}(y)$ for all $v \in \mathcal{V} = (0, 1)$. By Lemma 4.1, $\Pr[Y_d \leq y, V_z \leq v] = \int_0^v \Phi(a_{d,y} + b_{d,y} \Phi^{-1}(\tilde{v})) d\tilde{v}$, where $a_{d,y} = a_{Y_d, V_z}(y, v)$ and $b_{d,y} = b_{Y_d, V_z}(y, v)$

⁸If Y_1 and Y_0 have non-unit variances σ_1^2 and σ_0^2 , then RS requires $\text{Cov}(Y_1 - Y_0, \tilde{V}) = (\sigma_1 - \sigma_0)\rho_V$, which allows for selection on gains, but the relationship between $Y_1 - Y_0$ and $\tilde{V}$ is restricted. In this case, $\text{Cov}(Y_1 - Y_0, \tilde{V}) = \sigma_1\rho_{1V} - \sigma_0\rho_{0V}$ in general.

⁹The result still holds when the potential outcomes have non-zero means and non-unit variances.

do not depend on v . For $y < y'$ with $0 < F_{Y_d}(y)$ and $F_{Y_d}(y') < 1$, $v \mapsto \Pr[y < Y_d \leq y', V_z \leq v] = \int_0^v [\Phi(a_{d,y'} + b_{d,y'}\Phi^{-1}(\tilde{v})) - \Phi(a_{d,y} + b_{d,y}\Phi^{-1}(\tilde{v}))] d\tilde{v}$ is nondecreasing and its integrand is continuous. It follows that $\Phi(a_{d,y'} + b_{d,y'}\Phi^{-1}(v)) \geq \Phi(a_{d,y} + b_{d,y}\Phi^{-1}(v))$ and, because Φ is strictly increasing, this means that $(a_{d,y'} - a_{d,y}) + (b_{d,y'} - b_{d,y})\Phi^{-1}(v) \geq 0$ for all $v \in (0, 1)$. Since $\Phi^{-1}(v)$ ranges over $\mathbb{R}$, $b_{d,y'} = b_{d,y}$, and therefore $\rho_{Y_d}(y) = \rho_{Y_d}$, i.e. $\rho_{Y_d}(y)$ does not depend on y , because $\rho_{Y_d}(y)$ is a one-to-one function of $b_{d,y}$. Thus, (Y_d, V_z) has a Gaussian copula. The converse is immediate. Similarly, one can show that $F_{Y_d|V_z}(y | v) = \Phi(a_{d,y} + b_{d,y}\Phi^{-1}(v))$ for all $v \in (0, 1)$ and $z \in \{0, 1\}$ is equivalent to a Gaussian copula.

The key assumption underlying the above result is that $\mathcal{V} = (0, 1)$. If $\mathcal{V}$ is a proper subinterval of $(0, 1)$, the argument does not apply. This is why the sufficient condition in Section 2.3 is stated for $\mathcal{V} \subseteq (0, 1)$ and CLD only restricts the local dependence parameter at and around d -treatment thresholds and thus accommodates non-Gaussian copulas (e.g., Figure 1 and Section 4.3).

C.4 Further Details on Roy Models

First, we prove (4.8) in Section 4.3. By $Y_0 \perp\!\!\!\perp V \mid \Theta$, the law of iterated expectations, and (4.5),

$$\mathbb{E}[Y_0 1\{V \leq v\}] = \mathbb{E}[\mathbb{E}[Y_0 \mid \Theta] F_{V|\Theta}(v \mid \Theta)] = v \mathbb{E}[Y_0] + \ell_0(v) \mathbb{E}[Y_0 \Theta_0] + \ell_1(v) \mathbb{E}[Y_0 \Theta_1].$$

Similarly, $\mathbb{E}[\Theta_z 1\{V \leq v\}] = \mathbb{E}[\Theta_z F_{V|\Theta}(v \mid \Theta)] = v \mathbb{E}[\Theta_z] + \ell_0(v) \mathbb{E}[\Theta_z \Theta_0] + \ell_1(v) \mathbb{E}[\Theta_z \Theta_1] = \ell_z(v)$ by the moment normalizations in LT. Since $V \sim U(0, 1)$, dividing both displays by v gives $\mathbb{E}[\Theta_z \mid V \leq v] = \ell_z(v)/v$ and, with $\mathbb{E}[Y_0] = 0$, yields expression (4.8).

Next, we provide further interpretation of the control function $\lambda_0(v)$ under LT and prove (4.9) along the way. Consider (4.8). Defining $a_z := -\int_{-\infty}^{\infty} \Delta_z(F_{Y_0}(y)) dy$, (4.8) becomes

$$\lambda_0(v) = \frac{a_0 L_0(v) + a_1 L_1(v)}{v}, \quad (\text{C.1})$$

which can be derived as follows. Conditioning on Θ in (4.5) as above gives $\mathbb{E}[\Theta_z 1\{Y_0 \leq y\}] = \delta_z(F_{Y_0}(y))$, and hence $\mathbb{E}[\Theta_z 1\{Y_0 > y\}] = -\delta_z(F_{Y_0}(y))$ by $\mathbb{E}[\Theta_z] = 0$. Multiplying the pointwise identity $Y_0 = \int_0^{\infty} 1\{Y_0 > y\} dy - \int_{-\infty}^0 1\{Y_0 \leq y\} dy$ by Θ_z , taking expectations, and interchanging the order of integration (valid given $\mathbb{E}[|Y_0 \Theta_z|] < \infty$) yields $\mathbb{E}[Y_0 \Theta_z] = -\int_{-\infty}^{\infty} \delta_z(F_{Y_0}(y)) dy$. With $\ell_z = \kappa_z L_z$ and $\delta_z = \Delta_z/\kappa_z$, this equals a_z/κ_z , and the product of the payoff and selection factors is $(a_z/\kappa_z)(\kappa_z L_z(v)/v) = a_z L_z(v)/v$, delivering (C.1).

(C.1) shows that, under LT, two-channel sorting in the control function is determined by

the wage-rank loading a_z and the anchor $L_z(v)$. Note that the sign of $\lambda'_0(\pi_0(z))$ is determined by the relative magnitude of the mean wage of inframarginal individuals $\lambda_0(v)$ and the mean wage of marginal individuals $m_0(v) := E[Y_0 \mid V = v]$ evaluated at $\pi_0(z)$, because

$$\lambda'_0(\pi_0(z)) = \frac{m_0(\pi_0(z)) - \lambda_0(\pi_0(z))}{\pi_0(z)}, \quad (\text{C.2})$$

where, under [LT](#),

$$\lambda_0(\pi_0(z)) = \text{Cov}(Y_0, \Theta_z) E[\Theta_z \mid V \leq \pi_0(z)], \quad (\text{C.3})$$

$$m_0(\pi_0(z)) = \text{Cov}(Y_0, \Theta_z) E[\Theta_z \mid V = \pi_0(z)] + \text{Cov}(Y_0, \Theta_{1-z}) E[\Theta_{1-z} \mid V = \pi_0(z)]. \quad (\text{C.4})$$

While [\(C.2\)](#) holds for any joint distribution of (Y_0, V) , [LT](#) gives economic content to its two components, [\(C.3\)](#) and [\(C.4\)](#). By the anchoring [\(4.6\)](#), $E[\Theta_{1-z} \mid V \leq \pi_0(z)] = 0$, so the inframarginal mean wage [\(C.3\)](#) is the payoff-weighted prevalence of trait z alone among incumbent workers, whereas the marginal mean wage [\(C.4\)](#) combines the payoff-weighted prevalences of both traits among marginal entrants. Substituting [\(C.3\)](#) and [\(C.4\)](#) into [\(C.2\)](#) then yields [\(4.9\)](#), whose economic interpretation is given in the main text.

C.5 Details of Remark [4.1](#)

We formally show that a two-component Gaussian copula mixture violates [CLD\(i\)](#). For simplicity, consider the standard two-component Gaussian copula mixture

$$C_M(u, v) = \omega C(u, v; \rho_1) + (1 - \omega) C(u, v; \rho_2), \quad (u, v) \in (0, 1)^2, \quad (\text{C.5})$$

where $0 \leq \omega \leq 1$ and $\rho_1, \rho_2 \in (-1, 1)$. Since a convex combination of copulas is a copula, C_M is a valid copula. For each $(u, v) \in (0, 1)^2$, define $\rho_M(u, v)$ as the unique solution to

$$C(u, v; \rho_M(u, v)) = C_M(u, v). \quad (\text{C.6})$$

The question is whether

$$\rho_M(u, v_0) = \rho_M(u, v_1) \quad \text{for all } u \in (0, 1) \quad (\text{C.7})$$

can hold nontrivially when $v_0 \neq v_1$ where $v_z := \pi_d(z)$.

Proposition C.1 (Failure of [CLD\(i\)](#) for Gaussian mixture). *Let C_M be defined by [\(C.5\)](#) with $0 < \omega < 1$, $\rho_1 \neq \rho_2$, and $\rho_1, \rho_2 \in (-1, 1)$. Let $v_0, v_1 \in (0, 1)$ with $v_0 \neq v_1$. Then there*

exists $u^* \in (0, 1)$ such that $\rho_M(u^*, v_0) \neq \rho_M(u^*, v_1)$.

Remark C.1. *CLD(i) would only hold in degenerate cases: (i) $\omega \in \{0, 1\}$; (ii) $\rho_1 = \rho_2$; (iii) $v_0 = v_1$.* $\square$

C.6 Proof of Lemma 4.1

The result follows from the following property of the Gaussian copula:

$$C(u, v; \rho) = \int_0^v \Phi \left(\frac{\Phi^{-1}(u) - \rho \Phi^{-1}(\tilde{v})}{\sqrt{1 - \rho^2}} \right) d\tilde{v}. \quad (\text{C.8})$$

Indeed, by (C.8) and $V \sim U(0, 1)$,

$$\Pr[Y \leq y \mid V \leq v] = \frac{F_{Y,V}(y, v)}{F_V(v)} = \frac{1}{v} \int_0^v \Phi \left(\frac{\Phi^{-1}(F_Y(y)) - \rho_{Y,V}(y, v) \Phi^{-1}(\tilde{v})}{\sqrt{1 - \rho_{Y,V}(y, v)^2}} \right) d\tilde{v}.$$

Conversely, by (C.8) and uniqueness of the LGR

$$\begin{aligned} F_{Y,V}(y, v) = \Pr[Y \leq y \mid V \leq v] F_V(v) &= \int_0^v \Phi \left(\frac{\Phi^{-1}(F_Y(y)) - \rho_{Y,V}(y, v) \Phi^{-1}(\tilde{v})}{\sqrt{1 - \rho_{Y,V}(y, v)^2}} \right) d\tilde{v} \\ &= C(F_Y(y), v; \rho_{Y,V}(y, v)). \end{aligned}$$

C.7 Proof of Theorem 4.1

The result follows from Lemma 4.1 and noting that (i) $\rho_{Y_d, V_z}(y, v) = -b_{Y_d, V_z}(y, v) / \sqrt{1 + b_{Y_d, V_z}(y, v)^2}$ and (ii) $a_{Y_d, V_z}(y, v)$ only depends on v through $\rho_{Y_d, V_z}(y, v)$.

C.8 Proof of Theorem 4.2

At $v = \pi_0(z)$, (4.5), (4.6) and $\Delta_z(u) := C(u, \pi_0(z); r(u)) - u \pi_0(z)$ give

$$F_{Y_0, V}(y, \pi_0(z)) = F_{Y_0}(y) \pi_0(z) + \Delta_z(F_{Y_0}(y)) = C(F_{Y_0}(y), \pi_0(z); r(F_{Y_0}(y))).$$

By LGR (Lemma 2.1) and strict monotonicity of $r \mapsto \Phi_2(\cdot, \cdot; r)$ on $(-1, 1)$ at any interior $(F_{Y_0}(y), \pi_0(z))$, $\rho_{Y_0, V}(y, \pi_0(z)) = r(F_{Y_0}(y))$ for $z \in \{0, 1\}$ and $F_{Y_0}(y) \in (0, 1)$.

C.9 Proof of Proposition C.1

Let $a = \Phi^{-1}(u)$, $b_0 = \Phi^{-1}(v_0)$, and $b_1 = \Phi^{-1}(v_1)$. Since $v_0 \neq v_1$, we have $b_0 \neq b_1$. Fix $u \in (0, 1)$ or equivalently $a \in \mathbb{R}$. Define the curve

$$\Gamma_a = \{(x_a(\rho), y_a(\rho)) : \rho \in (-1, 1)\},$$

where $x_a(\rho) = \Phi_2(a, b_0; \rho)$ and $y_a(\rho) = \Phi_2(a, b_1; \rho)$. The point on this curve indexed by ρ records the two Gaussian copula slice values at v_0 and v_1 for the same local dependence parameter ρ . For the mixture copula, we have

$$C_M(u, v_0) = \omega x_a(\rho_1) + (1 - \omega)x_a(\rho_2),$$

$$C_M(u, v_1) = \omega y_a(\rho_1) + (1 - \omega)y_a(\rho_2).$$

Thus the pair $(C_M(u, v_0), C_M(u, v_1))$ is a chord point between the two points $(x_a(\rho_1), y_a(\rho_1))$ and $(x_a(\rho_2), y_a(\rho_2))$ on the curve Γ_a . If CLD(i) held at this u , then there would exist some $r = r(u) \in (-1, 1)$ such that $C(u, v_0; r(u)) = C_M(u, v_0)$ and $C(u, v_1; r(u)) = C_M(u, v_1)$, or equivalently

$$x_a(r) = \omega x_a(\rho_1) + (1 - \omega)x_a(\rho_2), \tag{C.9}$$

$$y_a(r) = \omega y_a(\rho_1) + (1 - \omega)y_a(\rho_2). \tag{C.10}$$

That is, the chord point would have to lie again on Γ_a .

We now show that this is not possible for a suitable interior value of u . Since $\frac{\partial}{\partial \rho} \Phi_2(a, b; \rho) = \phi_2(a, b; \rho) > 0$, $x_a(\rho)$ is strictly increasing in ρ . Thus, the curve Γ_a can be written as the graph $y = q_a(x)$ with $x = x_a(\rho)$, whose slope is

$$q'_a(x_a(\rho)) = \frac{y'_a(\rho)}{x'_a(\rho)} = \frac{\phi_2(a, b_1; \rho)}{\phi_2(a, b_0; \rho)}.$$

By the bivariate normal density, differentiating $\log q'_a(x_a(\rho))$ with respect to ρ gives

$$\frac{d}{d\rho} \log q'_a(x_a(\rho)) = \frac{(b_1 - b_0)\{a(1 + \rho^2) - \rho(b_0 + b_1)\}}{(1 - \rho^2)^2}. \tag{C.11}$$

Note that

$$\text{sign}(q''_a(x_a(\rho))) = \text{sign}\left(\frac{d}{d\rho} q'_a(x_a(\rho))\right) = \text{sign}\left(\frac{d}{d\rho} \log q'_a(x_a(\rho))\right),$$

where the second equality is because $q'_a(x_a(\rho)) > 0$ and the first equality is because

$$q''_a(x_a(\rho)) = \frac{\frac{d}{d\rho}q'_a(x_a(\rho))}{x'_a(\rho)},$$

but $x'_a(\rho) > 0$.

Now, suppose $\rho_1 < \rho_2$ without loss of generality, and let $I \equiv [\rho_1, \rho_2]$. Choose a large enough that $a(1 + \rho^2) - \rho(b_0 + b_1) > 0$ for all $\rho \in I$, which is possible because the function $\rho \mapsto \rho(b_0 + b_1)/(1 + \rho^2)$ is bounded on compact $I \subset (-1, 1)$. Since $b_1 - b_0 \neq 0$, equation (C.11) then has a non-zero sign on I . Hence q'_a is strictly monotone (i.e., q_a is strictly convex or strictly concave) on the interval between $x_a(\rho_1)$ and $x_a(\rho_2)$. Since the strict convex/concave graph q_a (i.e., the curve Γ_a) lies strictly below/above the interior of the chord, (C.9)–(C.10) cannot happen.¹⁰ Therefore CLD(i) fails.

D Supplements to Section 5

Here we discuss the asymptotic properties of the estimators $\hat{F}_{Y_d|X}$ and $\hat{\rho}_{Y_d;X}$ in Algorithms 5.1, 5.2, and 5.3 and the validity of the bootstrap. The asymptotic properties of the estimators of the target parameters in Algorithm 5.4 and the validity of the bootstrap then follow because all the functionals involved are Hadamard differentiable under standard regularity conditions (e.g., van der Vaart and Wellner, 1996; Chernozhukov et al., 2013). Specifically, the functional delta method implies that $\sqrt{n}(\hat{\delta}_u - \delta_u)$ converges in distribution to a mean-zero Gaussian process Z_δ and the functional delta method for the bootstrap implies that the bootstrap consistently estimates the limiting law (van der Vaart and Wellner, 1996, Chapter 3.9). Note that Hadamard differentiability of the inverse operator fails for discrete and mixed discrete-continuous outcome variables. In this case, one can perform inference on the QSF and QTE using the method proposed by Chernozhukov et al. (2020).

To state the formal results, we introduce some additional notation. Let $\tilde{\mathcal{Y}}$ be a compact subset of $\mathcal{Y}$ when $\mathcal{Y}$ is uncountable and let $\tilde{\mathcal{Y}}$ be equal to $\mathcal{Y}$ otherwise. Define $\tilde{\mathcal{D}}$ analogously. Let $\mathcal{AB} := \{(a, b) : a \in \mathcal{A}, b \in \mathcal{B}\}$ for sets $\mathcal{A}$ and $\mathcal{B}$. Denote the set of bounded functions on the set $\mathcal{A}$ by $\ell^\infty(\mathcal{A})$. Finally, let $\rightsquigarrow$ denote weak convergence.

Using arguments similar to those in Chernozhukov et al. (2025), one can show that the estimators $\hat{F}_{Y_d|X}$ and $\hat{\rho}_{Y_d;X}$ in Algorithms 5.1 and 5.2 satisfy functional central limit theorems

¹⁰Since x_a is strictly increasing, equation (C.9) also implies $r \in (\rho_1, \rho_2)$.

(FCLTs),

$$\begin{aligned}\sqrt{n} \left(\widehat{F}_{Y_d|X}(y|x) - F_{Y_d|X}(y|x) \right) &\rightsquigarrow Z_{F_{Y_d|X}(y|x)} \text{ in } \ell^\infty(\tilde{\mathcal{D}}\mathcal{X}\tilde{\mathcal{Y}}), \\ \sqrt{n} (\widehat{\rho}_{Y_d;X}(y;x) - \rho_{Y_d;X}(y;x)) &\rightsquigarrow Z_{\rho_{Y_d;X}(y;x)} \text{ in } \ell^\infty(\tilde{\mathcal{D}}\mathcal{X}\tilde{\mathcal{Y}}),\end{aligned}$$

where $Z_{F_{Y_d|X}(y|x)}$ and $Z_{\rho_{Y_d;X}(y;x)}$ are zero-mean Gaussian processes, and that the multiplier bootstrap is valid. We therefore focus on the estimators $\widehat{F}_{Y_d|X}$ and $\widehat{\rho}_{Y_d;X}$ in Algorithm 5.3, which have a different structure.

We start by introducing some additional notation. Let $G(u) := \phi(u)/[\Phi(u)\Phi(-u)]$ and define the expected Hessians

$$\begin{aligned}H(y) &:= -\mathbb{E}[G(B(D, Z, X)' \beta(y)) \phi(B(D, Z, X)' \beta(y)) B(D, Z, X) B(D, Z, X)'], \\ H(d) &:= -\mathbb{E}[G(B(Z, X)' \pi(d)) \phi(B(Z, X)' \pi(d)) B(Z, X) B(Z, X)'].\end{aligned}$$

We impose two assumptions. The first assumption is essentially a first-stage assumption ensuring that $a_{d,y;x}$ and $b_{d,y;x}$ are well-defined.

Assumption D.1. $|(B(1, x) - B(0, x))' \pi(d)| \geq c > 0$ for $(d, x) \in \mathcal{D}\mathcal{X}$.

The second assumption is a version of Condition DR in Chernozhukov et al. (2013). It ensures that the first-step coefficient estimators $\widehat{\beta}(y)$ and $\widehat{\pi}(d)$ satisfy FCLTs. We present the theoretical results for the case where the outcome is continuously distributed. Results for discrete outcomes can be obtained similarly.

Assumption D.2. (i) $\mathcal{D}$ is a compact interval in $\mathbb{R}$ and $\mathcal{X}$ is compact. (ii) The conditional densities $f_{Y|D,Z,X}(y | d, z, x)$ and $f_{D|Z,X}(d | z, x)$ exist and are uniformly bounded and uniformly continuous. (iii) $B(d, z, x)$ and $B(z, x)$ are bounded over $\tilde{\mathcal{D}} \times \{0, 1\} \times \mathcal{X}$ and continuous, $\mathbb{E}[\|B(D, Z, X)\|^2] < \infty$, and $\mathbb{E}[\|B(Z, X)\|^2] < \infty$. (iv) The minimum eigenvalues of $-H(y)$ and $-H(d)$ are bounded away from zero over $\tilde{\mathcal{Y}}$ and $\tilde{\mathcal{D}}$.

The following theorem establishes FCLTs for the estimators $\widehat{F}_{Y_d|X}(y|x)$ and $\widehat{\rho}_{Y_d;X}(y;x)$ in Algorithm 5.3 and the validity of the exchangeable bootstrap.

Theorem D.1. Consider the estimators $\widehat{F}_{Y_d|X}(y|x)$ and $\widehat{\rho}_{Y_d;X}(y;x)$ defined in Algorithm 5.3. Suppose that Assumptions D.1 and D.2 hold. Then,

$$\begin{aligned}\sqrt{n} \left(\widehat{F}_{Y_d|X}(y|x) - F_{Y_d|X}(y|x) \right) &\rightsquigarrow Z_{F_{Y_d|X}(y|x)} \text{ in } \ell^\infty(\tilde{\mathcal{D}}\mathcal{X}\tilde{\mathcal{Y}}), \\ \sqrt{n} (\widehat{\rho}_{Y_d;X}(y;x) - \rho_{Y_d;X}(y;x)) &\rightsquigarrow Z_{\rho_{Y_d;X}(y;x)} \text{ in } \ell^\infty(\tilde{\mathcal{D}}\mathcal{X}\tilde{\mathcal{Y}}),\end{aligned}$$

where $Z_{Y_d|X}(y|x)$ and $Z_{Y_d;X}(y;x)$ are zero-mean Gaussian processes defined in the proof in SA D.1, and the exchangeable bootstrap is consistent for estimating the limiting laws.

D.1 Proof of Theorem D.1

Step 1: FCLT for $(\hat{\beta}(y)', \hat{\pi}(d)')'$ and bootstrap validity. In this step, we establish FCLTs for $\hat{\beta}(y)$ and $\hat{\pi}(d)$ and the validity of the bootstrap, building on Chernozhukov et al. (2013).

Under Assumption D.2, Corollary 5.3 in Chernozhukov et al. (2013) implies that

$$\begin{aligned}\sqrt{n}(\hat{\beta}(y) - \beta(y)) &= H^{-1}(y) \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\Phi(B(D_i, Z_i, X_i)' \beta(y)) - I_i(y)}{\Phi(B(D_i, Z_i, X_i)' \beta(y)) \Phi(-B(D_i, Z_i, X_i)' \beta(y))} \phi(B(D_i, Z_i, X_i)' \beta(y)) B(D_i, Z_i, X_i) \\ &\quad + o_P(1), \\ \sqrt{n}(\hat{\pi}(d) - \pi(d)) &= H^{-1}(d) \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\Phi(B(Z_i, X_i)' \pi(d)) - J_i(d)}{\Phi(B(Z_i, X_i)' \pi(d)) \Phi(-B(Z_i, X_i)' \pi(d))} \phi(B(Z_i, X_i)' \pi(d)) B(Z_i, X_i) + o_P(1),\end{aligned}$$

and

$$\sqrt{n} \left(\begin{pmatrix} \hat{\beta}(y) \\ \hat{\pi}(d) \end{pmatrix} - \begin{pmatrix} \beta(y) \\ \pi(d) \end{pmatrix} \right) \rightsquigarrow \begin{pmatrix} Z_{\beta}(y) \\ Z_{\pi}(d) \end{pmatrix} \quad \text{in } \ell^\infty(\tilde{\mathcal{D}}\tilde{\mathcal{Y}})^{\dim(B(Z,X)) + \dim(B(D,Z,X))},$$

where $Z_{\beta}(y)$ and $Z_{\pi}(d)$ are mean-zero Gaussian processes, and that the bootstrap is valid.

Step 2: FCLT for $\hat{\rho}_{Y_d;X}(y;x)$ and $\hat{F}_{Y_d|X}(y|x)$ and bootstrap validity. In this step, we build on Step 1 to establish a FCLT for $\hat{\rho}_{Y_d;X}(y;x)$ and $\hat{F}_{Y_d|X}(y|x)$ and the validity of the bootstrap. Because all maps involved are Hadamard differentiable, the result follows from the functional delta method and the functional delta method for the bootstrap.

Under Assumption D.1, by Step 1 and the functional delta method,

$$\sqrt{n} \left(\begin{pmatrix} \hat{a}_{d,y;x} \\ \hat{b}_{d,y;x} \end{pmatrix} - \begin{pmatrix} a_{d,y;x} \\ b_{d,y;x} \end{pmatrix} \right) \rightsquigarrow \begin{pmatrix} Z_{a_{d,y;x}} \\ Z_{b_{d,y;x}} \end{pmatrix} \quad \text{in } \ell^\infty(\tilde{\mathcal{D}}\mathcal{X}\tilde{\mathcal{Y}})^2,$$

where

$$\begin{aligned}Z_{a_{d,y;x}} &= \frac{B(1,x)' \pi(d) B(d,0,x)' - B(0,x)' \pi(d) B(d,1,x)'}{(B(1,x) - B(0,x))' \pi(d)} Z_{\beta}(y) \\ &\quad + \left(\frac{B(d,0,x)' \beta(y) B(1,x)' - B(d,1,x)' \beta(y) B(0,x)'}{(B(1,x) - B(0,x))' \pi(d)} \right. \\ &\quad \left. - \frac{(B(d,0,x)' \beta(y) B(1,x)' \pi(d) - B(d,1,x)' \beta(y) B(0,x)' \pi(d)) (B(1,x) - B(0,x))'}{((B(1,x) - B(0,x))' \pi(d))^2} \right) Z_{\pi}(d)\end{aligned}$$

and

$$Z_{b_{d,y;x}} = \frac{(B(d, 1, x) - B(d, 0, x))'}{(B(1, x) - B(0, x))'\pi(d)} Z_\beta(y) + \frac{(B(d, 1, x) - B(d, 0, x))'\beta(y)(B(0, x) - B(1, x))'}{((B(1, x) - B(0, x))'\pi(d))^2} Z_\pi(d).$$

Define $\hat{\mu}_{d,y;x} := \hat{a}_{d,y;x}/\sqrt{1 + \hat{b}_{d,y;x}^2}$ and $\mu_{d,y;x} := a_{d,y;x}/\sqrt{1 + b_{d,y;x}^2}$. Then, by the functional delta method, we have

$$\sqrt{n}(\hat{\mu}_{d,y;x} - \mu_{d,y;x}) \rightsquigarrow Z_{\mu_{d,y;x}} \quad \text{in } \ell^\infty(\tilde{\mathcal{D}}\mathcal{X}\tilde{\mathcal{Y}}),$$

where $Z_{\mu_{d,y;x}} = Z_{a_{d,y;x}}/\sqrt{1 + b_{d,y;x}^2} - a_{d,y;x}b_{d,y;x}Z_{b_{d,y;x}}/(1 + b_{d,y;x}^2)^{3/2}$. Moreover, we have that

$$\sqrt{n}(\hat{\rho}_{Y_d|X}(y; x) - \rho_{Y_d|X}(y; x)) \rightsquigarrow Z_{\rho_{Y_d|X}(y; x)} \quad \text{in } \ell^\infty(\tilde{\mathcal{D}}\mathcal{X}\tilde{\mathcal{Y}}),$$

where $Z_{\rho_{Y_d|X}(y; x)} = -Z_{b_{d,y;x}}/(1 + b_{d,y;x}^2)^{3/2}$. Finally, another application of the functional delta method yields¹¹

$$\sqrt{n}(\hat{F}_{Y_d|X}(y|x) - F_{Y_d|X}(y|x)) \rightsquigarrow \phi(\mu_{d,y;x})Z_{\mu_{d,y;x}} =: Z_{F_{Y_d|X}(y|x)} \quad \text{in } \ell^\infty(\tilde{\mathcal{D}}\mathcal{X}\tilde{\mathcal{Y}}).$$

The validity of the bootstrap follows from the functional delta method for the bootstrap.

References

- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, AND A. GALICHON (2010): “Quantile and Probability Curves Without Crossing,” *Econometrica*, 78, 1093–1125.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, AND S. LUO (2025): “Distribution Regression with Sample Selection and UK Wage Decomposition,” *Journal of Political Economy*, 133, 3952–3992.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, AND B. MELLY (2013): “Inference on Counterfactual Distributions,” *Econometrica*, 81, 2205–2268.
- CHERNOZHUKOV, V., I. FERNÁNDEZ-VAL, B. MELLY, AND K. WÜTHRICH (2020): “Generic Inference on Quantile and Quantile Effect Functions for Discrete Outcomes,” *Journal of the American Statistical Association*, 115, 123–137.
- CHERNOZHUKOV, V. AND C. HANSEN (2005): “An IV model of quantile treatment effects,” *Econometrica*, 73, 245–261.

¹¹If rearrangement is used in the last step of Algorithm 5.3, this step additionally relies on Hadamard differentiability of the rearrangement operator established in Chernozhukov et al. (2010).

- DE CHAISEMARTIN, C. (2017): “Tolerating defiance? Local average treatment effects without monotonicity,” *Quantitative Economics*, 8, 367–396.
- DOS ANJOS, U. U. AND N. KOLEV (2005): “Representation of bivariate copulas via local measure of dependence,” RT-MAE 2005-3.
- GALE, D. AND H. NIKAIDO (1965): “The Jacobian matrix and global univalence of mappings,” *Mathematische Annalen*, 159, 81–93.
- HAN, S. AND E. J. VYTLACIL (2017): “Identification in a generalization of bivariate probit models with dummy endogenous regressors,” *Journal of Econometrics*, 199, 63–73.
- HECKMAN, J. J. AND E. J. VYTLACIL (2007): “Chapter 71 Econometric Evaluation of Social Programs, Part II: Using the Marginal Treatment Effect to Organize Alternative Econometric Estimators to Evaluate Social Programs, and to Forecast their Effects in New Environments,” in *Handbook of Econometrics*, ed. by J. J. Heckman and E. E. Leamer, Elsevier, vol. 6B, 4875–5143.
- IMBENS, G. W. AND J. D. ANGRIST (1994): “Identification and Estimation of Local Average Treatment Effects,” *Econometrica*, 62, 467–475.
- VAN DER VAART, A. W. AND J. A. WELLNER (1996): *Weak Convergence and Empirical Processes: With Applications to Statistics*, Springer New York, NY.
- VYTLACIL, E. J. (2002): “Independence, monotonicity, and latent index models: An equivalence result,” *Econometrica*, 70, 331–341.